RESEARCH MONOGRAPH

Do Budgeting Tools Help Low-Income Students Save?

A randomized experiment with OBMEP scholars, and what their reports can and cannot show

Gabriel Moreno Ribeiro

Advisor: Aaron Litvin, Ph.D., Harvard University

Fundação Estudar
Salvador, Bahia, Brazil

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Research monograph in behavioral economics. Revised and expanded version of the October 2025 paper and the December 2025 extended abstract.

Advisor
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Acknowledgments

This work began as a question I could not stop asking: why do people who know they should save so often end the month with nothing left? Turning that question into an experiment, and then into this monograph, took the help of many people.

My first thanks go to the 208 scholars of the Junior Scientific Initiation Program who agreed to take part. For three months they filled in forms, opened spreadsheets and apps, and told me how much of their stipend they had kept. Every number in this text exists because of them. I also thank their parents and guardians, who read the consent forms and trusted the study with their children's time.

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Gabriel Moreno Ribeiro

Abstract

In this work, I ask whether a simple, free budgeting tool changes how much public school students save out of a small, regular income. I randomized 208 scholars of the Junior Scientific Initiation Program (PIC) of OBMEP, who receive R$ 300 a month, within grade strata, to a control group or to one of three tools: a spreadsheet template, a budgeting app, or both together. From August to October 2025, each student reported how much they had saved in the month. In October, the control group reported saving 44.25 % of the income left after essential expenses, and the tool groups between 91.40 % and 101.07 %. Receiving any tool raised reported saving by 52.0 percentage points (95 % CI 44.2 to 59.8), about R$ 71 a month, and the result holds up under randomization inference, across 108 specifications, and within bounds for attrition. The rest of the evidence calls for caution. The three tools cannot be ranked: the experiment never had the power to do so. The 25 treated students who never logged a single expense reported saving as much in October as the other treated students, which contradicts the attention channel of the model I propose. And the reports do not behave like measures of a habit: net of the group mean, a student's saving in one month does not predict the next month's, even in the control group (intraclass correlation of –0.07), and the amounts are almost never round. In six simulated worlds calibrated to the data, the world in which the effect runs through tool use is rejected; with noisy reports, a real effect equal for everyone and an effect on reporting alone leave the same traces, and the data cannot tell them apart. I conclude that the tools greatly changed what students reported, and that this study cannot say how much of that is money saved. I propose a new experiment with three groups (control, attention placebo, and tool), with the balance of a bank account as the outcome, and I leave the analysis program written and rehearsed on synthetic data. This text, dated October 26, 2026, revises and expands the October 2025 paper and the December 2025 extended abstract, redoing every calculation from each student's data, with open code.

Keywords: saving; financial education; budgeting apps; present bias; randomized experiment; reliability of self-reports; Brazil

Resumo

Neste trabalho, pergunto se uma ferramenta simples e gratuita de orçamento muda quanto estudantes de escola pública poupam de uma renda pequena e regular. Sorteei 208 bolsistas do Programa de Iniciação Científica Jr. (PIC) da OBMEP, que recebem R$ 300 por mês, dentro de estratos de série, entre um grupo de controle e três ferramentas: um modelo de planilha, um aplicativo de orçamento e as duas juntas. De agosto a outubro de 2025, cada aluno informou quanto poupou no mês. Em outubro, o controle declarou poupar 44,25 % da renda que sobrava depois dos gastos essenciais, e os grupos com ferramenta, entre 91,40 % e 101,07 %. Receber qualquer ferramenta aumentou a poupança declarada em 52,0 pontos percentuais (IC 95 % de 44,2 a 59,8), cerca de R$ 71 por mês, e o resultado resiste à inferência por aleatorização, a 108 especificações e a limites para as desistências. O resto da evidência pede cautela. As três ferramentas não podem ser ordenadas: o experimento nunca teve poder para isso. Os 25 alunos tratados que nunca registraram um gasto declararam poupar em outubro o mesmo que os demais tratados, o que contradiz o canal de atenção do modelo que proponho. E os relatos não se comportam como medidas de um hábito: descontada a média do grupo, a poupança de um aluno num mês não prevê a do mês seguinte, inclusive no controle (correlação intraclasse de –0,07), e os valores quase nunca são redondos. Em seis mundos simulados, calibrados nos dados, o mundo em que o efeito passa pelo uso da ferramenta é rejeitado; com relatos ruidosos, um efeito real igual para todos e um efeito só de relato produzem os mesmos sinais, e os dados não distinguem um do outro. Concluo que as ferramentas mudaram muito o que os alunos declararam, e que este estudo não consegue dizer quanto disso é dinheiro poupado. Proponho um novo experimento com três grupos (controle, placebo de atenção e ferramenta), com o saldo de uma conta bancária como desfecho, e deixo o programa de análise pronto e ensaiado em dados sintéticos. Este texto, de 26 de outubro de 2026, revê e amplia o artigo de outubro de 2025 e o resumo expandido de dezembro de 2025, refazendo todas as contas a partir dos dados de cada aluno, com código aberto.

Palavras-chave: poupança; educação financeira; aplicativos de orçamento; viés do presente; experimento aleatorizado; confiabilidade de relatos; Brasil

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List of symbols

i, t	student; month (0 = baseline; 1, 2 and 3 = August, September and October 2025)
g, T_{gi}, T_i	group (G1 control, G2 spreadsheet, G3 app, G4 hybrid); indicator that student i was assigned to group g; indicator for any tool
y	monthly PIC stipend, R$ 300
e_i, R_i	the student's essential expenses at baseline; residual income, $R_i = y - e_i$ (R$)
S_{it}	reported saving in month t, as a percentage of baseline residual income
s_{it}, r_{it}	true saving and reported saving in the model of Chapter 4 (R$)
c, c^*	planned discretionary spending; its optimal value (R$)
m_i, a_{it}, a_{0i}	total of small unplanned purchases; share of them the student notices; that share without a tool
u_{it}, κ	intensity of tool use (logs per week); how much each unit of use raises attention
v, Δv	marginal value of one real saved; increase in that value when the goal becomes salient
β, δ	present bias factor; per-period discount factor (dimensionless, between 0 and 1)
ε_{it}, η_{it}	the month's shock to saving; reporting error
b	mean reporting deviation of the treated students (R$)
$Y_i(1)$, $Y_i(0)$	potential outcomes of student i with and without treatment
τ	average treatment effect
α, β_g, γ, μ_s	constant, effect of group g, covariate coefficients, and fixed effect of stratum s in the main regression
X_i	baseline covariates (age, gender, rural origin, laptop, essential expenses)
ρ	correlation of true saving across months; in ANCOVA, correlation between the outcome and the baseline
λ	reliability of the report: share of the variance of the reports that is signal
σ, σ^2	standard deviation and variance
n, n_1, n_2	sample size; sizes of two groups being compared
α (test)	significance level, 5 % in this work
z_q	quantile q of the standard normal distribution
d, g (Hedges)	difference between means in standard deviations
p	p-value of a test
SD, CI, pp	standard deviation; confidence interval; percentage points
SE, EP	standard error; EP is the symbol for the standard error in some formulas
ICC	intraclass correlation
MDE	minimum detectable effect
HC2	standard error robust to unequal variances across observations, in the version of MacKinnon and White (1985)
ANCOVA	regression of the final outcome on treatment, with the baseline outcome as a control
TOST	two one-sided tests of equivalence

CHAPTER 1

Introduction

1.1 Why this question

I grew up in Brazil and saw up close how families fall into debt. It was almost never because of one big mistake. It was the sum of decisions made in a hurry, with little money and little information: an installment that looked small, a loan taken out to cover another one, a bill that arrived before the paycheck. I began to study behavioral economics because I wanted to understand how people with little money decide what to do with it, and whether small, cheap changes in the way they look at that money can help.

To study this carefully, I needed a group of people with the same income, who received that money every month and whom I could follow. The Junior Scientific Initiation Program (PIC) of OBMEP offers exactly that. It is tied to the Brazilian Mathematics Olympiad for Public Schools (OBMEP) and pays a monthly stipend of R$ 300 to the medalists who take part. The stipend is the same for everyone: a middle school student and a high school student, in the city or in the countryside, receive the same amount, on the same schedule. For some of these students, the stipend may be the first regular income they manage on their own. From the point of view of someone who studies saving, the PIC works as a natural laboratory. The income is fixed, and what changes from one student to the next is what each one does with it.

The question also matters outside the PIC. In 2024 the federal government created Pé-de-Meia, which pays monthly incentives to low-income students in public high schools (Brasil, 2024), and the Base Nacional Comum Curricular, Brazil's national curriculum, treats financial education as a cross-cutting theme to be worked on throughout school (Brasil. Ministério da Educação, 2018). More and more Brazilian teenagers receive some regular money, and schools are being asked to teach them what to do with it. If a free tool and one hour of training changed the behavior of these young people, it would be a cheap policy, easy to take to many schools. If they did not, that would also be good to know before spending money on it.

The question of this work fits in one sentence: *Does giving a PIC scholar a simple, free tool to keep track of their own money, a spreadsheet, an app, or both, change how much of the stipend they save?*

1.2 Reasons to expect that it does, and to doubt it

There are good reasons to think it does, and Chapter 2 explains each of them step by step. The first is scarcity. When money is short, attention narrows to whatever is due today, and little is left for planning the rest of the month (Mani et al., 2013; Mullainathan and Shafir, 2013). The second is present bias: people give the present more weight than they themselves would want to give it when they think the matter over calmly, and so they put off saving until next month, every month (Laibson, 1997). The third is mental accounting. Money that has not been set aside for a purpose tends to get spent, while money with a label tends to stay where it is (Thaler, 1999). A budgeting tool touches all three. It makes visible the small purchases that would go unnoticed, like a snack on the way out of school or a phone top-up, it splits money into categories, and it puts a number on the saving goal. The app also sends weekly reminders to log expenses, and reminders alone have raised saving in field experiments (Karlan et al., 2016).

There are also good reasons to doubt it. Financial education courses, the most common policy in this area, change behavior very little. Pooling dozens of randomized evaluations, Kaiser et al. (2022) estimate effects on financial behavior of about a tenth of a standard deviation, and Fernandes et al. (2014) find even smaller effects, which fade over time. A tool is different from a class, but it has a problem of its own: it only helps if it is used. A spreadsheet that a student opens on the first day and then forgets makes nothing visible. If a tool's effect runs through use, those who use it more should save more, and that is something the data can confirm or refute.

Taken together, the two sides lead to a harder question than "does it work?". If the tool works, *through which path* does it work? A serious answer has to say what the data would show if each explanation were the right one, and what they would show if none were. This is the question the first version of this study failed to ask, and the one this version tries to answer.

1.3 What I did, and why I redid everything

To answer the question, I ran a randomized experiment in 2025. Of the 389 scholars contacted, 208 returned the interest and consent form; all of them had a smartphone and internet access, and none had ever used a budgeting tool in a systematic way. I randomized these students, within 7 grade strata, into four groups of 52: a control group, a group with a spreadsheet template in Google Sheets, a group with the budgeting app Mobilis, and a hybrid group, with both tools. Everyone was invited to a 60-minute online workshop on budgeting and saving goals, recorded for those who could not watch it live. The first part was the same for everyone. After it, the students in the three tool groups spent 30 minutes in separate rooms, learning to use their own tool. The intervention ran from August 1 to October 31, 2025. At the end of each

month, each student reported on an online form how much they had saved, and the outcome is that saving as a percentage of the income left over after essential expenses (Section 2.6). Of the 208, 203 made it to the end. Chapter 5 describes the design in detail.

I wrote the first version of this study (v1), a paper, in October 2025 (Ribeiro, 2025a), and an extended abstract in December 2025 (Ribeiro, 2025b). Neither was peer reviewed. In them, I presented the results with an analysis of variance, a chart of the monthly means, and a regression of saving on how often the tool was used. The conclusions were optimistic: all three tools raised saving, the hybrid group was the best, thanks to a "synergy" between spreadsheet and app, and engagement with the tool was the main driver of the effect. Two figures used illustrative values in place of the data.

This version goes back to each student's row in the spreadsheet and redoes everything from scratch, with public code. I did this for three reasons. The first is that the first version worked with summaries, such as means and an analysis of variance table, and never showed the data for each student; only by going back to the microdata could I check whether each number was right. The second is that the first version did not test its own explanation. It said the tools worked by making spending visible, but it never asked what the data would show if that explanation were wrong. The third is that the whole study rests on what the students said, and I had never checked how far those reports can be trusted.

The first step was an audit. I compared every numerical claim in the first version with the microdata. Of the 24 claims, 13 match, 5 differ in value without changing the conclusion, 4 point in the right direction but are not statistically significant, and 2 cannot be reproduced. Table C.1, in Appendix C, lists them one by one. All the descriptive results match to the second decimal place, which shows that the two versions use the same data; the problems are in the inferences. The most serious one is the relationship between use and saving. The first version reported that each additional weekly log raised saving by 12.1 percentage points. I tested both possible readings of that outcome, and neither gives a positive, significant slope (Section 7.3).

Beyond the audit, this version adds four things. A simple model of attention and saving, with predictions the data can reject (Chapter 4). Power calculations and equivalence tests, which say what this experiment could and could not detect. An examination of how reliable the reports are. And a proposal for a follow-up study, with the analysis program written and tested before any data are collected. The experiment was not preregistered, and almost everything that goes beyond the first version was done after the data existed. For that reason, these analyses should be read as exploratory.

1.4 What I found

The main result is large. In October, the control group reported saving 44.25 % of their residual income, and the three tool groups reported between 91.40 % and 101.07 %. Receiving any of the tools raised reported saving by 52.0 pp (95 % confidence interval from 44.2 to 59.8), about R$ 71 a month. The result holds up under everything I tried: none of 20,000 new random assignments of the groups produced a difference that large, the estimate stays large and significant across 108 different specifications, and the five dropouts cannot explain it (Chapter 6).

The rest of the evidence is less favorable, and it changes how that number should be read. The three tools cannot be ranked. The hybrid group had the highest mean, but its lead over the app group, 3.2 pp, is far below the smallest difference between two groups that this experiment could detect with 80 % power, 13.5 pp. Nor do the data show that the tools are equivalent: they may differ by a few points or by more than ten (Chapter 7).

Use of the tool does not track saving. The 25 treated students who never logged a single expense reported saving 96.6 % in October, the same as those who did log (96.7 %). In the model of Chapter 4, this is the most direct prediction of the idea that the tool works by making spending visible (Prediction P2), and it fails.

The reports look unreliable. If a student's saving has any regularity, someone who saves above average in one month should tend to save above average in the next. In the data, after removing the group means, the correlation between consecutive months is 0.00, and the intraclass correlation over the four months is –0.05. The amounts in reais are almost never round numbers, even though sums recalled from memory usually are. In October, 43 % of the treated students reported saving more than their entire baseline residual income. And the effect, 2.0 standard deviations of the control group, is about twenty times the typical effect of financial education (Chapter 8).

Put together, this leads me to a narrower conclusion than the one in the first version. The tools greatly changed what students *reported* saving, and this experiment cannot say how much of that is money actually put away. To understand what it could say, I built synthetic worlds: computer-generated data sets, with the same size and the same means as the study, in which I know the truth. In some, the tool works through attention; in others, every treated student gains the same amount, whether or not they use the tool; in others, nobody saves more and the treated students just report more. The real data do not look like the attention world, but they look equally like the other two (Section 8.6). Telling true saving apart from a change in reporting takes a different design, and Chapter 9 proposes one: measure saving in a bank account, without asking the student, and compare the tool with a placebo group that gets the same attention on a topic unrelated to money.

This conclusion does not say the tools are useless. It says the measure used does not let us know. It is possible that the students really did save more, and that the

reports are merely noisy. It is also possible that a good part of the effect lies in the way they filled in the form, for example out of a wish to show progress to the people who gave the training. A skeptical reader who starts from the typical effect in the literature and takes the reports at face value still ends up believing in an effect several times larger than that typical effect (Section 8.5). But that calculation assumes the reports are unbiased, and that is exactly what is in doubt.

1.5 Where the numbers in this text come from

No number in this text that comes from the data, from an estimate, or from a simulation was typed by hand. The Python programs in the `code/` folder read the students' original spreadsheet, do the calculations, draw the figures, build the tables, and write each number to a file, `results/numbers.json`. The LaTeX text reads that file and prints the numbers through macros. If a calculation changes, the number changes everywhere it appears, and none can be left out of date in some corner of the text. Appendix B shows the code.

Each figure and each table says, in its source line, where its numbers come from, using one of five labels. *Measured* is the value as the student reported it, read from the spreadsheet without any calculation, such as baseline essential expenses. *Estimated* is what I calculated from the data, such as an effect, a confidence interval, or a correlation. *Simulated* is what comes from computer-generated synthetic data, always identified as such, in programs 14 to 16. *Illustrative* is a parameter I chose to explain an idea, such as the size of present bias in the figures of Chapter 2, and it says nothing about the students. And *v1* is a value taken from the first version, quoted for comparison.

Anyone can redo everything with one command, from the root of the repository:

```
python3 code/reproduce_all.py
```

The program runs the others in the right order, each random draw with a fixed seed, without using the internet. At the end, it writes the SHA-256 hash of each generated file to `data/sha256_outputs.txt`, so that anyone who runs it again on the same machine can check that they got the same files, byte for byte.

The economic ideas I use come from the literature, and each chapter says where. What is mine is the design of the experiment, the analyses, and the way I read them. Chapter 10 keeps the two apart.

1.6 What each chapter does

The next three chapters lay the groundwork. Chapter 2 teaches, from scratch, the economics of saving that the rest of the text uses: income, consumption and saving,

the choice between today and tomorrow, present bias, mental accounting, limited attention, and scarcity. It also shows what a PIC scholar's budget looks like and what the literature says about the effect of financial education and of tools, so that the reader knows what size of effect to expect. Chapter 3 explains how a randomized experiment separates cause from coincidence: potential outcomes, random assignment, estimators, uncertainty, statistical power, multiple comparisons, randomization inference, equivalence tests, and measurement problems. Chapter 4 presents a simple model of attention and saving and draws five predictions from it, plus an alternative explanation in which only the report changes.

Chapter 5 describes the study: the PIC, the sample, the four groups, the outcome, the ethics, the estimation strategy, and the balance between groups. Chapter 6 gives the main result, month by month, and all the robustness checks. Chapter 7 asks which tool works best, whether using more means saving more, and who benefits, and sets the five predictions against the data. Chapter 8 asks how much of the effect is real saving: who reported and to whom, how persistent the reports are, saving above 100 %, round amounts, what a skeptical reader should conclude, and what the synthetic worlds show. Chapter 9 proposes the follow-up study, with the sample size calculation, the analysis plan, and a full rehearsal of that plan on synthetic data. Chapter 10 answers the question of this introduction with what the data allow us to say.

The appendices hold the longer derivations (Appendix A), the code (Appendix B), and the tables, among them the audit of the first version and a general table of the data (Appendix C).

CHAPTER 2

Foundations of the economics of saving

This chapter gathers the economic ideas the rest of the text uses, explained from the beginning. I do not assume the reader has studied economics. I start with the simplest part: what income, consumption and saving are, and why anyone would save. Then I show how economists describe the choice between spending today and spending tomorrow, and why real people depart from that description in predictable ways: they give too much weight to the present, they split money into mental accounts, and they cannot pay attention to everything. Next, I look at the budget of a PIC scholar, with the study's numbers, and explain the experiment's outcome. I close with what the literature says about the effect of financial education and of budgeting tools, because that tells us what size of effect it would be reasonable to expect, and with the Brazilian context. For a short introduction to behavioral economics, I recommend Laibson and List (2015) and the book by Thaler and Sunstein (2008); for a long review of choices over time, Ericson and Laibson (2019).

2.1 Income, consumption and saving

Income is the money a person receives in a period, for example in a month. For a PIC scholar, the most regular income is the R$ 300 stipend. **Consumption** is what the person spends in that period, on anything at all: a bus fare, a snack, a phone top-up, a book, a contribution to the household bills. **Saving** in the period is what is left:

$$s_t = y_t - c_t, \tag{2.1}$$

where y_t is income in month t, c_t is consumption, and s_t is saving. The equation is an accounting identity. It holds by definition and says nothing about behavior. A student who receives R$ 300 and spends R$ 240 has saved R$ 60 that month, whether the money is in an account, in a piggy bank, or in a wallet.

There is a distinction that will matter a great deal in Chapter 8. Saving in a month is a **flow**: it is how much was added to the reserve that month. The total put away so far, adding up all the months, is a **stock**, the balance. If a student saves R$ 60 in August, R$ 60 in September, and R$ 60 in October, October's saving is R$ 60, but

the balance at the end of October is R$ 180. When someone asks "how much did you save?", the answer can be either one, and the person answering does not always notice the difference.

2.1.1 Why save

To save is to give up consuming now in order to be able to consume later. There are three main reasons to do this, and all three show up in a scholar's life.

The first is shocks. Life brings expenses that cannot be foreseen: the phone falls and the screen cracks, the family needs help in a hard month, a school trip comes up. Someone with a reserve can absorb the shock without cutting essentials and without borrowing. Economists call this precautionary saving.

The second is goals. Some things cost more than a whole stipend, like a laptop, a course, or the fee for an olympiad in another city. To buy them without debt, one has to put the money together little by little. In this study's sample, every student had a smartphone, but only 86 of the 208 had a laptop; for the others, a computer is a concrete example of a goal.

The third is interest. Someone who lends money, for example by leaving it in a bank, gets back more than they lent. If the interest rate is 1 % a month, R$ 100 put away today become R$ 101 a month from now. Seen another way, interest is the *price of time*: it says how many reais tomorrow a person gets for giving up one real today. When interest rises, waiting pays off more, and anyone in debt pays more for having brought consumption forward. With stipend-sized amounts and over a few months, interest earns very little. For a teenager, the first two reasons weigh much more.

Behind all three lies a common idea. People generally prefer steady consumption to consumption that swings between plenty and want. Saving in good months and drawing down the reserve in bad ones keeps consumption more even over time. This idea, consumption smoothing, is the basis of the economic theory of saving.

2.2 Choosing between today and tomorrow

Every decision to save compares something gained now with something gained later. Economists call this **intertemporal choice**. The model most often used to describe it was proposed by Samuelson (1937): the person gives each future date less weight than the present, and that weight falls by the same proportion with each period that passes. With a **discount factor** δ between 0 and 1, the value the person assigns today to a sequence of consumption levels is

$$U_0 = u(c_0) + \delta\, u(c_1) + \delta^2 u(c_2) + \cdots \tag{2.2}$$

where $u(c_t)$ is the satisfaction that consumption c_t brings in period t. The weight of one period ahead is δ, that of two periods is δ^2, and so on. This is **exponential discounting**. If δ is close to 1, the person is patient: the future is worth almost as much as the present. If δ is small, the person is impatient. In this model, being impatient is not a mistake. It is a preference, like liking one flavor more than another.

Samuelson presented this formula as a simple hypothesis, not as a law of behavior, and he already noted that it was an arbitrary choice (Samuelson, 1937). Even so, it became the standard in economics for decades, largely because of a property I show in the example below.

Example 2.1 R$ 100 today or R$ 110 next week

A scholar can choose between receiving R$ 100 today or R$ 110 a week from now. Suppose they discount the future exponentially, with δ equal to 0.99 per week (an illustrative value). Today, next week's R$ 110 are worth $110\,\delta$ to them, that is, R$ 108.9. Since that is more than R$ 100, they prefer to wait.

Now change the question: R$ 100 in 52 weeks or R$ 110 in 53 weeks? Today, the first option is worth $100\,\delta^{52}$ and the second $110\,\delta^{53}$. Dividing both by δ^{52}, the comparison is again between 100 and $110\,\delta$, the same as before. Again they prefer to wait. And when the 52 weeks have gone by and the choice becomes "today or next week", the calculation will still be the same, and they will still wait.

The property the example shows is called **time consistency**. With exponential discounting, all that matters is the distance between the two dates, not how far they are from today. So the plan a person makes today for a year from now is the same plan they will want to follow when that day arrives. They never regret their own plan because they changed their mind about time. If in January they decide to put away half of December's stipend, in December they will still want to put it away.

Real people do not behave like this. Since Strotz (1955), economists have known that if the weight of the future does not always fall by the same proportion, plans stop being consistent: the person plans one thing and, when the time comes, prefers another. The review by Frederick et al. (2002) gathers decades of studies that tried to measure people's discount rates. The estimates vary enormously from one study to the next, and regularities appear that the exponential model does not explain, among them that impatience is greater when one of the options is immediate.

2.3 Present bias

Present bias is the tendency to give the current moment a weight out of proportion to all future moments. It is not just preferring now; the exponential model already allows that. It is that the difference between today and tomorrow weighs much more than the difference between a year from now and a year and a day from now.

The simplest way to write this down is the **quasi-hyperbolic** model, also called the beta-delta model, which Laibson (1997) used to study saving. It adds a second parameter, β, between 0 and 1, which multiplies the whole future:

$$U_0 = u(c_0) + \beta \left[\delta\, u(c_1) + \delta^2 u(c_2) + \delta^3 u(c_3) + \cdots \right]. \tag{2.3}$$

Between the present and the next period, the weight falls from 1 to $\beta\delta$. Between any two future periods, the weight falls only by δ, as in the exponential model. With $\beta = 1$, Equation (2.3) turns back into Equation (2.2). The smaller β, the stronger the present bias.

Figure 2.1 compares the two models. The gray line is the exponential weight, which falls slowly and without jumps. The blue line is the quasi-hyperbolic weight: it is 1 today, drops to $\beta\delta$ right after, and from then on falls at the same pace as the gray one. All of present bias is in that first step down.

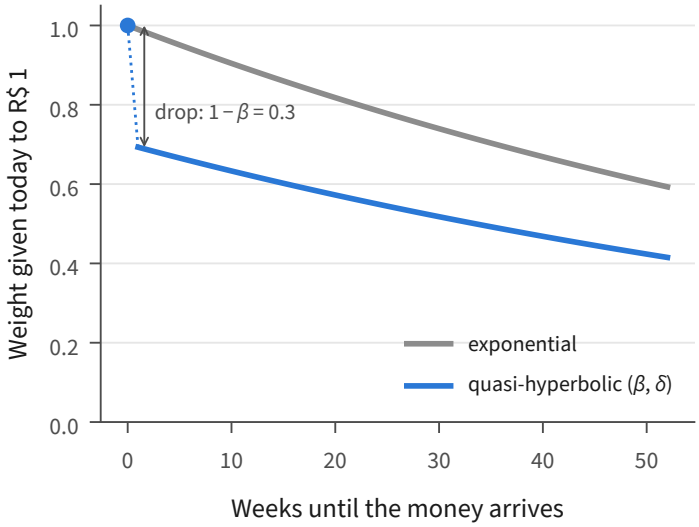


Figure 2.1. Weight a person gives today to R$ 1 received some weeks from now. Gray: exponential discounting, δ^t. Blue: quasi-hyperbolic discounting, which is 1 today and $\beta\delta^t$ afterward; the drop of $1 - \beta$ right after today is the present bias. Weekly parameters β equal to 0.7 and δ equal to 0.99.

Source: author's own work, illustrative (`code/14_foundations.py`).

2.3.1 Preference reversal, step by step

The step in Figure 2.1 produces something the exponential model forbids: **preference reversal**. I go back to Example 2.1, now with the same scholar having present bias, with β equal to 0.7 and δ equal to 0.99 per week.

First step: the choice for today. The R$ 100 today are worth 100, with no discount at all. Next week's R$ 110 are worth $\beta\delta \times 110$, that is, about R$ 76. They take the R$ 100 now.

Second step: the choice for a year from now. The R$ 100 in week 52 are worth $\beta\delta^{52} \times 100$, and the R$ 110 in week 53 are worth $\beta\delta^{53} \times 110$. Both are in the future, so both carry the factor β. Dividing both by $\beta\delta^{52}$, the comparison becomes 100 against $110\,\delta$, or R$ 108.9. They prefer to wait for the R$ 110.

Third step: the year goes by. In week 52, the choice that used to be "week 52 or 53" has become "today or next week", exactly the one in the first step. Now the R$ 100 are in the present and escape the factor β. They change their mind and take the R$ 100.

Nothing in the world changed between the second and third steps: the amounts and the gap between them are the same. The only thing that changed was where today sits. Figure 2.2 shows this for every distance: when the choice is far away, the blue line (waiting) lies above the orange one; when the earlier option reaches today, the orange line jumps over it. The general calculation, which says for which values of β and δ the reversal happens, is in Appendix A.3.

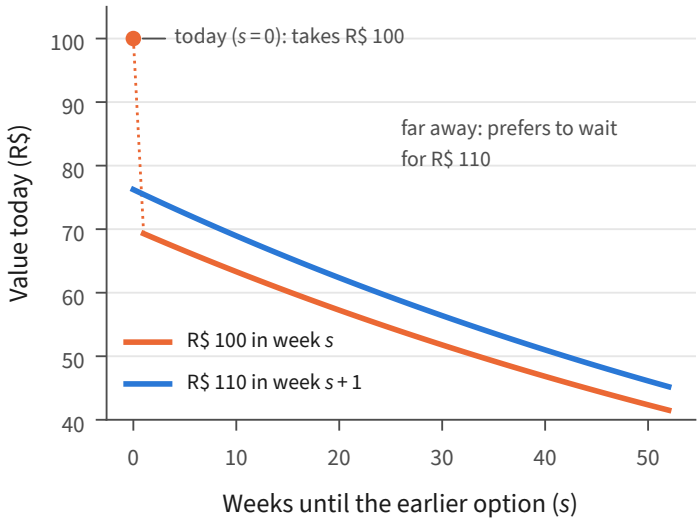


Figure 2.2. Value today of two options, as a function of how many weeks remain until the earlier option (s). Orange: R$ 100 in week s. Blue: R$ 110 in week $s + 1$. For any s from 1 on, the person prefers to wait for the R$ 110; at $s = 0$, the earlier option is no longer discounted by β and the person takes the R$ 100. Weekly parameters β equal to 0.7 and δ equal to 0.99.

Source: author's own work, illustrative (`code/14_foundations.py`).

For saving, the consequence is direct. In July, the scholar plans to put away half of August's stipend, because August's sacrifice and the later benefit both lie in the future. When August comes, the sacrifice has become the present and the benefit is still far off. They spend a little more than they had planned and promise to make up for it in September. In September, the same calculation repeats. No single month looks like a bad decision; added up, they form a pattern of putting things off. Laibson (1997) showed that this kind of preference helps explain why people who say they want to save end up saving little, and why they may value investments that are hard to withdraw from.

2.3.2 Naive and sophisticated

The reversal catches some people by surprise and not others. Strotz (1955) already distinguished the two situations, and O'Donoghue and Rabin (1999) studied them in detail. A **naive** person has present bias but does not know it: they believe that next month they will follow the plan they make today. So they put things off without worry, convinced that next time will be different. A **sophisticated** person knows that their tomorrow self will also prefer the present, and takes that into account when deciding. O'Donoghue and Rabin (1999) show that these two attitudes lead to different behavior, and that knowing about one's own bias does not always improve choices. But sophistication opens a door that naivety keeps shut: someone who knows they will change their mind may want to tie their own hands.

2.3.3 Commitment devices

A **commitment device** is a rule or a product that a person chooses today to restrict their own future choices. Two experiments became well known for showing that such devices work for saving. Ashraf et al. (2006) offered clients of a bank in the Philippines a savings account from which money could only be withdrawn after a date, or after reaching a goal, chosen by the client. The offer was made to randomly selected clients, and a little over a quarter of them took it up. A year later, the clients selected to receive the offer had larger balances than those in the control group. The authors also found that, among women, the account attracted more of those who showed time-inconsistent preferences in hypothetical questions, as if these people knew about their own problem. Thaler and Benartzi (2004) proposed to employees of American firms that they commit, in advance, to putting part of their future pay raises toward retirement. Since the commitment applied only to the future, accepting it cost nothing today; among those who signed up, the retirement saving rate rose several times over in a few years.

The tools in this study are not commitment devices. None of them locks the money away; a student can spend it all and record that they spent it. But they approach the same problem from another side. A goal recorded in the spreadsheet, with a "thermometer" that shows how much is still missing, makes the future benefit more visible at the moment of decision. In the model of Chapter 4, this is the salience channel.

2.4 Mental accounts

For traditional economic theory, money is money. Fifty reais from the stipend and fifty reais from a birthday present buy the same things, and a rational person should treat them the same way. Economists say money is *fungible*. People, however, do

not treat it like that. Thaler (1999) gave the name **mental accounting** to the set of operations people use to organize, evaluate, and keep track of their own money. They split it into accounts by source ("the stipend money", "the money Grandma gave me") and by use ("the snack money", "the laptop money"), they set budgets by category, and they are reluctant to take from one account to cover another, even when doing so would pay off.

> **Analogy: the kitchen jars**
>
> Many Brazilian families have kept money in jars or envelopes: one for the electricity bill, another for cooking gas, another for the street market. For a traditional economist, the jars make no sense, because the money is the same and could all be in one place. But they work as rules. Someone who looks at the electricity jar and sees it half full knows that money cannot be spent on anything else. Mental accounting is the same system of jars, only inside one's head.

Mental accounts break fungibility, but they can help people who find self-control hard. Thaler and Shefrin (1981) proposed describing a person as if two characters lived inside them: a *planner*, who thinks about the long run, and a *doer*, who lives in the moment and wants to spend now. The planner cannot give the doer orders directly, but can set up rules that limit what the doer does: setting aside part of the paycheck before seeing it, not touching the savings, not taking the card out of the house. Mental accounts are one of these rules. Setting aside "the laptop money" creates a psychological barrier against spending it on something else.

A budgeting tool makes these accounts explicit. The spreadsheet template used in the study split spending into categories (essentials, education, leisure, other), worked out the month's balance, and showed progress toward a goal. The app asked the student to create categories and monthly budgets. For a student who used to lump everything together, these categories can work like labeled jars. That is why mental accounting is among the reasons to expect the tools to help (Section 1.2).

2.5 Limited attention and scarcity

So far, I have assumed that a person knows how much they spend and merely chooses badly between today and tomorrow. There is an earlier problem: often they do not even know. Simon (1959) argued that people have **bounded rationality**. They cannot consider every option or process all the available information, so they use simple rules and settle for solutions that are good enough, instead of looking for the best one. Paying attention takes effort, and effort is scarce.

Gabaix (2019) reviews economic models of **inattention**, in which a person takes into account only some of the variables relevant to a decision and gives more weight to the most visible ones. Small, frequent expenses are the typical case of something

hard to see. A snack costing a few reais on the way out of school does not look like a financial decision, and so it is not treated as one. But twenty snacks in a month add up to a large share of a R$ 300 stipend. The model of Chapter 4 uses exactly this idea: part of the small purchases goes unnoticed and leaks out of the budget, and a tool that makes the student record every expense raises the share they notice (Section 4.4).

2.5.1 Scarcity

Mullainathan and Shafir (2013) took the idea of limited attention to poverty. Their thesis is that lacking a resource, whether money, time, or food, captures attention. Someone with little money focuses their mind on the most urgent problem, the bill due tomorrow, and less mental capacity is left for everything else, including planning. The authors call this narrowing "tunneling" and the available mental capacity "bandwidth".

> **Analogy: scarcity hijacks attention**
>
> Picture a student who has to pay for the week's bus fares and has exactly enough money for it. For days, part of their mind is taken up by that sum: they check their balance, work out whether it will be enough, think about what to do if it is not. It is as if scarcity hijacked part of their attention and kept it tied to the immediate problem. What is left for thinking about next month, the laptop goal, or the daily snack is less than they would have without that worry. The problem is not a lack of intelligence. It is a lack of free space.

Some experiments support this thesis. Mani et al. (2013) asked people at a shopping mall in the United States to think about a financial problem, such as a car repair, and then gave them cognitive tests. When the repair was expensive, lower-income people did worse on the tests than higher-income people; when it was cheap, there was no difference. In the same paper, sugarcane farmers in India did better on the tests after the harvest, when they had money, than before it. Shah et al. (2012) created scarcity in laboratory games, giving some participants fewer resources than others, and saw that those who were poorer in the game focused more on the immediate problem and borrowed more against the future, to the detriment of their final result. Haushofer and Fehr (2014) review evidence that poverty can lead to more short-term decisions, through stress among other paths.

Not everything holds up in the same way in other settings. Carvalho et al. (2016) randomly assigned low-income households in the United States to answer a survey before or after payday, when they have less or more money. Those who answered before payday were more impatient in choices about money, but there was no difference in performance on cognitive tests or in the quality of the decisions the authors measured. This does not refute the scarcity thesis, but it shows that the effect may depend

on how scarcity is created and measured. For this work, the lesson is to treat scarcity as a hypothesis to be tested, not as a given. Chapter 7 does that (Prediction P3).

2.5.2 Reminders and tools as aids to attention

If the problem is attention, one cheap answer is to remind people. Karlan et al. (2016) ran experiments with bank clients in Bolivia, Peru, and the Philippines who had opened savings accounts with a goal. Some received periodic reminders by text message or letter. The reminders raised the chance that a client would reach the saving goal, and they worked better when they mentioned the goal together with the account's financial incentive. To explain the results, the authors propose a model in which future expenses outside the routine stay out of "top of mind", get little attention, and therefore lose out to the expenses of the moment; the reminder brings them back.

In the study, the app and hybrid groups turned on weekly reminders. They reminded the student to *record* expenses, not to save. Even so, each reminder brought money back to the student's mind.

> **Analogy: the cognitive prosthesis**
>
> A prosthesis replaces or reinforces a part of the body that does not work as it should. The first version of this study described budgeting tools as a cognitive prosthesis (Ribeiro, 2025a). The spreadsheet adds up expenses, keeps the history, and shows the goal, tasks the student's memory and attention would do poorly, especially in a tight month. If scarcity hijacks attention, the tool gives some of it back, because the student no longer has to keep everything in their head.

The analogy is useful, but it has an implication worth keeping in mind. A prosthesis only helps the person who uses it. If the tools work as a prosthesis for attention, students who logged more expenses should have saved more, and those who never logged anything should have gained little. This is Prediction P2 of Chapter 4, and Chapter 7 shows that the data do not confirm it.

2.6 A PIC scholar's budget

With these ideas, I can look at the money of the students in the study. At baseline, before the intervention, each student reported how much they spent per month on essential expenses, those they considered obligatory, such as transport, food, or helping out at home. The mean was R$ 156.32. What is left of the stipend after that is the **residual income**:

$$R_i = R\$\,300 - e_i, \tag{2.4}$$

where e_i is the baseline essential spending of student i. Mean residual income was R$ 143.68. This is the part of the stipend over which the student really has a choice: they can spend it on leisure, on study materials, or on whatever they like, or they can put it away.

Figure 2.3 shows how, on average, the stipend was divided at baseline. Of the R$ 300, about R$ 156.32 went to essential expenses. Out of residual income, students reported saving on average about R$ 59, which is 20 % of the whole stipend. The rest of residual income went to other spending.

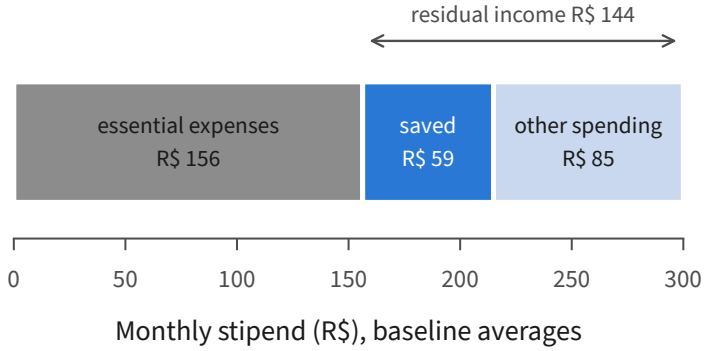


Figure 2.3. How the monthly R$ 300 stipend was divided, on average, at baseline: essential expenses (gray), amount saved (blue), and other spending (light blue). The arrow marks residual income, the stipend minus essential expenses. Means for the 208 randomized students.

Source: study data, measured (`code/14_foundations.py`).

2.6.1 Why the outcome is a share of residual income

The experiment's outcome, defined in the first version and kept here, is the percentage of initial residual income that the student reported having saved in the month:

$$S_{it} = 100 \times \frac{\text{amount saved in month } t}{R_i}. \tag{2.5}$$

Chapter 5 defines the outcome with full care (Equation (5.1)); here I want to explain the idea. Dividing by residual income puts all students on the same scale. A student who pays a bus fare every day and another who walks to school receive the same stipend, but they do not have the same freedom to save. Measuring saving as a share of what is left after essentials compares what each one did with the part they could control. At baseline, among the students who completed the study, the groups reported saving on average between 37.55 % and 45.89 % of residual income.

Example 2.2 the same saving, different percentages

Consider two students, with illustrative values. Ana spends R$ 200 a month on essentials; her residual income is R$ 100. Bruno spends R$ 100; his residual income is R$ 200. If both put away R$ 50 in a month, Ana saved 50 % of her residual income and Bruno 25 %. In reais, they saved the same; on the outcome's scale, Ana saved twice as much. Now suppose that in October Ana starts walking to school, cuts R$ 60 of bus fare, and puts away R$ 120. Since the denominator is still her baseline residual income, her October outcome is 120 %.

The example shows two properties of the outcome that come back later. The first is that it can exceed 100 %. This happens if the student cut essential spending after baseline, had some income besides the stipend, or reported something other than the month's saving, for example the accumulated balance, a stock in place of a flow (Section 2.1). The study did not measure essential spending after baseline, so it cannot tell these possibilities apart; Chapter 8 examines what the data allow us to say (Section 8.3). The second is that the same effect in reais looks larger, as a percentage, for someone with a small residual income, just because of the division. This matters when we ask whether scarcity increases the tools' effect (Prediction P3): a larger percentage effect among students on tight budgets may be nothing more than arithmetic. That is why I also present the outcome in reais.

There is also a chance difference between the groups. Mean essential spending ranged from R$ 149.21 in the app group to R$ 163.74 in the spreadsheet group, a difference produced by the random draw and taken into account by the estimates (Section 5.7).

2.7 What the evidence says about financial education and tools

Before looking at the results, it helps to know what size of effect would be reasonable to expect. The literature on financial education is large, and its conclusions are more modest than the enthusiasm around the subject suggests.

2.7.1 Knowing more about money and acting differently

Financial literacy is knowledge of basic concepts such as interest, inflation, and risk diversification. Lusardi and Mitchell (2014) show that it is low in many countries and that it is associated with better financial outcomes: people who know more plan more for retirement, save more, and take on less expensive debt. Among young people, the picture is no better. Lusardi et al. (2010) found that few young Americans answered three simple questions about these concepts correctly. In PISA 2018, which included an assessment of the financial literacy of 15-year-old students, Brazil scored below the OECD average (OECD, 2020).

Association, however, is not causation. People who know more about money may have grown up in families with more income, more schooling, or more of a habit of saving, and it would be these things, not the knowledge, that explain the behavior. To know whether *teaching* changes behavior, we need an experiment in which some people get the education and others do not, by random draw. Chapter 3 explains why.

2.7.2 What the experiments show

There are two large reviews of evaluations of financial education programs. Fernandes et al. (2014) gathered about two hundred studies and concluded that the interventions explain a very small part of the variation in financial behavior, that the effect is even smaller in low-income populations, and that it fades over the months. Kaiser et al. (2022) gathered only randomized evaluations, mostly more recent ones, and reached a somewhat more favorable conclusion: financial education has positive effects on behavior, on the order of a tenth of a standard deviation, comparable to those of other educational interventions. Some also argue that financial education is largely ineffective and that the money spent on it would be more useful in regulating financial products (Willis, 2011).

The study closest to this one, in Brazil, is that of Bruhn et al. (2016). Using a random draw among almost 900 high schools, they evaluated a program that built financial education into the curriculum. The program raised students' financial knowledge and, in the short run, improved some saving and budgeting behaviors. But it also increased the use of expensive credit, a reminder that more knowledge does not automatically lead to better decisions.

2.7.3 What a tenth of a standard deviation means

The **standard deviation** (SD) measures how spread out the values of a variable are around the mean; Chapter 3 defines it carefully. Measuring an effect in standard deviations makes it possible to compare studies with different outcomes, one in reais, another in percentages, another in number of deposits. In this study, the standard deviation of the control group's saving in October is 26.4 percentage points of residual income. An effect of a tenth of a standard deviation, the typical size the literature finds, would correspond here to about 2.6 points. This literature-sized effect is the one I use as a reference in the following chapters, for example to calculate the experiment's power (Section 3.5) and the starting point of a skeptical reader (Section 8.5). Keeping this number in mind helps in reading the result of Chapter 6: the estimated effect is 52.0 points, or 2.0 standard deviations of the control group.

2.7.4 Tools, apps and the design of saving

There is less evidence on tools than on classes, and nearly all of it comes from adults. Carlin et al. (2023) used the launch of a mobile app for a platform that gathers the user's financial information, in Iceland, and found that, after the app made this information easier to check, users went on to pay fewer fees for insufficient funds. The launch was not randomized, and the causal reading depends on assumptions about who adopted the app. Gargano and Rossi (2024) studied users of a saving app in which only a group of testers, chosen at random, could set goals. Comparing how the two groups evolved, they found that setting goals raised the saving rate, without lowering saving outside the app. In both cases, the users were adults who were already using the service on their own.

Other experiments show that small changes in the way money is kept can have large effects. Dupas and Robinson (2013) offered people in Kenya simple ways to put money aside for health expenses. Giving them a safe place to keep the money, a metal box with a padlock, was enough to raise saving for health. Adding a label, setting the money aside for a purpose, which is the mental accounting of Section 2.4 turned into an object, helped only in some cases, such as when the money was reserved for emergencies. Kast et al. (2018) studied saving groups in Chile in which participants announced goals and followed one another's progress, and found large increases in the number of deposits and in balances. Together with the reminders of Karlan et al. (2016) and the commitment devices of Ashraf et al. (2006) and Thaler and Benartzi (2004), these studies suggest that the way saving is organized and brought to mind may matter as much as knowledge.

Most of these studies have something this one lacks: they measure saving in the records of banks, of apps, or of the programs themselves, not by asking the person. That difference carries weight, as Chapter 8 shows.

2.8 The Brazilian context

The PIC is a scientific initiation program for OBMEP medalists. OBMEP is a national mathematics competition for public school students. Besides the mathematics study, the program pays a monthly stipend of R$ 300. The stipend is the same for every participant, which makes the PIC a rare case: a large group of public school students, from several regions and grades, with the same regular income. The medalists are not typical students. They were selected for their performance in mathematics, and they may handle numbers and spreadsheets more easily than the average student. This limits how far the results of this study carry over to other young people (Section 5.1).

Pé-de-Meia, created in 2024, is a federal program that encourages low-income students in public high schools to stay in school (Brasil, 2024). It pays monthly amounts to students who meet the program's conditions, on a much larger scale than

the PIC, and part of the money can only be withdrawn after finishing high school. With it, a large share of the country's low-income teenagers began to receive money regularly, and the question of what they do with it has become more important for public policy.

The Base Nacional Comum Curricular includes financial education among the contemporary themes that should run across subjects, rather than forming a subject of its own (Brasil. Ministério da Educação, 2018). The first version of this study argued that schools still lack materials and tested methods for this (Ribeiro, 2025a). The literature in Section 2.7 suggests caution: classes alone change behavior little, and tools and changes in the way money is kept look more promising, although the evidence on them, especially among teenagers, is thin.

2.9 What to take into the next chapters

This chapter gave the reader the pieces the next ones will put together. Saving means trading consumption today for consumption tomorrow, and the standard model, with exponential discounting, describes that trade as a choice that is consistent over time. Real people depart from this model in three ways that matter here. Present bias makes them put off saving month after month, even when they want to save. Mental accounts make them treat money differently depending on its label, which can help protect it. And limited attention, which scarcity can squeeze further, lets many small expenses go unnoticed. A budgeting tool can, in principle, act on all three. Chapter 4 turns this intuition into predictions that the data can reject.

On the side of evidence, the lesson is one of modesty. Financial education programs usually have effects of about a tenth of a standard deviation, which here would be some 2.6 points of residual income. An effect much larger than that is not impossible, but it calls for an explanation, and it calls for asking whether the measure is capturing what we want to measure. Chapter 3 explains the statistical tools for asking that question.

CHAPTER 3

Foundations of experiments and causal inference

The previous chapter explained why a budgeting tool *could* help a scholar save. This chapter explains how one finds out whether it *actually* helps. The question looks simple, but answering it takes a small set of statistical ideas that come up in every later chapter: what a causal effect is, why randomly assigning the groups solves a problem no other comparison solves, how the effect is estimated, how much uncertainty is left in the estimate, how large the study needs to be, and when a number reported by the student deserves trust. I write for someone who has never taken a statistics course, and each idea comes with an example of a PIC scholar and their R$ 300 a month. For the same material in more depth, I recommend the book by Imbens and Rubin (2015), on causal inference, the one by Angrist and Pischke (2009), more oriented toward economics, and the practical guides by Duflo et al. (2007) and Gerber and Green (2012) for field experiments. The longer calculations are in Appendix A.

3.1 The causal question and potential outcomes

Think of a scholar, Ana, who received the spreadsheet at the start of August and in October reported having saved 60 % of her residual income. Did the spreadsheet make her save more? To answer, I would need to know how much Ana would have saved in October *if she had not received the spreadsheet*. That second number will never exist: October happened only once, and in it Ana had the spreadsheet.

The language statisticians use for this situation is that of **potential outcomes**, proposed by Splawa-Neyman (1990) for agricultural experiments (the original text, in Polish, is from 1923) and extended by Rubin (1974) to studies of any kind.

> **Definition 3.1 potential outcome**
>
> For each student i, $Y_i(1)$ is the outcome they would have if they received the tool, and $Y_i(0)$ is the outcome they would have without it. Both numbers exist before the random draw, as characteristics of the student; the draw only decides which one will be seen.

The **causal effect** of the tool on student i is the difference

$$\tau_i = Y_i(1) - Y_i(0). \tag{3.1}$$

If Ana would save 60 % with the spreadsheet and 45 % without it, the effect for her is 15 percentage points. The problem is that, for each student, I observe one of the two numbers and never the other. What I see is

$$Y_i = T_i\, Y_i(1) + (1 - T_i)\, Y_i(0), \tag{3.2}$$

where T_i equals 1 if the student was assigned to a tool and 0 if they stayed in the control group. Imbens and Rubin (2015) call this the **fundamental problem of causal inference**: the effect of a tool on a specific person cannot be observed, not even with the best data.

Since the individual effect is out of reach, the question changes its target. Instead of the effect on Ana, I look for the average effect on a group of students.

Definition 3.2 average treatment effect

The **average treatment effect** (ATE) in a group of N students is the mean of the individual effects,

$$\tau = \frac{1}{N} \sum_{i=1}^{N} \big[Y_i(1) - Y_i(0) \big]. \tag{3.3}$$

The mean also involves numbers I do not see, but it has one advantage over the individual effect: it can be estimated. The idea is to use different students for each side of the calculation. If one group receives the tool and another does not, the mean of $Y(1)$ in the first and the mean of $Y(0)$ in the second can serve as estimates of the means for everyone. This only works if the two groups are alike in everything that matters, and Section 3.2 shows how to make sure of that.

Figure 3.1 shows the idea with ten made-up students. Each student has a circle, $Y(0)$, and a square, $Y(1)$, and the vertical distance between them is that student's effect. In these synthetic data I know both numbers for each student, something that never happens in real life, so I can calculate the true average effect: 18.5 points. Then I randomly assigned five students to the tool. For those assigned I see only the square; for the others, only the circle. The difference between the two observed means is 46.4 points, well over twice the true effect.

There was nothing wrong with the draw. The error comes from bad luck. With only five students on each side, the draw put into the tool group students 4 and 9, who would save a lot even without it, and left in the control group students 5, 6, and 10, who would save little. The observed difference mixes the tool's effect with this starting difference between the groups. With ten students, a single draw can be far

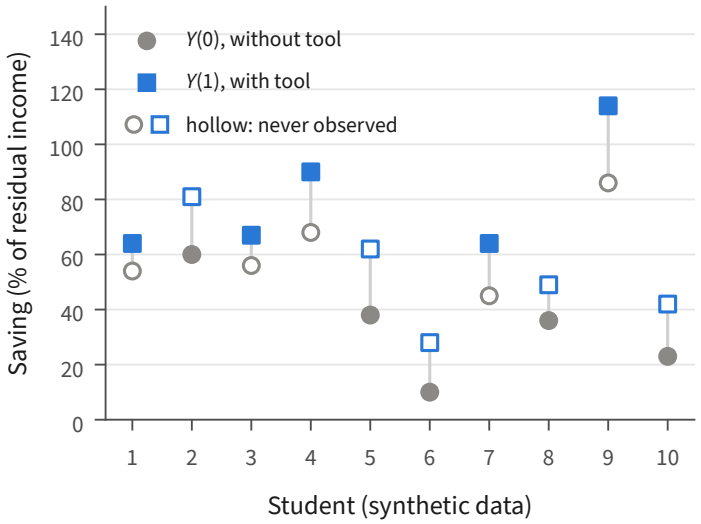


Figure 3.1. The two potential outcomes of ten made-up students. Gray circles: saving without the tool, $Y(0)$; blue squares: saving with the tool, $Y(1)$; the line between them is each student's effect. Filled markers are the ones observed after a draw that assigned five students to the tool; hollow markers would never be seen.

Source: synthetic data, simulated (`code/14_foundations.py`).

off; with hundreds, the accidents on each side tend to cancel out. That is why sample size matters so much, and I come back to it in Sections 3.4 and 3.5.

3.1.1 Why comparing users and non-users is not enough

Before the experiment, the most obvious way to answer the question would be to find scholars who already use a budgeting spreadsheet on their own and compare them with those who do not. Suppose the users save 20 points more. Does that show the spreadsheet works?

It does not, because those who choose to use a spreadsheet differ from those who do not. An organized student, who already writes everything down in a notebook and thinks about the phone they want to buy six months from now, is more likely to adopt a spreadsheet *and* to save more, with or without it. In terms of potential outcomes, the observed difference breaks into two parts:

$$\begin{aligned} &\mathbb{E}[Y_i \mid \text{uses}] - \mathbb{E}[Y_i \mid \text{does not use}] \\ &\quad = \underbrace{\mathbb{E}[Y_i(1) - Y_i(0) \mid \text{uses}]}_{\text{effect on users}} \\ &\qquad + \underbrace{\mathbb{E}[Y_i(0) \mid \text{uses}] - \mathbb{E}[Y_i(0) \mid \text{does not use}]}_{\text{selection bias}}. \end{aligned} \tag{3.4}$$

The symbol $\mathbb{E}[\cdot \mid \cdot]$ is the mean among the students who meet the condition after the bar. The second term, the **selection bias**, is how much more the users would

save than the non-users *even without the spreadsheet*. For the organized student, it is positive, and the comparison exaggerates the effect. For a student who only looks for a spreadsheet when the money runs out before the end of the month, it can be negative, and the comparison hides the effect. Without knowing the sign and size of this term, the raw difference says nothing about the tool (Angrist and Pischke, 2009).

The same problem comes back inside the experiment itself. The treated students chose how much to use their tool, so the relationship between use and reported saving, which the first version of this study read as causal, is subject to the selection bias of Equation (3.4). I discuss this in Section 7.3.

3.2 Random assignment

The solution to selection bias is to take away from the student, and from the researcher, the choice of who receives the tool. Fisher (1935) argued that the treatments in an experiment should be assigned by a random draw, and the argument is the same almost a century later.

> **Analogy: a coin for each student**
>
> Imagine that, for each scholar, I tossed a coin: heads, they get the spreadsheet; tails, they stay in the control group. The coin does not know whether the student is organized, lives in the countryside, spends a lot on bus fares, or has a brother who borrows money. So the organized students split between the two groups in roughly the same proportion as the disorganized ones, and the same goes for those who spend a lot or a little on snacks, for those who do or do not have a laptop, and for any other characteristic, including those I never thought of measuring. If in the end the two groups save different amounts, the only thing that systematically separates them is the coin, that is, the spreadsheet.

In the language of potential outcomes, random assignment makes T_i independent of $\big(Y_i(0), Y_i(1)\big)$. As a result, the mean of $Y(0)$ in the control group is, on average over the possible draws, equal to the mean of $Y(0)$ in the treated group, and the selection bias of Equation (3.4) is zero. The phrase "on average over the draws" matters. A particular draw can come out unbalanced, like the one in Figure 3.1. What the random draw guarantees is that the imbalance has no direction: it does not systematically favor any group, and it shrinks as the sample grows.

In the study there was not one coin per student. A coin can, by bad luck, give 60 heads and 44 tails, and I wanted exactly 52 students in each of the four groups. So I drew the groups within strata, as I explain next. The logic is the same as with the coin: nobody chose anyone's group.

3.2.1 Stratification

To **stratify** is to split the sample into blocks of similar students before the draw and to draw within each block. In the study, the blocks are the school grades, 7 in all. Within each grade, a quarter of the students went to each group. The result is that the four groups have exactly the same makeup by grade: no draw can put all the students in the last year of high school into the control group. Bruhn and McKenzie (2009) compared ways of randomizing in development economics experiments and recommend stratifying on the variables that best predict the outcome, with the caveat that the analysis must take the strata into account, as I do in Equation (5.2).

Figure 3.2 shows what stratification does with the study's real data. I took the true grades and essential expenses of the 208 students and redid the draw 5,000 times, in two ways: a simple draw, without strata, and a draw within grades, like the one I ran. In each draw, I calculated the difference between the mean of the spreadsheet group and that of the other three groups. In the left panel, which shows mean grade, the simple draw produces differences with a standard deviation of 0.26 grade; the stratified draw always produces zero (0.00), by construction.

The right panel teaches the other half of the lesson. For essential expenses, the standard deviation of the difference is R$ 4.7 in the simple draw and R$ 4.6 in the stratified one: practically the same. Stratifying by grade only balances what moves together with grade, and spending on bus fares and snacks hardly depends on grade. In 28 % of the simple draws, the difference in essential spending between the spreadsheet group and the others exceeded R$ 5. In the stratified draw, it did so in 28 %: almost the same, as the standard deviation already suggested. The real draw landed in that range. The spreadsheet group started with mean essential spending of R$ 163.74, well above the overall mean of R$ 156.32, and the app group with R$ 149.21 (Section 5.7). Had I also stratified by essential spending, this imbalance would have been smaller; it is one of the lessons I carry into the follow-up study (Section 9.1).

3.2.2 Checking balance

Since a particular draw can come out unbalanced, every experiment compares the groups at baseline, before any treatment. This is Table 5.2 in Chapter 5. For each characteristic (age, gender, rural origin, owning a laptop, essential expenses, and so on), one tests whether the means of the four groups are equal.

This table is often misread. If the draw was honest, each test at 5 % has a 5 % chance of flagging a "significant" difference by pure chance. With eight characteristics tested, the chance that at least one gives $p < 0.05$ is $1 - 0.95^8$ if the tests are independent, about one in three. Finding one difference in eight tests is therefore no surprise, and it is not a sign that the draw failed. In the study, that is exactly what happened: essential expenses differed ($p = 0.04$), and a joint test, which asks whether all the characteristics together predict who received a tool, did not reject ($p = 0.81$). The

right response to an imbalance like this is not to redo the draw, but to include the characteristic as a control in the regression, which Section 3.3 explains.

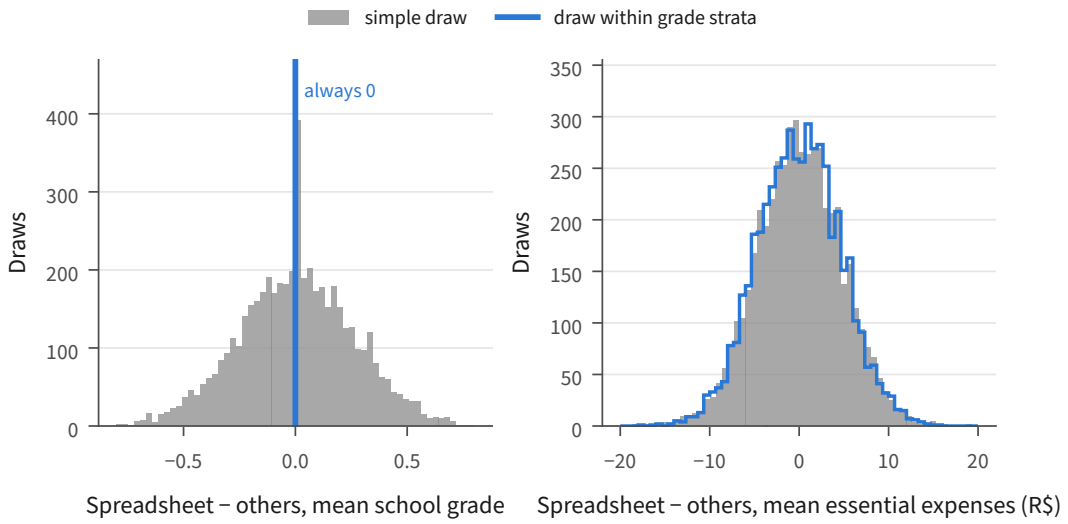


Figure 3.2. Difference between the mean of the spreadsheet group and that of the other three groups in 5,000 redone draws over the real characteristics of the 208 students. Gray: simple draw. Blue: draw within grade strata, as in the study. Left: mean grade; with strata, the difference is always zero. Right: initial essential expenses, in reais; the two histograms almost overlap.

Source: student characteristics measured at baseline; draws simulated (`code/14_foundations.py`).

3.3 How to estimate the effect

3.3.1 Difference in means

With the groups randomly assigned, the simplest estimator of the average effect is the difference between the mean of the treated group and that of the control group:

$$\hat{\tau} = \bar{Y}_1 - \bar{Y}_0, \tag{3.5}$$

where $\bar{Y}_1$ is the mean saving of the n_1 treated students and $\bar{Y}_0$ that of the n_0 control students. The hat on $\hat{\tau}$ signals that it is an estimate, computed from the data, of the true number τ. Because of random assignment, $\hat{\tau}$ is **unbiased**: if the experiment were repeated many times, with new draws, the mean of the estimates would equal τ (Imbens and Rubin, 2015). Figure 3.1 showed a single draw that missed; the mean over many draws would hit the target.

3.3.2 Regression with group indicators

The same calculation can be done with a linear regression. With four groups, I create three indicator variables: T_{2i} equals 1 if the student is in the spreadsheet group, T_{3i} in the app group, and T_{4i} in the hybrid group; the control group is the one where all three are zero. The regression

$$Y_i = \alpha + \beta_2 T_{2i} + \beta_3 T_{3i} + \beta_4 T_{4i} + \varepsilon_i \tag{3.6}$$

gives $\hat{\alpha}$ equal to the control mean and each $\hat{\beta}_g$ equal to the difference between the mean of group g and that of the control group. Without other variables, the regression adds nothing to Equation (3.5). Its advantage is that it accepts more variables on the right-hand side, and that is what column (1) of Table 6.2 does before the following columns add controls.

3.3.3 ANCOVA: making use of the baseline

Before the intervention, each student reported how much they saved, their baseline saving, Y_{i0}. This information is not needed for the estimate to be unbiased, because the random draw already takes care of that. But it can make the estimate more precise. **ANCOVA** (analysis of covariance) is the regression that includes the baseline as a control:

$$Y_i = \alpha + \sum_{g=2}^{4} \beta_g T_{gi} + \gamma\, Y_{i0} + \varepsilon_i. \tag{3.7}$$

The intuition is this. Part of the difference between two students in October comes from old differences: a student who already saved a lot in July probably saves a lot in October. That part has nothing to do with the tool, but it increases the variation in the data and, with it, the uncertainty of the estimate. Controlling for the baseline takes this old variation out of the calculation and leaves only the new. How much it takes out depends on the correlation ρ (not to be confused with the coefficient γ) between baseline and outcome. The variance of the ANCOVA estimator is approximately

$$\mathrm{Var}(\hat{\tau}_{\mathrm{ANCOVA}}) \approx (1 - \rho^2)\ \mathrm{Var}(\hat{\tau}_{\mathrm{dif}}), \tag{3.8}$$

as I derive in Appendix A.6. Figure 3.3 shows the curve. With $\rho = 0.5$, three quarters of the variance remain; with $\rho = 0.7$, about half, which is the same as doubling the sample size without recruiting anyone. McKenzie (2012) shows that when the correlation between rounds is low, ANCOVA is much more precise than comparing each student's change (difference in differences), and that it pays to measure the outcome more often.

In the study, ANCOVA barely changed the precision (Table 6.2), and the reason is telling: a student's reported saving in one month has almost no correlation with the

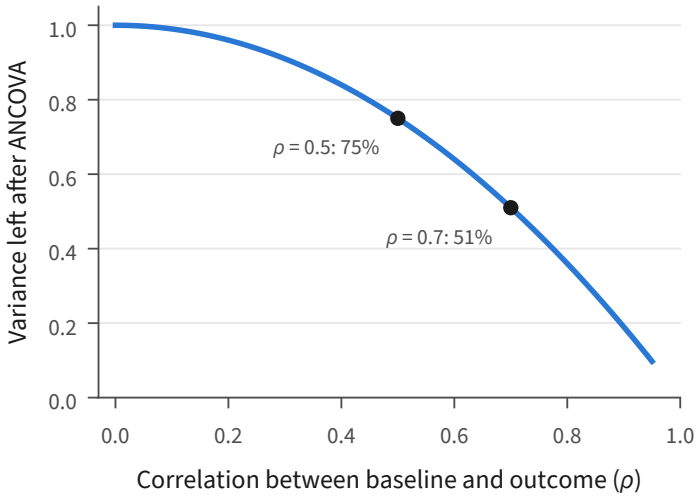


Figure 3.3. Share of the estimator's variance that remains after controlling for the baseline, $1 - \rho^2$, as a function of the correlation ρ between baseline and outcome. The dots mark $\rho = 0.5$ and $\rho = 0.7$.

Source: formula in Equation (3.8) (`code/14_foundations.py`).

next month's. With ρ near zero, Equation (3.8) says there is no gain. This same fact is the starting point of Chapter 8.

3.3.4 Robust standard errors

Every estimate comes with a standard error, which measures how much it would vary from one draw to another (Section 3.4). The classical regression formula assumes the spread of the data is the same in every group. In the study, it is not: in October, the standard deviation of reported saving was 26.4 points in the control group and 20.3 in the spreadsheet group. When the spread changes from one group to another, the data are **heteroskedastic**, and the classical standard error can come out too small or too large.

The way out is a **robust standard error**, which estimates the spread from each student's own residuals, without assuming it is the same for everyone. I use the HC2 version, one of the small-sample corrections studied by MacKinnon and White (1985). It slightly inflates the residuals of the students who pull hard on the fitted line, and, in a regression with only the group indicators, it exactly reproduces the standard error of the difference in means with different variances in the two groups, the one in Equation (3.9). Lin (2013) answered a critique by David Freedman of the use of regressions in experiments. He showed that, in large samples, robust standard errors remain valid even if the relationship between the controls and the outcome is not linear, and that, with interactions between treatment and controls, adjustment does not make precision worse. All the regressions in Chapters 6 and 7 use HC2.

3.3.5 Intention to treat and the effect on those who used the tool

Of the students assigned to a tool, 25 never logged a single expense. What should be done with them? The standard answer is to keep them in the group they were assigned to. The resulting estimate is the **intention-to-treat** effect: the effect of *offering* the tool and the training, not of using them. It may seem a less interesting question, but it is the one a public program would face, because no program can force every scholar to open the spreadsheet. And it is the only one the random draw protects. Removing from the treated group those who did not use the tool would mean selecting the treated by their own behavior, and the selection bias of Equation (3.4) would come back.

The effect on those who actually used the tool can be estimated with additional assumptions, for example that the offer affects saving only through use (Angrist and Pischke, 2009). In the study, that assumption is exactly what is in doubt, because the students who never logged anything reported the same gains as the others (Section 7.3). So I report only the intention-to-treat effect.

3.3.6 Attrition and Lee bounds

5 students left the study before October, and for them there is no outcome. If attrition were the same across groups and unrelated to saving, it would be enough to ignore them. The risk is that they are not: if, say, control students who saved little dropped out more often, the remaining control group would look thriftier than it is, and the effect would come out smaller.

Lee (2009) proposed a way to bound this risk without assuming anything about why people left. The idea, in words: if the treated group lost fewer students than the control group, it has "extra students", who may have stayed only because of the treatment. I do not know who they are. In the worst case for my conclusion, they are the ones who reported the most; in the best case, the ones who reported the least. So I drop that share of extra students from the treated group, first from the top of the distribution and then from the bottom, and recompute the effect in both situations. The result is an interval, the **Lee bounds**, that contains the true effect under weak assumptions. In the study, the treated groups lost 1.9 % and the control group 3.8 %; the bounds are in Section 6.4.

3.4 Uncertainty: standard error, interval and p-value

If I redid the draw of the 208 scholars, the groups would contain other students and the difference in means would come out a little different. This variation from one draw to another is **sampling variability**, and it is what made the estimate in Figure 3.1 miss by so much. The **standard error** (SE) is the standard deviation of this variation:

it says how far, typically, an estimate lies from the true value. For the difference between two means, it is

$$\mathrm{SE}(\hat{\tau}) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0}}, \tag{3.9}$$

where σ_1 and σ_0 are the standard deviations of the outcome in the two groups. The formula shows the two ways to reduce uncertainty: have less dispersed outcomes (which is what ANCOVA does) or have more students. Since n sits inside a square root, quadrupling the sample only cuts the standard error in half.

The 95 % **confidence interval** combines the estimate and the standard error:

$$\mathrm{CI}_{95\%} = \hat{\tau} \pm 1.96 \times \mathrm{SE}(\hat{\tau}). \tag{3.10}$$

The number 1.96 comes from the normal distribution: 95 % of the area under the curve lies within 1.96 standard deviations of the center. The correct reading of the interval is about the procedure: if the experiment were repeated many times and I computed an interval in each repetition, 95 % of those intervals would contain the true effect. In the study, the joint effect of the three tools has a 95 % CI from 44.2 to 59.8 points (Section 6.2).

Definition 3.3 *p*-**value**

The *p*-**value** is the probability of getting, by chance alone, an estimate at least as far from zero as the one observed, *assuming the true effect is zero*. This assumption is the **null hypothesis**.

A small p says the estimate would be rare if the tool did nothing. By convention, a result is called **statistically significant** when $p < 0.05$. The p-value is the most misunderstood measure in statistics, so I say plainly what it is *not*. It is not the probability that the null hypothesis is true: that probability requires a prior belief, the subject of Section 3.9. It does not measure the size of the effect: a tiny effect can have a small p in a huge sample. And a large p does not show that the effect is zero, only that the data cannot tell it apart from zero (Section 3.8). Finally, the p-value presumes the analysis was decided before looking at the data; when it was not, the p-value loses its meaning, as I explain in Section 3.6.

3.4.1 The tests that appear in the study

The *t*-**test** compares the means of two groups: it divides the difference by its standard error and asks whether the ratio is too large to be chance. I use the version that does not assume equal variances in the two groups, which is the one in Equation (3.9). To compare four groups at once, the first version of the study used one-way **ANOVA**, which tests whether all the means are equal by comparing the variation between groups with the variation within them; the result is the F statistic. Welch (1951)

proposed a version of ANOVA that does not assume equal variances. The test of Kruskal and Wallis (1952) asks the same question using the ranks of the data (each student's position, from smallest to largest) instead of the values, and so it is not thrown off by a few extreme values. The three tests agree in the study (Section 6.2).

3.5 Power and minimum detectable effect

An experiment can fail in two ways. It can declare an effect where there is none, a **type I error**, whose probability is the level of the test, α, usually 5 %. And it can fail to find an effect that exists, a **type II error**. The probability of avoiding the second error has a name of its own.

Definition 3.4 power

The **power** of a test is the probability that it rejects the null hypothesis when the true effect has a specified size. It depends on the size of the effect, the spread of the outcome, the number of students, and α.

Figure 3.4 shows the idea. The gray curve is the distribution of the estimate, divided by its standard error, if there is no effect; the blue one, if there is. The test rejects when the estimate falls to the right of the dashed line. The red area, under the gray curve to the right of the line, is half of the chance of rejecting by mistake; the other half lies in the left tail, which the figure does not mark. The blue area to the right of the line is the power. In this illustrative figure, the true effect lies 2.8 standard errors from zero, and power is 80 %, the value usually required when planning a study.

Analogy: the lens and the detail

Think of the experiment as a lens pointed at the scholars' saving. A weak lens shows the difference between a group that saves R$ 30 and another that saves R$ 100, but not the difference between R$ 95 and R$ 100: the two blurs run together. Zooming in, here, means adding students or reducing the noise in the measures. Power tells us whether the lens I built has enough zoom for the detail I want to see. Not seeing the detail through a weak lens does not prove it is not there.

When planning a study, the question is usually asked the other way round: with the students I have, what is the smallest effect I can detect with 80 % power? This number is the **minimum detectable effect** (MDE). For two groups with n_1 and n_2 students and standard deviation σ,

$$\text{MDE} = \left(z_{1-\alpha/2} + z_{\text{power}}\right) \sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}, \tag{3.11}$$

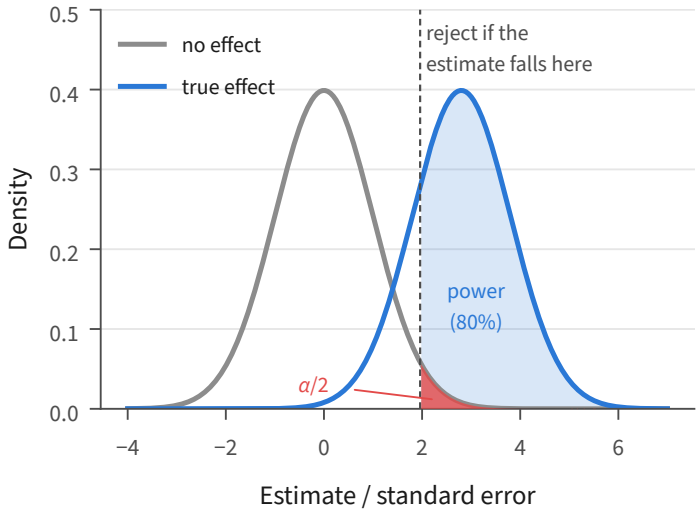


Figure 3.4. Distribution of the estimate divided by its standard error when there is no effect (gray) and when the true effect lies 2.8 standard errors from zero (blue). To the right of the dashed line the test rejects the null hypothesis. Red area: half of the chance of a false positive, $\alpha/2$. Blue area: the power.

Source: illustrative (`code/14_foundations.py`).

where z_q is the point of the normal distribution that leaves a share q of the area to its left. With $\alpha = 5\,\%$ and 80 % power, $z_{0.975} \approx 1.96$ and $z_{0.80} \approx 0.84$, and the sum is 2.8, the same distance as in Figure 3.4. The derivation is in Appendix A.7. The product of σ and the square root is the standard error of Equation (3.9) with equal variances, so the formula says something simple: one can only reliably detect an effect of at least 2.8 standard errors.

In the study, the within-group standard deviation of October saving is 24.0 points, and each group has about 51 students. With these numbers, the MDE for comparing *two tools with each other* is 13.5 points. This value does not come from Equation (3.11): the program computes the exact power of the t-test, which uses the t distribution instead of the normal, and searches for the smallest effect with 80 % power. Since the t distribution has slightly fatter tails, the normal formula gives a slightly smaller number. Differences smaller than this between the spreadsheet, the app, and the hybrid may exist without the experiment seeing them, and that is why Chapter 7 cannot rank the tools (Section 7.1). The same calculation, done in reverse, gives the number of students the follow-up study needs (Section 9.2).

3.5.1 Effect size in standard deviations

Is an effect of 10 percentage points large or small? It depends on how much the students already vary among themselves. To compare studies with different outcomes, one divides the effect by the standard deviation of the outcome. The result is **Cohen's** d (Cohen, 1988):

$$d = \frac{\bar{Y}_1 - \bar{Y}_0}{\sigma}. \tag{3.12}$$

Cohen (1988) suggested calling a d of 0.2 small, one of 0.5 medium, and one of 0.8 large, with the warning that these labels are only useful when nothing better is available. In financial education, something better is available: the review of randomized experiments by Kaiser et al. (2022) finds effects on behavior of about a tenth of a standard deviation, and the one by Fernandes et al. (2014) finds even smaller effects.

One way to picture d is through the overlap between the distributions of the two groups. If both are normal with the same spread, the share of area they have in common is

$$\text{overlap} = 2\,\Phi\left(-\frac{|d|}{2}\right), \tag{3.13}$$

where Φ is the cumulative distribution function of the standard normal. Figure 3.5 shows the two cases that matter for this monograph. With $d = 0.1$, the typical effect in the literature, the curves overlap by 96 %: the groups are almost indistinguishable, and only a large sample tells them apart. With $d = 2$, close to the effect reported in this study (2.0 standard deviations of the control group), the overlap drops to 32 %: the typical student with a tool reports saving more than almost the entire control group. An effect twenty times that of the literature is not impossible, but it calls for an explanation, and Chapter 8 looks for one.

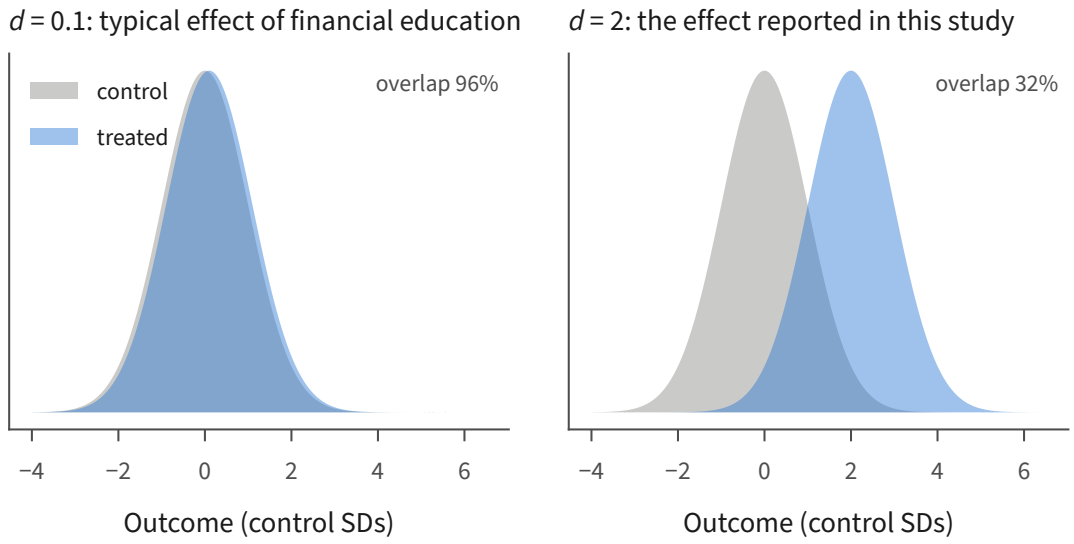


Figure 3.5. Normal distributions of the outcome in the control group (gray) and among the treated (blue), in standard deviations of the control group. Left: $d = 0.1$, the typical effect of financial education on behavior; overlap of 96 %. Right: $d = 2$, the size of the effect reported in this study; overlap of 32 %.

Source: formula in Equation (3.13) (`code/14_foundations.py`).

3.6 Many comparisons

When there is no effect at all, a test at 5 % is wrong one time in twenty. Someone who runs twenty tests should expect one false positive. The study has six comparisons between pairs of groups (control against each tool, and the tools against each other) and six subgroup splits (girls and boys, rural and urban, and others). If the six pairwise comparisons were independent and no difference existed, the chance that at least one gives $p < 0.05$ would be $1 - 0.95^6$, about one in four. The same reasoning as in the balance check (Section 3.2) applies here, except that now the false positive would be read as a discovery.

There are two families of correction. The first controls the chance of making *any* false positive in the set of tests. The method of Holm (1979) does this in steps: I sort the m p-values from smallest to largest, compare the smallest with α/m, the next with $\alpha/(m-1)$, and so on, stopping at the first one that fails. In the study, the difference between the hybrid and the spreadsheet has $p = 0.03$ on its own, but $p = 0.08$ after Holm's correction for the six comparisons (Section 7.1). The second family, from Benjamini and Hochberg (1995), is less demanding: it controls the expected *share* of false discoveries among the results declared significant. I use it for the subgroup splits, where the smallest p, 0.07, becomes 0.37 after the correction.

3.6.1 The forking paths

There is a subtler problem than the number of tests reported. When analyzing data, a researcher makes dozens of small decisions: the outcome in percentages or in reais, with or without the extreme values, with which controls, in which sample. Each decision is reasonable. But if they are made after looking at the data, even in good faith, the path tends to bend toward the direction that gives cleaner results, and the final p-value does not account for the paths not taken. Statisticians call this the garden of forking paths.

There are two defenses. The first is to decide the analysis before seeing the data and to record that decision in a **preregistered analysis plan**. The second, when that was not done, is to show all the reasonable analyses at once. The **specification curve** of Simonsohn et al. (2020) does this: it estimates the effect under every defensible combination of choices and shows the distribution of the estimates. If the effect appears in almost all of them, the result does not hinge on a hand-picked path. In the study, the 108 specifications give effects between 1.2 and 3.8 standard deviations of the control group (Section 6.4).

I need to be honest about what was decided beforehand. The first version of this study was not preregistered. It defined the outcome and announced an ANOVA, t-tests, a regression with group indicators and covariates, a regression of saving on tool use, and comparisons by gender, age, and origin. I redo all of these analyses. Everything else in this monograph, including the model of Chapter 4 and the analysis

of how reliable the reports are, was added after the data were collected and should be read as exploratory. The follow-up study of Chapter 9 starts from a registered plan (Section 9.3).

3.7 Randomization inference

The *p*-values of Section 3.4 use the normal distribution, an approximation that works well in large samples with well-behaved data. In an experiment, there is an alternative that relies on no approximation at all, and that comes straight from the idea of Fisher (1935): use the random draw itself as the source of uncertainty.

The reasoning goes like this. Suppose the tool had no effect at all on any student. Then each student's saving would be the same in any group, and the draw would only have decided the labels. I can therefore redo the draw on the computer, keeping each student's saving fixed, and recompute the difference in means. In the study, I shuffle the groups within each grade, among the students who completed the study, so that each grade keeps the number of students it had in each group. Repeating this N times, I get the distribution of differences that chance would produce if there were no effect. The randomization *p*-value is the share of redone draws, counting the real draw as well, that produced a difference at least as large as the one observed:

$$p = \frac{1 + K}{N + 1}, \tag{3.14}$$

where K is the number of redone draws with a difference at least as large as the one observed. Even if no redone draw comes close to the observed difference, the smallest possible value is $1/(N + 1)$, because the real draw always counts. In the study, with 20,000 redone draws, this minimum is 0.00005, and that was the value obtained (Section 6.4).

Young (2019) redid, in this way, the analyses of dozens of experimental papers published in economics journals and found that some of the results that were significant under the usual tests stop being significant. Athey and Imbens (2017) recommend randomization inference as the natural way to analyze experiments. Its advantage is that it needs no assumption about the shape of the distribution of saving; its limitation is that it tests the hypothesis of zero effect for *every* student, which is stronger than that of zero average effect.

3.8 Absence of evidence and evidence of absence

When a test does not reject, it is tempting to conclude there is no difference. In the comparison between the hybrid and the app, the study finds $p = 0.50$. This does not

show that the two tools are the same. It shows only that, with the power the study has, a difference of that size cannot be told apart from chance. Lack of evidence for a difference is not evidence that there is none.

To claim that two tools are *equivalent*, the test has to be turned around. First, I choose a margin m: the largest difference I would consider of no practical importance, for example 10 percentage points of residual income. Then I run two one-sided tests, the **TOST** of Schuirmann (1987): one tests whether the true difference is greater than $-m$, the other whether it is less than $+m$. I declare equivalence only if both reject. In practice, this is the same as looking at the 90 % confidence interval of the difference and asking whether it fits entirely within the band from $-m$ to $+m$ (Lakens, 2017).

In words, the picture is a band around zero with an interval drawn over it. There are three situations. If the interval lies entirely outside the band, the tools differ by more than the margin. If it lies entirely inside, they are equivalent. If it crosses the edge of the band, the study cannot decide. For the hybrid against the app, the 90 % interval runs from –4.8 to 11.3 points: it fits in a band of ±15 points, but crosses the band of ±10. With this experiment, I do not know whether the tools differ by a few points or by more than ten (Section 7.2).

3.9 What a skeptic should conclude

The p-value answers an indirect question: how rare the data would be if there were no effect. Often the question of interest is a different one: given what I saw, what size of effect should I believe in? To answer it, Bayesian statistics starts from an initial belief, the **prior**, and combines it with the data to arrive at a final belief, the **posterior**.

In the simplest case, the prior is a normal distribution with mean μ_0 and standard deviation s_0, and the estimate $\hat{b}$ has standard error SE. The posterior is again normal, with mean

$$\mu_{\text{post}} = w\,\hat{b} + (1 - w)\,\mu_0, \qquad w = \frac{s_0^2}{s_0^2 + \text{SE}^2}, \tag{3.15}$$

and with variance $1/(1/s_0^2 + 1/\text{SE}^2)$, smaller than that of either source (Appendix A.8). The final mean is a weighted average of what I believed and what the data say, and the weight of each side is its precision. A very wide prior (large s_0) gives almost all the weight to the data; a narrow prior resists more.

This calculation matters here because of the size of the effect. A reader who knows the financial education literature does not start from zero: they expect something like a tenth of a standard deviation, 2.6 points on this outcome. They can represent that expectation with a prior centered on this value and see where the data take them. With an estimate of 52.0 points and a standard error of 4.0, a prior whose own standard deviation is a tenth of a standard deviation ends up at a posterior mean of 17.7 points, still several times the literature's effect. I do this calculation with other

widths in Section 8.5. It has a limitation I must not forget: it assumes the estimate measures what it claims to measure. If the reports have a bias that shifts all the treated students in the same direction, no prior corrects for it, and the next section prepares the ground for that problem.

3.10 What is measured and what we want to measure

Everything so far assumes that the outcome measures well what I care about. In the study, I want to know how much the scholar saved. What I have is how much they *said*, on a monthly form, they saved. The two can differ in two ways: through noise, which scatters the reports around the true value with no direction, and through bias, which pushes them to one side.

3.10.1 Reports and what pulls a report to one side

People who answer questions about their own behavior tend to give the answer that seems most acceptable, which is called **social desirability bias**. Saving is seen as a virtue, so a student may round their saving up without even noticing. In an experiment, there is also the **experimenter demand effect**: the participant tries to guess what the researcher expects and acts, or answers, accordingly. A student who spent half an hour learning to use a saving spreadsheet with the researcher in person knows very well which answer would please. de Quidt et al. (2018) measured this effect by telling participants explicitly what the researcher expected them to do, and found it small in online experiments. In the study, however, treated students reported their saving directly to the person who trained them, and the control group had much less contact with the intervention.

Being observed can also change behavior itself, not just the report. Zwane et al. (2011) showed that being surveyed about a subject can change later decisions about it. The idea that mere observation changes behavior became known as the Hawthorne effect, after studies at an American factory in the 1920s; Levitt and List (2011) reanalyzed the original data and found little evidence of the effect that made them famous, and the systematic review by McCambridge et al. (2014) concludes that effects of taking part in research exist, but that little is known about when they appear and how large they are. For the study, the point is that both the report and the behavior may have changed because of the attention the treated students received, and not because of the tool itself.

3.10.2 Reliability

Noise calls for a different tool. Suppose the report of student i in month t is the true saving plus an error, $r_{it} = s_{it} + \eta_{it}$, with the error independent of everything else.

Definition 3.5 reliability

The **reliability** of a measure is the share of the variance of the reports that comes from the true signal:

$$\lambda = \frac{\mathrm{Var}(s)}{\mathrm{Var}(s) + \mathrm{Var}(\eta)}. \tag{3.16}$$

It equals 1 if there is no reporting error and approaches 0 when the error dominates.

Analogy: the jumpy scale

Imagine a bathroom scale that, each time someone steps on it, shows the true weight plus or minus a randomly drawn number of up to 10 kilograms. It is not rigged: on average, it gets it right. But if you weigh yourself today and tomorrow, the two numbers will have almost nothing to do with each other, even if your weight has not changed. And if you weigh a whole class, the scale will still show who is heavier on average, but the differences between similar classmates will be lost in the noise. A scale like this has low reliability. A good scale, weighing the same person twice in a row, shows almost the same number.

The analogy suggests how to measure reliability without knowing the true value: measure the same thing twice and see how much the two measures agree. This is the **test-retest correlation**. With several measures per person, one uses the **intraclass correlation** (ICC), which is the share of the total variance due to stable differences between people; Shrout and Fleiss (1979) describe its various forms. If a student's true saving has correlation ρ from one month to the next and the reporting errors are independent across months, the correlation between the reports of two months is

$$\mathrm{corr}(r_{it}, r_{it'}) = \rho\,\lambda, \tag{3.17}$$

as I derive in Appendix A.4. Noise always pulls the observed correlation toward zero, and this is called **attenuation**. Equation (3.17) also works in reverse: if I observe the correlation between months and make a reasonable assumption about ρ, I get an upper limit for the reliability of the reports.

For a scholar, ρ should not be small. Someone who spends R$ 100 a month on bus fares in August still spends it in September; someone in the habit of putting part of the stipend away does not lose that habit from one month to the next. In the study, however, the correlation between reports in consecutive months, after removing the group means, is 0.00. By Equation (3.17), this is only compatible with reliable

reports if true saving is almost random from one month to the next. Chapter 8 starts from this number and examines what it implies, together with other clues about the reports (Section 8.2).

One last remark ties this section back to the start of the chapter. Reporting noise that does not depend on the group creates no bias in the difference in means: it only makes it less precise. Reporting bias that depends on the group, such as a wish to show progress that only the treated students would have, goes straight into the estimate, and no random draw, robust standard error, or randomization test detects it. Random assignment guarantees that the groups were alike before the intervention; it does not guarantee that they answer the questions the same way after it.

3.11 Where each tool goes

The ideas in this chapter come back in this order. Potential outcomes and random assignment support the causal reading of all of Chapter 6, and baseline balance is in Section 5.7. The ANCOVA regression with stratum fixed effects and HC2 errors is the main specification of Section 6.2; Lee bounds, randomization inference, and the specification curve make up Section 6.4. Power, the minimum detectable effect, Holm's correction, and TOST explain why Chapter 7 can neither rank the tools nor show them to be equal, and selection bias explains why the relationship between use and saving in Section 7.3 is not causal. Reliability, attenuation, and the skeptic's prior are the tools of Chapter 8. The power calculation with ANCOVA sets the size of the follow-up study in Chapter 9.

One limitation remains that none of these tools solves. A well-run experiment says what the average effect was for *these* students, in *this* setting. Deaton and Cartwright (2018) point out that carrying this number over to another population requires understanding why the effect happened, not just that it happened. PIC scholars are medalists in a mathematics olympiad, all with the same stipend and a smartphone. What holds for them may not hold for Pé-de-Meia students, and that is a question this study does not answer.

CHAPTER 4

A model of saving with attention and reporting

4.1 Why write a model

The idea behind this study fits in one sentence: budgeting tools help people save. The trouble with that sentence is that almost no result could contradict it. If the students with a spreadsheet save more, it is confirmed. If they save the same, I can say that three months was too short, that the workshop was too brief, or that the students did not use the tool properly. An explanation that fits any result explains nothing, because it takes no risk.

I write a model to force the sentence to take a risk. A model, here, is a description in equations of how a student decides what to do with the stipend, and of the routes by which a tool could change that decision. It is small on purpose: one student, one month, two choices. I do not claim it is true in every detail. What I want is for it to say clearly what else should show up in the data if the sentence is right, so that I can look and see whether it does. If the tool helps because it makes spending visible, the students who used it more should have saved more. If it helps because it keeps the goal in view, use matters less. If it only changed what the students wrote on the form, other marks should appear. Each of these stories leaves a different signature in the data, and the model is there so I can write those signatures down before looking.

The chapter follows that order. Section 4.2 describes the student's budget and the leaks, the small purchases that slip past the plan. Section 4.3 shows how the student chooses how much to spend. Section 4.4 adds the tool, with two channels, and works out its effect on saving. Section 4.5 draws five predictions from that effect and says where in the following chapters each one is tested. Section 4.6 adds a model of how the student reports saving, because the study measures the report and not the money, and shows what each explanation predicts. Section 4.7 lists what the model leaves out. The full derivations, checked with symbolic algebra, are in Appendix A.

4.2 The student's budget and the leaks

Student i receives the PIC stipend, y = R$ 300 a month. Part of it goes to **essential spending**, e_i: the bus fare to school, the share of the phone plan the student pays, some school supplies. These are expenses the student does not choose month by month. What is left is the **residual income**,

$$R_i = y - e_i, \tag{4.1}$$

the money the student actually decides about. At the study's baseline, essential spending averaged R$ 156.32 and average residual income was R$ 143.68 (measured; Section 5.4). The outcome of the experiment is the month's saving as a percentage of R_i, with R_i fixed at baseline.

With the residual income, the student does two things. They plan **discretionary spending**, c: the Friday snack, a movie ticket, a game, a present. These are purchases they think about before making. Whatever is not spent is saved. But not all money leaves this way. Some small purchases happen without the student ever really deciding: the soda on the way out of school, the candy in the cafeteria line, one extra real of phone credit, the delivery app's fee. Each is small, and so none of them seems to matter. I call the total of these unplanned purchases in the month m_i.

The point of the model is that the student does not see all of them. They notice a fraction a_{it}, between 0 and 1, which I call **attention**. The noticed part, $a_{it}m_i$, enters the plan: the student counts it as one more chosen expense, inside c, where it competes with the snack and the movie. The unnoticed part, $(1 - a_{it})m_i$, leaves the budget without going through the plan. That is the **leak**. Adding everything up, the month's saving is

$$s_{it} = R_i - c_{it} - (1 - a_{it})\, m_i + \varepsilon_{it}. \tag{4.2}$$

The last term, ε_{it}, is a **shock** for the month, positive or negative, that the student does not control: a birthday that calls for a present, a phone that breaks, some money a grandmother gave. I assume it has mean zero and is independent from one month to the next.

> **Analogy: the leaky bucket**
>
> Think of residual income as a bucket of water the student fills at the start of the month. They take water out with a cup when they decide to spend; that is c. But the bucket has small holes in the bottom, and the water that drips through them is m_i. The student sees some of the holes and covers them with a finger, counting that water as part of what they were already going to take out. The holes they do not see keep dripping. At the end of the month, the water left over is the saving.

The idea that people perceive only part of a variable that matters for a decision, and act as if the rest were not there, is the simplest way to model inattention (Gabaix,

2019). It also connects with scarcity: when money is short, attention goes to the bill due today, and less is left for the small purchases (Mullainathan and Shafir, 2013) (Section 2.5).

4.3 What the student wants and how much they spend

To know how much the student chooses to spend, I need to say what they value. I write the month's satisfaction as

$$U = \log c + \beta v\, s. \tag{4.3}$$

The first term is the pleasure of spending now. The logarithm says this pleasure grows more and more slowly: the first snack of the month is worth a lot, the tenth is worth little. The second term is the value of money put aside. Each real saved is worth v, a number that sums up what the money is for (new earphones, a trip with friends from class, a fund for college). This value is constant: the hundredth real saved is worth the same as the first, a simplification I discuss in Section 4.7.

The factor β, between 0 and 1, is **present bias** (Section 2.3). The benefit of saving comes later, and a person with present bias gives everything that comes later an extra discount, on top of the ordinary discount for waiting (Laibson, 1997). With $\beta = 1$ there is no bias; the smaller β is, the more the student pulls money toward now. In this month, βv is what a saved real is worth from today's point of view.

Substituting Equation (4.2) into Equation (4.3), the student chooses c to maximize $\log c + \beta v\,[R_i - c - (1 - a_{it})m_i + \varepsilon_{it}]$. Taking the derivative with respect to c and setting it to zero gives the **first-order condition**

$$\frac{1}{c} = \beta v. \tag{4.4}$$

In words: the student spends up to the point where the last real spent on snacks gives exactly as much as the last real put in the piggy bank. If the snack still gave more, it would pay to spend one more real; if it gave less, it would pay to keep that real. The left side falls as c rises and the right side is fixed, so the two meet at a single point,

$$c^* = \frac{1}{\beta v}. \tag{4.5}$$

To be sure this point is a maximum and not a minimum, I look at the second derivative: $-1/c^2$, negative for every c, and equal to $-(\beta v)^2$ at c^*. Satisfaction rises up to c^* and falls after it (Appendix A.1 has the algebra).

Equation (4.5) has two properties that I use later. First, c^* grows when β falls: a student more tied to the present plans to spend more. Second, c^* does not depend on R_i. Every extra real of residual income goes to saving, because the value of keeping

it is constant and the pleasure of spending has already fallen to that level. At the optimal spending, saving is

$$s_{it} = R_i - \frac{1}{\beta\nu} - (1 - a_{it})\, m_i + \varepsilon_{it}. \tag{4.6}$$

There is one case where this calculation does not hold. If residual income minus the leak is smaller than the desired spending, $R_i - (1 - a_{it})m_i < c^*$, the student cannot spend c^* without borrowing, and the model does not allow borrowing. So they spend everything they have and plan to save nothing; the month's saving is just the shock. Economists call this a **corner solution**: the best choice sits at the edge of what is possible, not at the point where the two satisfactions are equal. It is more likely for students with small residual income, large leaks or strong present bias. For a student at the corner, valuing the future more changes nothing as long as they stay at the corner, and this comes back in Prediction P3.

4.4 The tool: attention and salience

A budgeting tool can change Equation (4.6) in two ways.

The first is attention. Recording an expense forces the student to look at it. Someone who writes the four-real soda into the spreadsheet starts to see that hole in the bucket. I write attention as

$$a_{it} = a_{0i} + \kappa\, u_{it}, \tag{4.7}$$

where a_{0i} is the attention the student already had without a tool, u_{it} is **intensity of use** (in the study, expense logs per week; Section 7.3) and κ says how much each extra weekly log raises the fraction noticed. Since a_{it} cannot exceed 1, the sum has a ceiling: a student who already noticed almost everything has little to gain.

> **Analogy: a lamp over the holes**
>
> Recording expenses is like switching on a lamp over the bottom of the bucket. The more often the student switches on the lamp, the more holes they see and the more holes they can cover. A student who got the lamp and never switched it on is still in the dark, and their bucket drips as before.

The second way is salience. The spreadsheet shows the saving goal and how far the student is from it, on a "thermometer"; the app sorts expenses into categories and monthly budgets; and the app and hybrid groups received weekly reminders to record their expenses. Seeing the goal every week can make the future feel closer, which in the model raises the value of each saved real from ν to $\nu + \Delta\nu$, with $\Delta\nu > 0$. Reminders of this kind have already increased saving in field experiments, precisely by bringing future expenses to the front of the mind (Karlan et al., 2016). Salience

does not require the student to record anything: it is enough that they set a goal and look at the tool now and then.

Comparing Equation (4.6) before and after the tool, with the same shock in both, the effect on saving is

$$\Delta s_{it} = \underbrace{\kappa\, u_{it}\, m_i}_{\text{attention}} + \underbrace{\frac{1}{\beta\nu} - \frac{1}{\beta(\nu+\Delta\nu)}}_{\text{salience}}. \tag{4.8}$$

The first term comes from the leak: the part of m_i that no longer escapes, κu_{it}, becomes saving. The second comes from planned spending: with the future worth more, c^* falls from $1/(\beta\nu)$ to $1/(\beta(\nu+\Delta\nu))$, and the difference is saved. Putting the two fractions over a common denominator, the salience term becomes

$$\frac{1}{\beta\nu} - \frac{1}{\beta(\nu+\Delta\nu)} = \frac{\Delta\nu}{\beta\,\nu\,(\nu+\Delta\nu)}, \tag{4.9}$$

which is positive whenever $\Delta\nu > 0$.

Equation (4.9) says one more thing. β is in the denominator, so the salience term is larger when β is smaller: students with more present bias gain more from the same light on the future. The reason is that they start from higher planned spending, and the same proportional rise in the value of saving cuts more reais from a larger amount. The derivative of the term with respect to β is $-\Delta\nu/[\beta^2\nu(\nu+\Delta\nu)]$, always negative (Appendix A.2). It would be an interesting prediction to test, but I cannot: the study did not measure any student's present bias.

Example 4.1 Two students and a spreadsheet

The values in this example are illustrative, chosen to keep the arithmetic simple; none of them was measured. A student has residual income R = R$ 150 and leaks of m = R$ 40 a month, of which they notice half, $a_0 = 0.5$. They value saving with $\nu = 1/60$ and have $\beta = 0.8$. By Equation (4.5), planned spending is $c^* = 60/0.8$ = R$ 75. The leak is $(1 - 0.5) \times 40$ = R$ 20, and expected saving is 150 – 75 – 20 = R$ 55.

The student gets the spreadsheet. With $\kappa = 0.05$ per weekly log and $u = 5$ logs a week, attention rises to 0.5 + 0.25 = 0.75 and the leak falls to 0.25×40 = R$ 10: the attention term is worth 0.25×40 = R$ 10. The goal on the screen moves ν from 1/60 to 1/48, and planned spending falls to 48/0.8 = R$ 60: the salience term is worth 75 – 60 = R$ 15. By Equation (4.9), $\Delta\nu = 1/48 - 1/60 = 1/240$ and $\beta\nu(\nu+\Delta\nu) = 0.8/(60 \times 48) = 1/3600$, which again gives 3600/240 = R$ 15. Saving goes to 150 – 60 – 10 = R$ 80, a gain of 10 + 15 = R$ 25, or from 36.7 % to 53.3 % of residual income.

A second student is identical to the first except for one thing: more present bias, $\beta = 0.6$. Without a tool, this student plans to spend 60/0.6 = R$ 100 and saves 150–100–20 = R$ 30. With the same spreadsheet, used the same way, planned spending falls to 48/0.6 = R$ 80, and the salience term is worth R$ 20, five reais more than for the first student. The attention term is the same, R$ 10, because it does not depend on β. This student's saving doubles, from R$ 30 to R$ 60.

Had the first student received the spreadsheet and never logged anything ($u = 0$), they would gain only the salience part: R$ 15 instead of R$ 25. In both cases, residual income minus the leak (R$ 130 before, R$ 140 after) stays above planned spending, so neither student is at the corner.

The example shows the most useful thing Equation (4.8) implies for the experiment: the two channels leave different marks. The gain through attention grows with use and disappears for those who do not use the tool. The gain through salience is the same for heavy users and non-users, and it varies with present bias, which I do not observe.

4.5 Five predictions

Equation (4.8), together with Equation (4.6), yields five predictions. I state each one and, right after, where and how the data test it. Chapter 7 gathers the verdict on all five in Table 7.4.

> **Prediction P1**
>
> *Receiving a tool increases saving.*

Both terms of Equation (4.8) are positive or zero, so the average Δs among the treated is positive. This is the weakest prediction: any story in which the tool helps makes it. The test is the comparison of reported saving in October between the treated students and the control group, with and without baseline saving as a control (Section 6.2). Since the random draw decided who got a tool, this difference is causal. But it is a difference in *reported* saving, and Section 4.6 shows that a pure reporting effect also satisfies this prediction.

> **Prediction P2**
>
> *Among the treated, saving grows with use: $\partial \Delta s / \partial u = \kappa m_i > 0$. Students who never log gain only through salience, and should be closer to the control group than active users.*

This is the prediction that separates the two channels and the most direct test of the idea behind the study. Section 7.3 tests it in two ways: the slope of the change in saving on logs per week, and the comparison between the 25 treated students who never logged an expense and those who did. Neither comparison is causal in the strict sense, because each student chose how much to use the tool, and Section 7.3 discusses what this allows me to conclude. If the slope is zero and those who never logged gain as much as the rest, the attention channel is left without support, and what remains is salience or something all the treated students had in common.

Prediction P3

The effect in reais varies with the budget. If scarcity lowers initial attention a_{0i}, there is more to gain when R_i is small; if leaks m_i grow with the budget, there is more to gain when R_i is large. The sign of the gradient tells which force dominates.

The two forces come from different places in Equation (4.8). The attention term is proportional to m_i, so a student with more money to leak has more to recover. Initial attention enters through the ceiling: the attention gain cannot exceed $1 - a_{0i}$, and a student with low initial attention, as scarcity would lead one to expect (Mani et al., 2013; Mullainathan and Shafir, 2013), has more room. There is also the corner solution, more common with small budgets, where salience loses its force. The model does not say which of these forces wins; it says that the effect in reais should change with R_i and that the direction of the change is informative. The test is in Section 7.4, which estimates the effect by tercile of residual income. One caution: the study's outcome is saving as a percentage of R_i, and an effect that is constant in reais looks larger, in percentage terms, for students with small R_i. That is why the test has to be done in reais.

Prediction P4

Students who already saved a large fraction at baseline have less room to improve, so the effect should be smaller for them.

In the model, saving a lot without a tool means having small leaks, high attention or low planned spending. In all three cases there is little left for the tool to fix: the attention ceiling is close, and the salience term is smaller when c^* is already small. Section 7.4 compares the effect across terciles of initial saving. The comparison is between treated and control students within each tercile, which protects against regression to the mean (the tendency of those who reported extreme values one month to report values closer to the average the next), since it hits both groups equally.

Prediction P5

Since β, m_i and a_{0i} are traits of the student, saving should be correlated from one month to the next.

In Equation (4.6), the only thing that changes from month to month is the shock ε_{it} (and, for the treated, use). Residual income, present bias, the size of the leaks and initial attention belong to the student. A student who saves above their group's average in August should, in general, save above average in September. If students differ in these traits, the correlation between months is positive; it is zero only if everyone is the same and saving varies by luck alone. Section 8.2 tests the prediction with the correlation between consecutive months of deviations from the group mean,

and with the intraclass correlation. Unlike the other four, this prediction says nothing about the effect of the tool; it is about saving itself. It is useful for another reason, which comes up in the next section: it is the prediction most sensitive to errors in reporting.

4.6 What the student reports

So far the model talks about s_{it}, true saving. The study does not observe it. It observes what the student wrote on the monthly form (Sections 3.10 and 5.4). To think clearly about this gap, I write a model of the report:

$$r_{it} = s_{it} + b\,T_i + \eta_{it}. \tag{4.10}$$

Here r_{it} is reported saving, T_i equals 1 for treated students and 0 for the control group, and there are two sources of difference between the report and the truth.

The term b is a **reporting bias** among the treated: a gap, in the same direction for all of them, between what they saved and what they reported. It could come from wanting to show progress to the researcher who trained them, from knowing the study was about saving, or from reporting the balance the tool shows instead of the month's saving. Online experiments tend to find small demand effects (de Quidt et al., 2018), but here the students reported directly to the person who ran the workshop. The term η_{it} is the **reporting error**: the random difference between the true value and the reported one, coming from a weak memory, a guess, a wrong sum. I assume it has mean zero, is independent across months, and does not depend on group or on use.

The two terms act in very different ways. I start with the bias. The difference in means between treated and control becomes

$$\mathbb{E}[r \mid T = 1] - \mathbb{E}[r \mid T = 0] = \mathbb{E}[\Delta s] + b, \tag{4.11}$$

so b enters the estimated effect in full. An experiment in which nobody saved one real more, but the treated reported b reais more, shows an effect of b. Prediction P1 cannot tell the two apart. The slope on use, however, does not change: b is the same for all treated students, whether each uses the tool a lot or a little, and its derivative with respect to u is zero. Among the treated, the relation between report and use is still κm_i (Appendix A.5). This assumes the bias does not depend on use. If the most active users wanted to show more, the bias would create a slope even with no saving at all; so a pure reporting bias cannot be ruled out by a positive slope, but it is consistent with a zero slope.

The error η does not move the means, because it has mean zero. It acts on persistence. Suppose true saving has correlation ρ from one month to the next, and let λ

be the **reliability** of the reports, the fraction of their variance that comes from true saving:

$$\lambda = \frac{\mathrm{Var}(s)}{\mathrm{Var}(s) + \sigma_\eta^2}. \tag{4.12}$$

Then the correlation between the reports for two months is

$$\mathrm{corr}(r_{it}, r_{it'}) = \rho\,\lambda, \tag{4.13}$$

as Appendix A.4 shows. The reason is simple: noise increases the variance of each month but does not increase what the two months have in common, because the August error has nothing to do with the September one. If true saving has $\rho = 0.5$ and only a fifth of the variance of the reports is signal ($\lambda = 0.2$), the observed correlation will be $0.5 \times 0.2 = 0.1$, even if each student's true saving is quite stable. With large noise, Prediction P5 fails in the reports even when it holds in saving. The bias b, by contrast, does not affect this correlation: it is a constant within the treated group and disappears when each student is measured relative to their group's mean.

Equation (4.13) can also be read backwards, and that is how it is used in Chapter 8. If I observe a correlation r and assume a value for ρ, reliability is at most r/ρ. Taking the upper limit of the correlation's 95 % interval, 0.08, as its ceiling, and with $\rho = 0.5$ (illustrative), reliability would be at most 0.16 (estimated; Section 8.2).

Putting Equation (4.8) and Equation (4.10) together, each explanation leaves a different pattern in four things the data can show. Table 4.1 summarizes these patterns. In every row, I assume the reporting bias is the same for all treated students and that the reporting error does not depend on group.

Table 4.1. What each explanation predicts for four patterns in the data. In the first row, the tool works through attention (with or without salience); in the second, every treated student gains the same amount, through salience, whether or not they use the tool; in the third, nobody saves more and the treated only report more. Persistence of the reports is the correlation between consecutive months of deviations from the group mean.

Explanation	Effect on the mean	Slope on use	Never logged *versus* logged	Persistence of reports
Attention	positive, $\mathbb{E}[\kappa u m]$ plus salience	positive, κm	those who never logged are closer to control	$\rho\lambda$: positive if noise is small
Salience, equal effect	positive, $\Delta v/[\beta v(v + \Delta v)]$	zero	equal, both above control	$\rho\lambda$: positive if noise is small
Reporting only	positive, equal to b	zero	equal, both above control	$\rho\lambda$: near zero if noise is large

Source: prepared by the author, from Equations (4.8), (4.10) and (4.13).

The table carries an uncomfortable lesson. The last two rows are identical in the first three columns: a salience effect that reaches all treated students equally and an effect that exists only in the reports produce the same difference in means, the same zero slope on use, and the same equality between those who logged and those who

did not. The fourth column does not separate the channels; it measures how reliable the reports are, whatever the channel. A reporting-only explanation sits naturally with noisy reports, but salience with noisy reports would give the same pattern. With this study's data, then, the best I can do is test attention. Separating salience from reporting requires an outcome that does not depend on what the student writes, such as a balance checked by someone else, and that is what the design in Chapter 9 aims for.

This is exactly the question the synthetic worlds of Chapter 8 ask through simulation (Section 8.6). Each world generates monthly reports for the same number of students, with the same means and the same variance as the data, combining the three channels of Table 4.1 with two reliabilities: reports equal to true saving, or reports in which the signal share of the variance is only 0.10. True saving has persistence 0.5 between months (illustrative), and each world is simulated 1,000 times. The real data are then placed next to these worlds, to see which one they resemble.

4.7 What the model leaves out

The model is static. It looks at one month at a time, as if the student decided from scratch every month, with no regard for what they saved before or what they plan to do with the money set aside. A student who has already put together enough for the earphones may save less the next month, and the model does not see that. There is also no borrowing: the student cannot spend more than they have, which is reasonable for teenagers with R$ 300, but it makes the corner solution more common than it would be in a model with credit.

The value of saving is constant, and this makes planned spending independent of income (Equation (4.5)). It is a strong simplification: in practice, a student with higher residual income probably spends a little more, and does not only save more. I chose it because it gives the tool's effect a closed form, Equation (4.8), and the qualitative predictions do not depend on it. Attention and salience are reduced forms: a and ν are labels for anything that makes the student see their spending more or value the future more. A goal written in the spreadsheet also works as a mental account, money set aside for a purpose (Thaler, 1999), and the model folds that effect into $\Delta\nu$ without telling it apart. Being asked about saving every month may, by itself, change behavior or the report (Zwane et al., 2011). Since the control group answered the same forms, this effect reaches both groups and does not show up in the difference between them; only the part that is larger among the treated enters the estimated effect, in s if it changes behavior, in b if it changes only the report.

Finally, none of the model's parameters was measured. The study did not measure β, ν, κ, m_i or a_{0i}; it measured use, u_{it}, and the report, r_{it}. The numbers in Example 4.1 are illustrative. The model does not say how large the effect should be. It says what

shape it should have: growing with use, changing with the budget, shrinking for those who already saved, and coming with persistent reports. Chapters 6 to 8 compare that shape with the data.

CHAPTER 5

The experiment

The previous chapters explained why a budgeting tool might change how much a teenager saves (Chapters 2 and 4) and how a random draw makes it possible to measure that effect (Chapter 3). This chapter tells what I actually did: who took part, how the students were split into groups, what each group received, how saving was measured, which ethical safeguards were taken and which were missing, and how the data are analyzed in the chapters that follow. I end by checking whether the random draw produced similar groups and whether attrition could have spoiled that similarity.

The facts about the design, the workshop, the tools, the dates, the ethics and the data collection come from the first version (Ribeiro, 2025a) and from the spreadsheet of the 208 students. The analyses follow the revised 2026 paper, which redoes the first version's calculations from that spreadsheet. Where the first version (v1) and the revised paper disagree, I say which one I follow and why. Where a detail was not recorded, I say it does not exist, instead of filling it in with what would be reasonable.

5.1 The PIC and the R$ 300 stipend

The Brazilian Mathematics Olympiad for Public Schools (OBMEP) is a national competition for public school students, from 6th grade to the end of high school. Medalists can join the Junior Scientific Initiation Program (PIC) of OBMEP, which pays a monthly stipend of R$ 300. The sample came from this group of scholars.

The stipend is the same for everyone. A 7th grader in a small inland town and a girl in her last year of high school in a state capital receive the same amount, on the same schedule. For some of them this may be the first regular income they manage on their own. For a study about saving, this helps in three concrete ways.

First, nominal income does not vary across students. In a study with adults, those who earn more almost always save more, and the income gap gets mixed up with everything one wants to measure. Here, everyone starts from the same R$ 300, and what varies is what each student has to pay with that money: bus fare, a snack at school, help at home, phone credit. I call what is left of the stipend after these essential expenses the **residual income**. It is the part of the money over which the student has a real choice, and the outcome is defined on it (Section 5.4).

Second, the stipend arrives every month, on a known date and in a known amount. This makes it possible to ask "how much did you save from this month's stipend" with a clear reference point, which would be hard with irregular income, such as odd jobs or an allowance that changes.

Third, the program has its own channels for talking to scholars, and that is how the invitation reached them. An intervention that works in this setting could, in principle, be repeated through the same channels, without building anything new.

The study also connects with larger policies. The federal Pé-de-Meia program, created in 2024, pays monthly incentives to low-income students in public high schools, on a much larger scale than the PIC (Brasil, 2024), and Brazil's national curriculum guidelines include financial education among their cross-cutting themes (Brasil. Ministério da Educação, 2018). If a simple tool helped PIC scholars save, the next question would be whether it would also help those students. PIC scholars, however, are not a sample of that population, and I return to this limit in the next section.

5.2 Who took part

Through the program's channels, 389 active PIC scholars were contacted. Of these, 208 returned the interest and consent form, a little over half, and they make up the sample. To take part, a student had to meet four criteria: have an active scholarship; have regular internet access and a smartphone; never have used budgeting spreadsheets or apps in a systematic way; and not have had formal financial education.

The first two criteria are practical: without internet and a phone, the student could not use the app or fill in the monthly forms. The last two make sure the tools were new to everyone. A student who already tracked spending in a spreadsheet of their own would gain little from another one, and mixing experienced students with beginners would make the groups less comparable.

These criteria also have a cost. Those who answered the invitation chose to answer, and I do not know how the 208 who returned the form differ from those who did not. Besides, OBMEP medalists are not typical public school students: they are good at math, were selected by a hard exam, and already take part in a program that demands commitment. None of this affects the comparison *between* the groups, because the random draw was held after selection, within the sample. But it limits who the results apply to. An effect found here says something about PIC scholars willing to join a study about money, and only indirectly about Pé-de-Meia students.

Figure 5.1 summarizes the sample. It is 62.0 % female (129 girls and 79 boys) and 27.4 % rural (57 students from rural areas and 151 from urban ones). The average age is 15.3 years, ranging from 11 to 18. All have a smartphone, since that was a criterion; 86 also have a laptop, 23 have a tablet, and 6 have all three devices. Grades run from

6th grade to the third year of high school, and the largest group is in the first year of high school.

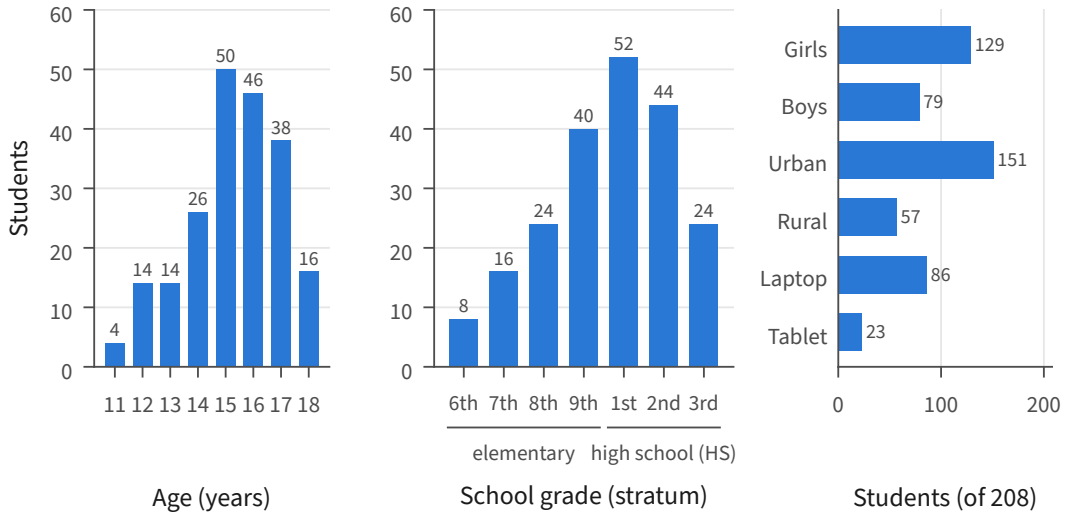


Figure 5.1. The sample (n = 208). Left: number of students by age, in years. Center: number of students by grade, from 6th grade to the third year of high school (HS); each grade is a stratum of the random draw. Right: gender, origin (urban or rural) and devices; the device categories overlap, because a student can have both a laptop and a tablet, and all have a smartphone.

Source: study data, measured (registration form; `code/13_figures.py`).

Essential spending was reported on the registration form. On average it was R$ 156.32 a month, which leaves an average residual income of R$ 143.68. In other words, a typical student spent a little over half the stipend on what they considered necessary and had a little under half to divide between spending and saving.

5.3 The random draw and what each group received

5.3.1 The random draw

The 208 students were randomly assigned to four groups of 52: control, spreadsheet, app and hybrid (spreadsheet plus app). The draw was **stratified** by grade, that is, carried out separately within each grade (Section 3.2). There are 7 grades, so there are 7 strata. The number of students in each grade is divisible by four, so each group got exactly a quarter of every grade. The grade composition of the four groups is therefore identical.

Analogy: a draw by grade

Imagine that the students in each grade write their names on slips of paper and put them in an urn for their grade only. From the 6th grade urn, the slips come out one by one and go, in turn, to control, spreadsheet, app and hybrid; then the same is done with the 7th grade urn, and so on. Within each urn, who lands in which group is pure chance, like a coin toss for each student. But no grade can end up concentrated in a single group, because each urn is split evenly.

There is a practical reason to stratify by grade. Grade is tied to age, and age is probably tied to how much a student spends and how much they manage to put aside. A simple draw, without strata, could by bad luck put more high school students in one group than in another. With strata, this source of difference disappears by construction, and the estimates become a little more precise (Bruhn and McKenzie, 2009).

The first version does not describe how the draw was run (for instance, with which software or seed) nor when, relative to the workshop. But it does say that, at the workshop, the treated students were split into rooms, one per tool, and for that each student already needed a group. From this I deduce that the draw came before the workshop. This is my inference, not a recorded fact.

5.3.2 The workshop

Before the intervention, I ran a 60-minute online workshop for the 208 students, and the recording was later made available on Google Drive for those who could not attend live. The study did not record who watched live, who saw only the recording, or who saw neither. The script had four parts.

The workshop opened with five minutes of welcome. Next came a fifteen-minute common session, the same for all groups, control included. It covered what a budget is, the difference between income and expenses, the difference between essential and discretionary spending (“what I need” and “what I want”), short-, medium- and long-term saving goals, and a simple explanation of present bias, the same bias as in Section 2.3.

After that, the treated students spent thirty minutes in separate rooms, one per tool, learning to use it. In the spreadsheet group, students learned to copy the template and fill in entries. In the app group, to create an account and record expenses. In the hybrid group, both, with a suggested routine for using them together. The script does not say what the control group did during those thirty minutes. The last ten minutes covered daily expense tracking, the monthly forms and how to ask for help.

Whether the control group took part in the common session is a point on which the two versions of the work disagree.[1] This matters for reading the results. The control group was not left with nothing: it received the same short lesson on budgets and goals. The difference between treated and control students, then, does not measure the effect of "financial education plus a tool" against "nothing". It measures the effect of everything the treated got and the control group did not: the tool, the thirty minutes in the separate room, the instruction to record every expense and, in the groups with the app, the weekly reminders.

5.3.3 The spreadsheet

The spreadsheet group received a Google Sheets template with three tabs. The *entries* tab had one row per transaction, with the date, a description, a category picked from a list, the type (income or expense) and the amount. The categories covered income (the PIC stipend and other), essential spending (transportation, food, help at home), education (books, courses, supplies), leisure (outings, snacks, games) and other expenses.

The *monthly summary* tab did the math on its own: total income, total expenses and the balance, which is the month's saving, plus spending in each category in reais and as a percentage of the total. The *dashboard* tab turned these numbers into pictures: a pie chart of spending by category, a bar chart of income against expenses, and a "thermometer" that filled up as the student got closer to the saving goal they had chosen themselves.

The design follows the idea in Section 2.6: a student who writes down an R$ 8 snack and sees, at the end of the month, the "leisure" slice of the pie chart starts to notice expenses that used to go unseen. The thermometer puts a number on the goal, which stops being a vague wish.

5.3.4 The Mobilis app

The app group was told to install Mobilis, a personal finance app, and received a step-by-step guide to creating expense categories, setting monthly budgets (spending limits) and turning on reminder notifications. Students with the app received automatic weekly reminders to *record* their expenses, that is, a phone notification reminding the student to write down what they had spent. The study did not record how many students followed each step of the guide. These were not messages saying "save". The distinction matters because, in the model of Chapter 4, the reminder

1 The first version describes the control group both as "no intervention" and as taking part in the workshop held with the 208 students (Ribeiro, 2025a). The workshop script, in the v1 appendix itself, indicates that the control group attended the common session. I follow the script.

works through attention (the student records more and sees more) and not through a direct appeal to save.

5.3.5 The hybrid group

The hybrid group received both tools, with a suggested routine: record expenses in the app every day, since it is quick and always in your pocket, and review the month in the spreadsheet, which shows the big picture. Because it used the app, this group also received the weekly reminders.

All three treated groups were told to record every expense, ideally on the same day or at the moment of purchase. The intervention ran from August 1 to October 31, 2025, three full months. Figure 5.2 shows the students' path from first contact to analysis.

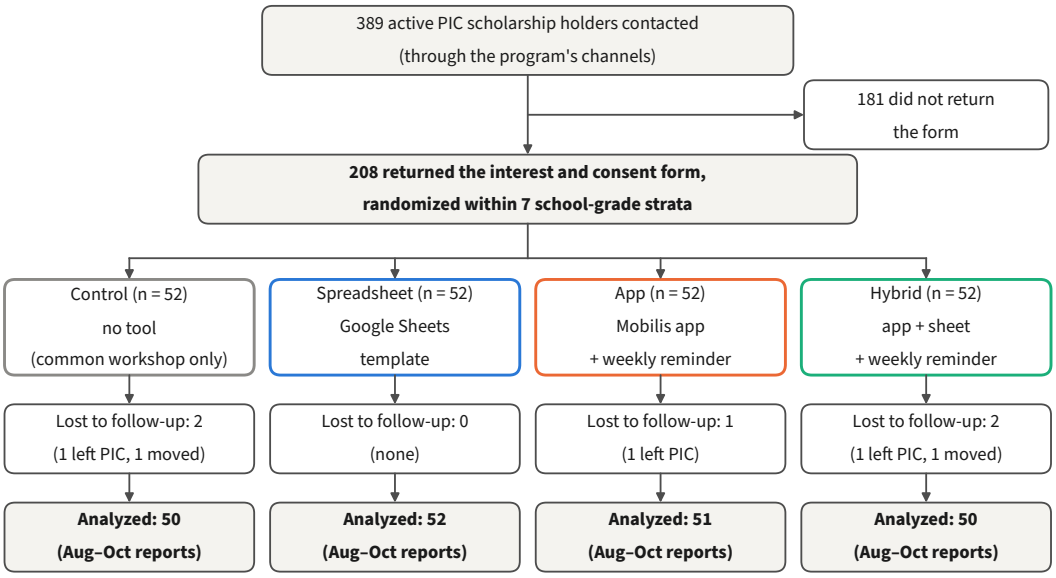


Figure 5.2. Participant flow, from contact to analysis. The 208 students who returned the form were randomly assigned within 7 grade strata to four groups of 52 (gray outline: control; blue: spreadsheet; orange: app; green: hybrid). The third row says what each group received; the fourth gives each group's dropouts and the reason; the last, the students with reports from August to October who enter the analyses.

Source: study data, measured (`code/13_figures.py`).

5.4 What was measured

5.4.1 The monthly forms

At the end of August, September and October 2025, all students, in all four groups, filled in a simple online form saying how much they had saved that month. This is

the study's only measure of saving, and it is a *report*: the number the student wrote. Nobody checked that number against a bank statement, a piggy bank or an account. The first version says students reported the amount saved, but the study spreadsheet stores the percentage, with two decimal places, and I do not know whether each student typed the value in reais or as a percentage.

The treated students were also asked to send, along with the form, a screenshot or a summary from the tool. These screenshots gave the count of *logs per week*, the measure of tool use that Chapter 7 analyzes. The logs are used as they appear in the spreadsheet. For 25 treated students the value is zero, and the file does not distinguish a true zero (the student logged nothing) from a missing screenshot (the student logged but did not send the image). The first version also reports no comparison between reported saving and what appeared in those screenshots, and the screenshots are not among the files I have.

Besides the three months of the intervention, there is a **baseline**, measured before August: the student's characteristics and essential spending, reported on the registration form, and initial saving, on the same scale as the outcome. The baseline serves two purposes: checking whether the random draw produced similar groups (Section 5.7) and making the estimates more precise (Section 5.6).

5.4.2 The outcome

The **outcome** is the variable the experiment tries to change. Here, it was defined in the first version as the fraction of initial residual income that the student saved in the month:

$$S_{it} = 100 \times \frac{\text{saved by } i \text{ in month } t}{R\$\,300 - e_i}, \tag{5.1}$$

where i indexes the student, t the month (0 for baseline; 1, 2 and 3 for August, September and October) and e_i is student i's essential spending at baseline. The denominator, $R\$\,300 - e_i$, is initial residual income, R_i, and does not change from month to month. The main outcome is S_{i3}, October saving.

Example 5.1 One student's outcome

Take an illustrative student, Júlia, who at registration reported spending R$ 150 a month on bus fare and snacks. Her residual income is $R\$\,300 - R\$\,150 = R\$\,150$. If in September she reports having saved R$ 60, the outcome is $100 \times 60/150 = 40$, that is, 40 % of residual income. Another student, with essential spending of R$ 250 and residual income of R$ 50, who saved the same R$ 60, would have an outcome of 120 %.

The example shows the two properties of the outcome that weigh most on how the results are read. The first is that it measures saving relative to what the student had to decide about. For Júlia, R$ 60 is a moderate share of what is left; for the other

student, it is more than everything that is left. The first version chose this measure precisely to compare students with different budgets more fairly.

The second is that S_{it} can exceed 100. Because the denominator is fixed at baseline, a student can report more than 100 % if they cut essential spending after registration, if they had some income beyond the stipend (a birthday present, an odd job), or if they reported something other than the month's saving, for example the balance accumulated since August. The study did not measure essential spending after baseline, so the data cannot separate these explanations. Chapter 8 returns to this point, because in October many treated students reported more than 100 %.

I also present the same outcome in reais: S_{it} times residual income, divided by 100. The spreadsheet has a column in reais only for October, and that column is just the percentage times residual income. For baseline, August and September, I do the same calculation. Since the percentage has two decimal places, the recovered value in reais is off by at most about one centavo. The two scales answer different questions. The percentage says how much of the student's room for choice turned into saving; reais say how much money that represents, and they are the natural scale for comparing with the stipend or with other programs.

5.4.3 Attrition

Five students left the study before the end: 2 from the control group, 0 from the spreadsheet group, 1 from the app group and 2 from the hybrid group. Three left the PIC and two moved away (Figure 5.2). None of these reasons has an obvious link to the tools. That leaves 203 **completers**, the students with all three monthly reports, and the estimates are based on them. Section 5.7 checks whether dropping out is linked to group or to initial saving.

5.4.4 Timeline

Table 5.1 lists the stages of the study with the dates the first version gives. Stages without an exact date appear only in order, and the order of the random draw and the workshop is the inference explained in Section 5.3.

5.5 Ethics

The students and, for minors, their parents or guardians signed a consent form that described the study's aims, procedures, risks, benefits, and the right to withdraw at any time without penalty. Data were collected under pseudonymous codes, in the first version's "Participante-G*x*-N*y*" format, with no names in the monthly reports,

Table 5.1. Timeline of the study and of the versions of the work.

When	What
Before August 2025	Invitation to 389 scholars; interest and consent form; baseline
Before August 2025	Random draw of the 208 students in 7 grade strata (before the workshop, by inference)
Before August 2025	60-minute online workshop; recording shared
Aug. 1 to Oct. 31, 2025	Intervention: treated groups use the tools
End of Aug., Sept. and Oct. 2025	Monthly saving forms; screenshots from the treated
October 2025	First version of the paper (v1)
December 2025	Extended abstract
October 2026	Revised paper and this monograph

Source: prepared by the author, with the dates from v1 (Ribeiro, 2025a,b); the last row is the date of this version. The exact dates of the invitation, the random draw and the workshop were not recorded.

and were kept with access restricted to me during the study. The study spreadsheet contains no names, schools or contact details.

There are two things the first version does not report: approval of the project by a research ethics committee, and registration of the trial in a public registry before data collection. I have no approval number or registration number to present here, and I will not write as if I did.

In Brazil, research with human subjects follows Resolução n.º 466/2012 of the Conselho Nacional de Saúde (the National Health Council) (Brasil. Conselho Nacional de Saúde, 2012). It organizes ethical review into a system made up of research ethics committees (CEPs) and the National Research Ethics Commission (CONEP), and says that research with human subjects must be submitted to this system. The researcher is responsible for presenting the project to a CEP, or to CONEP, and for waiting for approval before starting the research. For participants who are minors, the rule is twofold: the parent or guardian consents, and the participant also gives their own assent, on an assent form written in language they can follow. The resolution also announces complementary rules for research in the humanities and social sciences (item XIII.3); here I rely only on the text of 466/2012. The study had part of this, the form signed by students and guardians (the first version does not mention a separate assent form), but it lacks documented prior review.

What does this mean for this monograph? The study's risks seem small: the intervention did not touch the students' money, and the control group received the same common lesson as everyone else. But ethical review does not exist only for large risks. It is the moment when someone from outside asks, before data collection, whether the form explains the study well, whether the data can be published, and how participants will be protected. Without it, two questions remain open. The first is whether the signed form allows publishing the de-identified spreadsheet, as this work does in its repository; if it does not, the spreadsheet should come out of the public repository and be available only on request, with the code unchanged. The second is whether the students, reporting their saving to the same researcher

who trained them, felt free to report a low number, a question of ethics and also of measurement, which Chapter 8 examines.

The lack of preregistration has a statistical consequence. Without a plan published before data collection, the reader cannot know which analyses were chosen before seeing the data. That is why, in Section 5.6, I separate what the first version already described from what was added later, and I treat the second group as exploratory.

The follow-up study I propose in Chapter 9 fixes both gaps (Section 9.5): submission to a CEP before recruitment, consent and assent forms that ask explicit permission to share de-identified data, and registration of the trial and the analysis plan in a public registry before the random draw.

5.6 Empirical strategy

5.6.1 What I want to estimate

The parameter of interest is the effect of being *assigned* to each tool on reported saving in October. The word "assigned" is important. A student in the app group who never opened the app stays in the app group in the analysis. This is the *intention-to-treat* effect of Section 3.3: it answers what happens when the tool is *offered*, not what happens to those who use it. It is the relevant effect for a public program, which can offer a tool but cannot make anyone use it.

Since there is an outcome only for the 203 completers, the estimate is the intention-to-treat effect *among completers*. If the five dropouts had been drawn at random, this effect would be the same as for the whole sample. I cannot guarantee that, so Chapter 6 uses two procedures that show how much the estimate could change because of attrition: the bounds of Lee (2009) and a worst-case imputation.

5.6.2 The main regression

The main specification is a regression of October saving on the group indicators, controlling for initial saving and some student characteristics. This kind of regression, with the initial outcome as a control, is called **ANCOVA** (Section 3.3):

$$S_{i3} = \alpha + \sum_{g=2}^{4} \beta_g T_{gi} + \rho\, S_{i0} + X_i'\gamma + \mu_{s(i)} + \varepsilon_i. \tag{5.2}$$

Each term has a job. S_{i3} is student i's saving in October, by Equation (5.1). T_{gi} equals 1 if the student was assigned to group g and 0 otherwise, with $g = 2$ for the spreadsheet, $g = 3$ for the app and $g = 4$ for the hybrid; control ($g = 1$) is the reference and has no indicator. So α is, roughly, the expected saving of a control student, and

β_g is the effect of group g in percentage points of residual income: how much more, on average, a student assigned to tool g saves than a student assigned to the control group. The three β_g are the numbers the study is after.

S_{i0} is initial saving, on the same scale, and ρ measures how much it helps predict October saving. X_i is the vector of covariates: age, gender, rural origin, owning a laptop, and initial essential spending; γ are their coefficients. $\mu_{s(i)}$ is a **fixed effect** for the student's stratum $s(i)$, that is, a separate intercept for each grade, which absorbs any average difference in saving between grades. Finally, ε_i is what is left: everything that makes a student's saving differ from the prediction.

I also estimate a *pooled* version, in which the three indicators are replaced by a single one, T_i, equal to 1 for any tool. Its coefficient is the effect of receiving some tool, without distinguishing which, and it is the study's most precise estimate, because it compares the 153 treated completers with the 50 in control.

5.6.3 Why control for anything, if the groups were drawn at random

The controls are not needed for the estimate to be unbiased. Since the random draw decides who gets each tool, on average the groups are equal in everything, measured or not, and the simple difference in means already estimates the effect (Section 3.1). I include the controls for three reasons.

The first is precision. If initial saving or essential spending help predict October saving, taking them out of the error term reduces the variance of ε_i and narrows the confidence interval. Section 3.3 shows that the gain depends on the correlation between the initial and the final outcome; Chapter 6 shows that in these data the gain is small, and Chapter 8 explains why.

The second is to correct imbalances the draw produced by chance. Essential spending, and with it residual income, ended up different across groups (Section 5.7), and controlling for it takes that difference out of the comparison.

The third is to respect the design. When the draw is stratified, the usual advice is to include the strata in the regression (Bruhn and McKenzie, 2009), and that is what the grade fixed effects do. In an experiment, the bias this adjustment can introduce shrinks with sample size and is small in large samples, and robust standard errors remain valid (Lin, 2013).

5.6.4 Standard errors

The standard errors are **heteroskedasticity-robust**, in the HC2 version (MacKinnon and White, 1985). Heteroskedasticity means that the spread of the error is not the same for everyone. Here that is to be expected: the groups may have different standard deviations, and students with different budgets may have more or less variable saving. The ordinary standard error assumes equal spread and can come out too small. HC2

makes no such assumption and was proposed for small samples like this one, in which the original robust version tends to understate uncertainty.

5.6.5 Month-by-month dynamics

The main regression uses October only. To see how the effect built up over the three months, I estimate an **event study** on the student-month panel, with one observation per student per month (baseline, August, September and October):

$$S_{it} = a_i + \lambda_t + \sum_{g=2}^{4} \sum_{k=1}^{3} b_{gk}\, T_{gi}\, \mathbb{1}[t = k] + e_{it}. \tag{5.3}$$

Here a_i is a student fixed effect, which absorbs everything that is constant for the student over time, including their level of saving at baseline. λ_t is a month fixed effect, which absorbs what changes for everyone at once, such as a month with more school expenses. $\mathbb{1}[t = k]$ equals 1 in month k. The coefficient b_{gk} is then the difference between group g and the control group in month k, net of the difference that already existed at baseline. It is a **difference in differences**, month by month.

Since each student appears four times, errors for the same student may be correlated from one month to the next (McKenzie, 2012). So the standard errors are **clustered by student**: the uncertainty calculation treats the student's four observations as one block, not as four independent pieces of information. I also estimate the pooled version, with a single coefficient for "any tool, August to October".

5.6.6 Checks on inference

Four checks go with the estimates.

Randomization inference (Section 3.7) redoes the random draw 20,000 times on the computer, always within the grade strata, and recalculates the difference between treated and control in each redone draw (Athey and Imbens, 2017; Fisher, 1935; Young, 2019). The *p*-value is the fraction of redone draws that produce a difference as large as the one observed. This test relies on no assumption about the distribution of saving, only on the random draw itself.

The six comparisons between pairs of groups (spreadsheet against control, hybrid against app, and so on) are adjusted with the method of Holm (1979), which controls the chance of at least one false positive among the six. The six subgroup splits (gender, age, origin and others) are adjusted with the false discovery rate control of Benjamini and Hochberg (1995), which controls the expected share of false positives among the results declared significant (Section 3.6). Both corrections appear in Chapter 7.

Attrition is handled with the bounds of Lee (2009) and with the worst-case imputation, described in Chapter 6.

5.6.7 What was planned and what came later

No analysis plan was published before data collection. The closest thing to a plan is what the first version describes in its methods section: the outcome in Equation (5.1), a one-way ANOVA comparing the four groups, *t*-tests between pairs, a regression with group indicators and covariates, a regression of saving on frequency of tool use, and comparisons by gender, age and origin. I repeat all of these analyses, and Appendix C checks every v1 number against the data.

Everything else was added after the data were collected and, in part, after the first version was written: the model of Chapter 4 and its predictions, the event study, randomization inference, the specification curve, the Lee bounds, the quantile effects, the equivalence tests and all the analyses of report reliability. These analyses should be read as *exploratory*. That does not mean they are wrong. It means I chose them knowing, in broad terms, what the data looked like, and a careful reader should weigh their results with more caution than those of an analysis announced in advance.

5.7 Did the random draw produce similar groups?

A random draw produces equal groups *on average*, if it were repeated many times. In a single draw, with 52 students per group, small differences are always left over by chance. Checking **balance** means looking at those differences and asking whether they are the size chance produces or larger. Table 5.2 compares the four groups at baseline, with all 208 assigned students, including those who later dropped out.

Grade is identical across the four groups, as it should be. That leaves nine rows, but two of them, essential spending and residual income, are the same test, which gives eight distinct tests. One characteristic differs at the 5 % level: essential spending, which is higher in the spreadsheet group and lower in the app group (p of 0.04). Since residual income is R$ 300 minus essential spending, it differs by the same amount in the opposite direction. The app group also has more rural students than the others, but that difference is not significant at 5 %.

One rejection in eight tests is no surprise. If the groups were perfectly comparable, each test would still have a 5 % chance of rejecting by chance, and with eight tests the chance of at least one rejection is about one in three. The better question is whether the covariates, taken together, help predict who got a tool. Two regressions answer this. In the first, assignment to any tool is regressed on the seven covariates that do not define strata, and the test that all their coefficients are zero gives p of 0.81. In the second, a multinomial model predicts the group among the four, and

Table 5.2. Baseline characteristics by group, with the 208 assigned students. Group means; rows with % are shares of students. The last column is the *p*-value of a robust *F*-test that the four means are equal. Grade is equal by construction, because the draw was held within each grade. Residual income is R$ 300 minus essential spending, so the two rows have the same *p*. The joint test regresses assignment to any tool on the seven covariates that do not define strata; the multinomial test does the same for the four groups at once.

	Control (n = 52)	Spreadsheet (n = 52)	App (n = 52)	Hybrid (n = 52)	p (F-test)
Age (years)	15.27	15.23	15.33	15.29	0.99
School grade (6–12)	9.63	9.63	9.63	9.63	1.00
Female	59.6%	69.2%	55.8%	63.5%	0.53
Rural	25.0%	21.2%	42.3%	21.2%	0.07
Owns a notebook	36.5%	38.5%	42.3%	48.1%	0.65
Owns a tablet	17.3%	9.6%	5.8%	11.5%	0.30
Essential expenses (R$)	157.86	163.74	149.21	154.47	0.04
Residual income (R$)	142.14	136.26	150.79	145.53	0.04
Baseline saving (% of residual)	41.70	45.89	39.10	37.42	0.16
Baseline saving (R$)	59.23	62.92	57.67	54.67	0.58
Joint test, 7 covariates → any treatment: p = 0.81; multinomial LR test: p = 0.85					

Source: study data, measured (registration form) and estimated (`code/02_balance_attrition.py`).

the likelihood ratio test gives p of 0.85. Neither comes close to rejecting: knowing a student's characteristics does not help guess their group.

The p-value depends on sample size; a large difference may not be significant in a small group. That is why I also look at the **standardized difference**: the difference between a treated group's mean and the control mean, divided by the standard deviation of the characteristic. It does not depend on sample size, and a common rule of thumb treats absolute values below 0.1 as negligible. Figure 5.3 shows the standardized differences for each group with a tool. The largest in absolute value is 0.39 standard deviation, for the share of rural students in the app group. Close behind, in the same group, come tablet ownership (fewer students with a tablet than in the control group) and essential spending and residual income.

Several differences exceed 0.1, which is common with groups of 52 students. What they call for is exactly what Equation (5.2) does: include those characteristics as controls, so that a chance imbalance is not confused with an effect of the tool. Chapter 6 shows that the estimates barely change when the controls go in, which is the expected sign of a random draw that worked.

5.7.1 Attrition

Losing students along the way can spoil a well-run random draw. If, for example, control students who saved little dropped out more often, the remaining control group would seem to save more than it really does, and the effect would come out

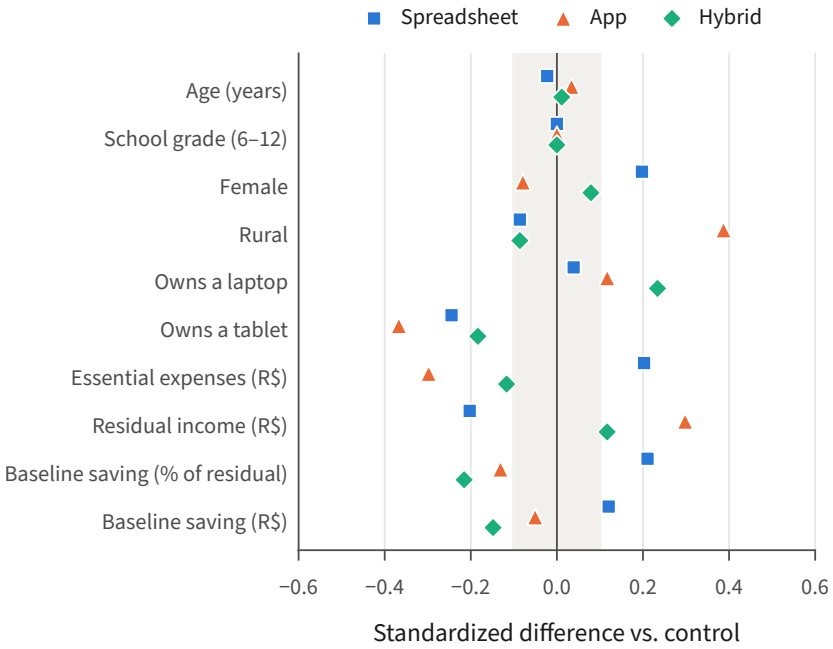


Figure 5.3. Standardized difference between each group with a tool and the control group at baseline: difference in means divided by the standard deviation of the characteristic. Blue squares: spreadsheet; orange triangles: app; green diamonds: hybrid. The shaded band marks differences smaller than 0.1 in absolute value. Grade sits at zero by construction.

Source: study data, estimated (`code/13_figures.py`).

underestimated. This problem is called **attrition**, and there are two questions to ask about it.

The first is whether the attrition rate differs between groups. In the control group, 3.8 % dropped out; among the treated, 1.9 %. Fisher's exact test, suited to small counts like these, gives p of 0.60: the difference is consistent with chance.

The second is whether those who dropped out were different from those who stayed. At baseline, dropouts reported saving 41.3 % of residual income, and completers 41.0 % (p of 0.96, by the Mann-Whitney test). The reasons for dropping out (leaving the PIC, moving to another city) also have no obvious link to saving or to the tools.

With only five dropouts, these tests have little power: they would not rule out moderate differences. The stronger point is a different one. Five students out of 208 are too few to change an effect of the size Chapter 6 finds, even in the worst case, and Section 6.4 shows this with numbers.

In short, the random draw produced comparable groups, with a chance imbalance in essential spending that the regression controls for, and dropouts were few and do not look selective. The comparison between groups, the subject of the next chapter, can be read as the causal effect of the assignment on what the students reported. What this chapter does not settle is the nature of the outcome: a number reported by the student, with no outside check. That question is left for Chapter 8.

CHAPTER 6

Main results

This chapter answers the study's first question: did receiving a budgeting tool change how much the students reported saving? I start with what the data show without any model, the average of each group month by month. Then I move to the regressions of Chapter 5, which give the size of the effect with its confidence interval, and to the event study, which shows how the effect built up from August to October. Finally, I put the result through a series of checks: redone random draws on the computer, more than a hundred alternative ways of doing the calculation, attrition under the worst case, and the effect in each part of the distribution.

The short answer is yes, and by a wide margin. What this chapter does not answer, and I come back to this at the end, is whether reported saving corresponds to money actually put aside, and which tool works best.

6.1 What happened month by month

Figure 6.1 shows the average of each group at baseline and in the three months of the intervention, and Table 6.1 gives the same numbers with standard deviations. At baseline, the four groups were close: the control group reported saving 41.27 % of residual income, and the spreadsheet, app and hybrid groups 45.89 %, 39.21 % and 37.55 %. The differences among them are the size that chance produces (Section 5.7).

As early as August, the first month, the three groups with a tool pulled away from the control group: 67.45 %, 70.37 % and 75.97 %, against 42.33 %. In September they rose to 80.59 %, 89.64 % and 91.33 %. In October they reached 91.40 %, 97.82 % and 101.07 %. The hybrid group's October average is above 100 %: on average, these students reported having saved more than all the residual income they had at baseline.

The control group, meanwhile, hardly moved. It went from 41.27 % at baseline to 44.25 % in October, a change of 3.0 percentage points that a paired *t*-test cannot distinguish from zero (*p* of 0.47). This can be read in two useful ways. First, nothing suggests that the common part of the workshop, which the control group also got, changed reported saving on its own (the study did not record who attended the workshop, and v1 does not say whether the baseline was measured before it). Second, there was no general movement, such as a month of unusual expenses, pushing all students in the same direction. Whatever changed, changed only in the treated groups.

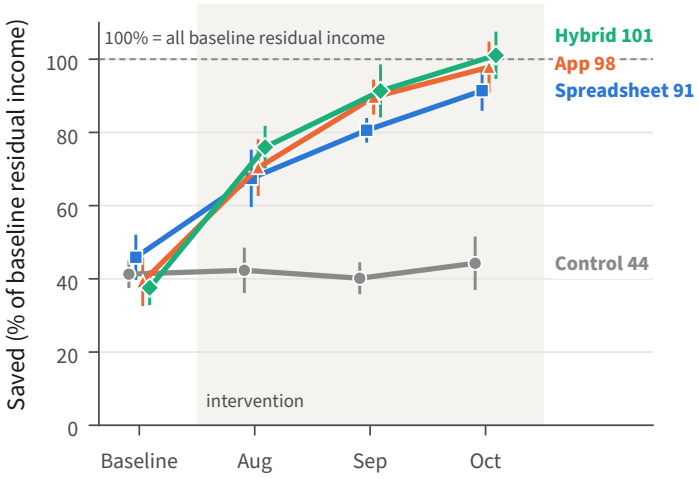


Figure 6.1. Average reported saving by group and month, as a percentage of initial residual income (Equation (5.1)), with 95 % confidence intervals. Gray: control; blue: spreadsheet; orange: app; green: hybrid. The shaded band marks the three months of the intervention, and the dashed line marks 100 %, saving all of initial residual income. The numbers on the right are the October means.

Source: study data, measured (monthly forms) and estimated (`code/13_figures.py`).

Table 6.1. Reported saving, in % of initial residual income, by group and month: mean and, in parentheses, standard deviation. Completers only.

	Control	Spreadsheet	App	Hybrid
Baseline	41.27	45.89	39.21	37.55
	(13.5)	(22.7)	(24.4)	(17.1)
Aug.	42.33	67.45	70.37	75.97
	(22.4)	(28.8)	(28.2)	(20.9)
Sept.	40.16	80.59	89.64	91.33
	(15.7)	(12.4)	(17.4)	(26.1)
Oct.	44.25	91.40	97.82	101.07
	(26.4)	(20.3)	(25.5)	(23.3)

Source: study data, measured (monthly forms; `code/04_panel.py`).

The standard deviations help put the differences in proportion. Within each group, in October, the standard deviation ranges from 20.3 to 26.4 points. The gap between the control group and the treated groups is about 52.0 points, or roughly 2.0 times the control group's standard deviation. Differences like this are rare in social experiments, and Section 6.2 comes back to this point.

6.1.1 Student by student

Averages can hide a lot. A group can have a high average because three students reported huge values while the other forty-odd stayed where they were. Figure 6.2 shows each student in October, and the picture is different: the whole distributions shifted. The control median is 40.0 %, and the treated groups' medians are 91.1 %, 97.9 % and 104.4 %, very close to the means. There is overlap between control and

treated, but it is small: the few control students above 90 % are exceptions, and so are the few treated students below 60 %.

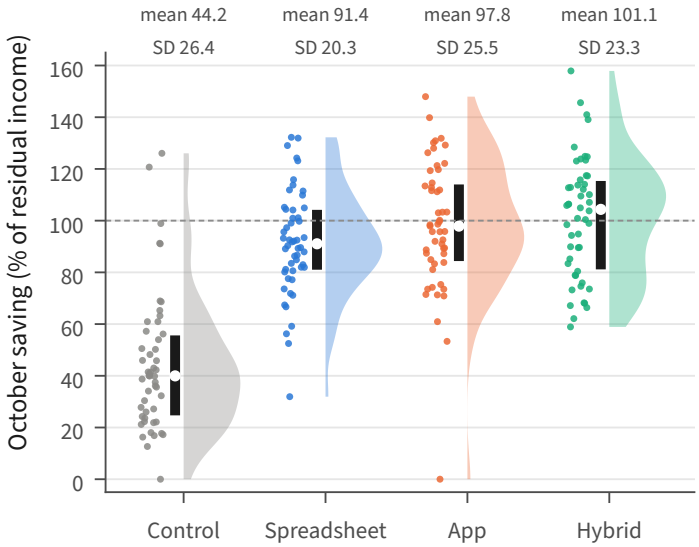


Figure 6.2. Reported saving in October, student by student, in % of initial residual income. Each dot is a student; beside it, the estimated density of the group's distribution. The black bar is the interquartile range (25th to 75th percentile), and the white dot is the median. At the top, each group's mean and standard deviation (SD). The dashed line marks 100 %. Colors as in Figure 6.1.

Source: study data, measured (monthly forms; `code/13_figures.py`).

The figure also shows that many treated students ended up above the 100 % line: 43 % of the treated, against 4 % of the control group. As I explained in Section 5.4, this is possible if the student cut essential spending or had other income, but it is also a sign that some may have reported something other than the month's saving. I examine this possibility in Section 8.3.

Figure 6.3 follows each student across the four time points. In the control group, the lines cross in every direction around a flat average. In the three treated groups, almost all lines rise from baseline to October. The gain, then, is widespread and does not come from a few extreme values. But the figure shows something else, which will matter in Chapter 8: the individual lines are very irregular. A student who reports a lot one month may report little the next, and the ranking of students within a group changes almost completely from one month to another (Section 8.2).

6.1.2 In reais

Percentages of residual income are an abstract scale. Figure 6.4 translates the averages into reais per month. At baseline, the four groups reported saving similar amounts: R$ 59 in the control group and R$ 63, R$ 58 and R$ 55 in the spreadsheet, app and hybrid groups. In October, the control group reported R$ 65, and the treated groups R$ 123, R$ 148 and R$ 148. For comparison, the whole stipend is R$ 300: a typical

treated student reported putting aside between two fifths and half of it in October, against a little over a fifth in control.

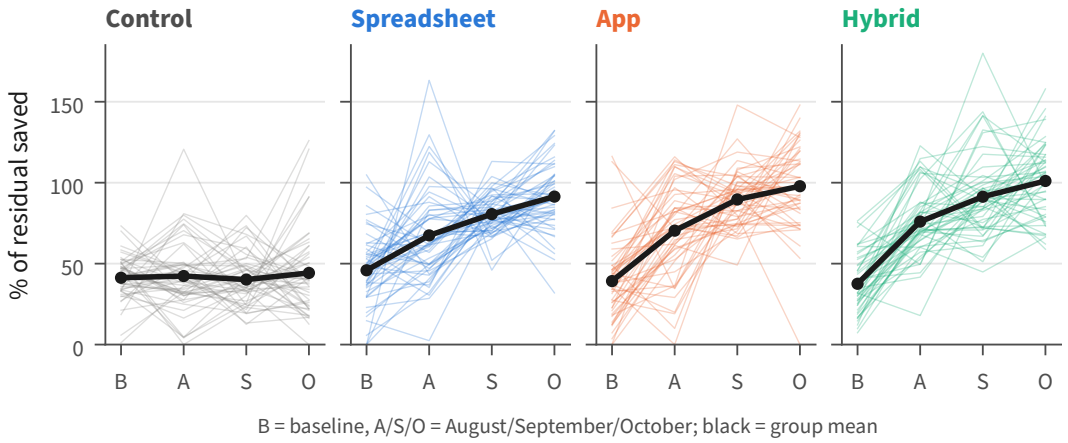


Figure 6.3. Trajectory of each completer (thin lines), in % of initial residual income, at baseline (B) and in August (A), September (S) and October (O), with the group mean (thick black line). One panel per group, with the colors of Figure 6.1.

Source: study data, measured (monthly forms; `code/13_figures.py`).

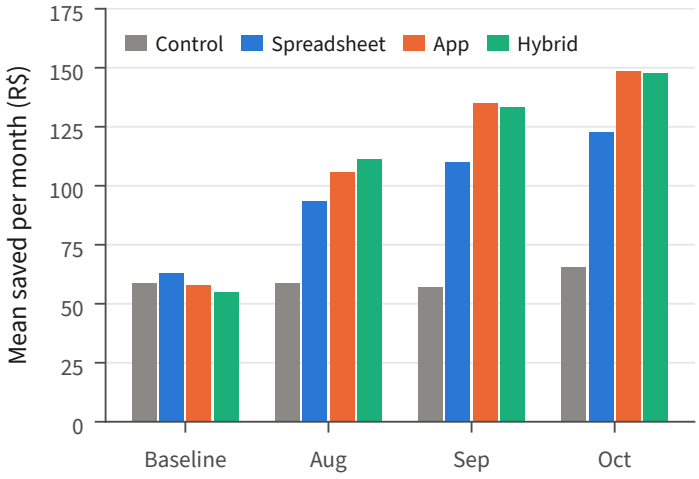


Figure 6.4. Average reported saving in reais per month, by group, at baseline and in the three months of the intervention. Colors as in Figure 6.1. All values are the reported percentage times each student's initial residual income; the spreadsheet has a column in reais only for October, computed the same way.

Source: study data, measured (monthly forms) and estimated (`code/13_figures.py`).

In reais, the order of the tools shifts a little: the app group ties with the hybrid group. This happens because the app group had, by chance, the highest average residual income (Table 5.2), and the same percentage is worth more reais in a larger budget. It is a first warning that the comparison between tools is fragile, the subject of Chapter 7.

6.2 The size of the effect

The averages already tell the story, but they do not say how much uncertainty there is in them. Table 6.2 estimates Equation (5.2) in four versions, one per column. Each cell gives the effect of one group relative to the control group, with the HC2 standard error in parentheses.

6.2.1 Column by column

Column (1) has no controls at all. The coefficients are simply the difference between each group's October average and the control average: 47.1 points for the spreadsheet (standard error 4.7), 53.6 for the app (5.2) and 56.8 for the hybrid (5.0). In an experiment, this is the most transparent estimate, because it depends on no choice of specification. All three effects are about ten times their standard errors.

Table 6.2. Effect of the tools on reported saving in October. Ordinary least squares (OLS) regressions, one per column, with HC2 standard errors in parentheses. Columns (1) to (3): saving in percentage points of initial residual income. Column (4): reais saved in October, controlling for initial saving in reais. Covariates: age, gender, rural origin, laptop ownership and essential spending; FE: fixed effects. The *p* rows test equality between tools. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

	% of residual income saved			R$ saved
	(1)	(2)	(3)	(4)
Spreadsheet	47.15***	47.43***	46.86***	61.59***
	(4.67)	(4.69)	(4.70)	(6.93)
App	53.57***	53.45***	53.75***	75.59***
	(5.16)	(5.20)	(4.88)	(7.76)
Hybrid	56.82***	56.60***	55.68***	77.44***
	(4.97)	(5.01)	(4.89)	(7.35)
Baseline outcome	–	✓	✓	✓
Covariates, grade FE	–	–	✓	✓
Control mean	44.25	44.25	44.25	65.48
p: Spreadsheet = App = Hybrid	0.07	0.11	0.16	0.05
p: Hybrid = App	0.50	0.51	0.67	0.79
R^2	0.48	0.48	0.52	0.61
Observations	203	203	203	203

Source: study data, estimated (`code/03_main_effects.py`).

Column (2) adds initial saving. The effects become 47.4, 53.4 and 56.6, practically the same. The standard errors barely change either, and the R^2 stays at 0.48, the same as in column (1). This is surprising. In Section 3.3, ANCOVA gains precision when initial saving helps predict final saving, and for a teenager's saving one would naturally expect that: someone who saved a lot before probably keeps saving a lot. Here, initial saving does not help predict October saving. Section 8.2 shows that this holds for any pair of months, and discusses what it says about the reports.

Column (3) is the main specification, the full Equation (5.2), with covariates and grade fixed effects. The effects are 46.9, 53.7 and 55.7 points. The R^2 rises a little, to 0.52. The imbalance in essential spending (Section 5.7) was not distorting the comparison: going from column (2) to column (3), which adds that control and the others, moves each estimate by less than one point.

Column (4) measures the outcome in reais. The effects are R$ 61.6, R$ 75.6 and R$ 77.4 a month, on a control average of R$ 65.48.

The bottom rows compare the tools with each other. The test that the three effects are equal gives p between 0.05 and 0.16, depending on the column, and the test that hybrid and app are equal gives p between 0.50 and 0.79. None rejects at 5 % in the columns in percentage points; only in column (4), in reais, does the test that all three are equal reject at 5 %, and only barely ($p = 0.05$). Chapter 7 deals with this comparison in detail; here it is enough to note that every tool beats the control group by a wide margin, and that the differences among them are small next to that margin.

6.2.2 The effect of receiving any tool

Since the three tools have similar effects, the question the data answer best is the pooled one: how much does reported saving change for someone who receives some tool? Estimating Equation (5.2) with a single treatment indicator, the effect is 52.0 pp, with a 95 % confidence interval from 44.2 to 59.8. In reais, it is R$ 71.29 a month (95 % CI from R$ 59.19 to R$ 83.39), about R$ 855 a year if the effect stayed at the same level for twelve months, which the study did not observe.

To give a sense of scale, the effect amounts to an increase of 117 % over the control average in October: reported saving more than doubled. Measured in control standard deviations, the usual scale for comparing different studies (Section 3.5), the effect is 2.0 standard deviations. The typical effect of financial education on financial behavior is about a tenth of a standard deviation (Kaiser et al., 2022), which on this scale would be 2.6 pp. The effect here is about twenty times larger. That does not prove it is wrong, but it is a reason to look at the data more carefully, and it is the starting point of Section 8.5.

The confidence interval deserves a careful reading. It does not say that the true effect lies between 44.2 and 59.8 points with 95 % probability; it says that a procedure producing intervals like this one captures the true value in 95 % of experiments (Section 3.4). In practice, what matters here is that the lower limit, 44.2 pp, is already a huge effect. Even the most pessimistic reading that sampling uncertainty allows is that reported saving nearly doubled. The doubt about this result, then, does not lie in the sample size or in chance but, as I discuss at the end of the chapter, in what the reported number measures.

Example 6.1 The effect for a typical student

Go back to the illustrative student of Example 5.1, Júlia, with residual income of R$ 150, close to the sample average (R$ 143.68). For her, each percentage point is worth R$ 1.50. Suppose, also for illustration, that she saved 40 % of residual income, close to what the control group reports: R$ 60 a month. An effect of 50 points, rounded from the estimate, would take her to 90 %, or R$ 135 a month, R$ 75 more. Over twelve months that would be R$ 900 more, the equivalent of three monthly stipends.

The pooled estimate in reais, R$ 71.29, is not exactly this product, because each student has their own residual income and the regression in reais weights students differently. But the order of magnitude is the same: the reported effect is about a quarter of the stipend each month.

6.2.3 The first version's ANOVA

The first version compared the four groups with a **one-way ANOVA**, which tests whether the four October means are equal. ANOVA splits the total variation in saving into two parts: the variation *between* groups, which is the sum, over groups, of the number of students times the squared distance between the group mean and the overall mean; and the variation *within* groups, which is the spread of students around their own group's mean. The F statistic compares the two, each divided by its degrees of freedom. If the means were equal, F would be close to 1.

With the microdata, $F(3, 199) = 61.75$, with a p-value of practically zero. The share of total variation that lies between groups, η^2 (eta squared), is 0.48: almost half of everything that makes a student report more or less saving in October is explained by the group they landed in.

The first version reported $F = 75.43$, with a between-group sum of squares of 88,541.3 and a within-group sum of squares of 77,864.5. The data give 106,150.8 and 114,029.4. The v1 between-group sum cannot be right, and one does not even need to open the spreadsheet to see it: it depends only on the means and the group sizes, and the first version's own table of means and sizes implies 106,151, exactly what the data give. The v1 means match the data; the error is in the calculation made from them. The verdict does not change, because both F statistics are enormous, and Appendix C records the discrepancy in the audit table (Table C.1).

Classical ANOVA assumes that the four groups have the same variance and that saving is normally distributed within each group. Levene's test does not reject equal variances (p of 0.56). Even so, I checked with two alternatives that drop these assumptions. The ANOVA of Welch (1951), which allows different variances, gives $F = 53.0$ (with 109.7 degrees of freedom in the denominator). The test of Kruskal and Wallis (1952), which uses only the ranking of the values and not their distribution, gives $H = 82.7$. All three agree.

6.3 How the effect built up

Table 6.2 uses October only. The event study of Equation (5.3) uses all four time points and answers a different question: in which month the effect appeared, and whether it kept growing. Figure 6.5 shows the coefficients b_{gk}, the difference between each treated group and the control group in each month, net of the difference that already existed at baseline.

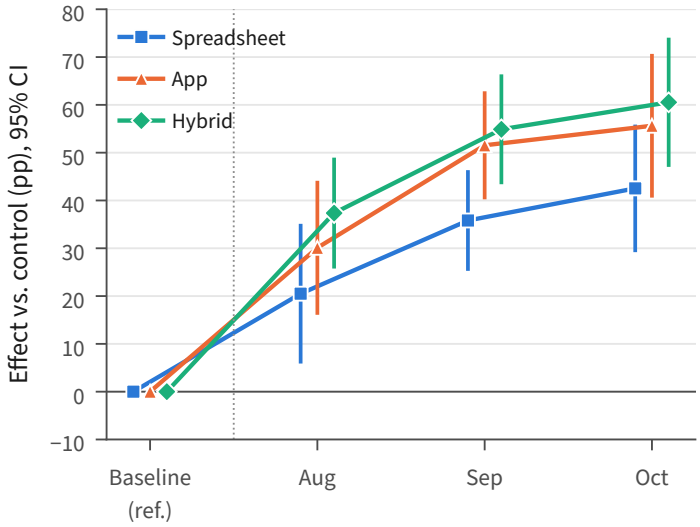


Figure 6.5. Event study (Equation (5.3)): effect of each tool relative to the control group, in percentage points of initial residual income, by month, with baseline as the reference (zero by construction). Student and month fixed effects; 95 % confidence intervals with errors clustered by student. Blue: spreadsheet; orange: app; green: hybrid.

Source: study data, estimated (`code/04_panel.py` and `13_figures.py`).

The effect is already large in August: 20.5 points for the spreadsheet, 30.1 for the app and 37.4 for the hybrid. In September it rises to 35.8, 51.5 and 54.9, and in October to 42.5, 55.6 and 60.5. From August to October, the effect grows by a further 22.0 points for the spreadsheet (p of 0.01), 25.5 for the app (p of 0.009) and 23.2 for the hybrid (p of 0.006). In all three groups this growth is statistically significant.

But it slows down. Pooling the three tools, 55 % of the October effect was already there in August and 90 % in September. The first month brought a little over half the effect; the second, most of what was missing; the third, a small addition. The curve seems to be approaching a plateau, but three points are not enough to say where it would settle, nor whether the effect would last after October, once the reminders stopped and the study ended. This study cannot answer that question, which is why the follow-up in Chapter 9 plans for at least a year.

The October effects from the event study are not identical to those in column (3) of Table 6.2, and the difference is instructive. The event study nets out the whole baseline difference, as in a difference in differences. ANCOVA nets out only the part of the baseline that helps predict October, and since that part is almost nil in these

data, it nets out practically nothing. The spreadsheet group started a little above control (45.89 % against 41.27 %), so its effect in the event study (42.5) is below its ANCOVA effect (46.9). The hybrid group started below (37.55 %), and its effect is above (60.5 against 55.7). Both estimates are valid in an experiment; ANCOVA is the more precise one when the initial outcome predicts the final one poorly (McKenzie, 2012).

The pooled version of the event study, with a single coefficient for any tool in any month from August to October, gives 43.1 pp (95 % CI from 35.9 to 50.3). It is smaller than the 52.0 pp from ANCOVA because it is the average of the three months, and August, with the effect only halfway there, pulls the average down.

That the effect showed up so quickly deserves a comment. In the model of Chapter 4, the attention channel depends on the student recording expenses for some time before seeing where the money leaks. A jump of half the effect in the very first month is consistent with that, but also with any explanation that acts right away, such as the salience of the goal or a change in how students answer the form. The dynamics alone do not separate these stories.

6.4 Does the result hold up?

An effect this size calls for checks. Each of the sections below attacks a different assumption behind the main estimate: that the standard error is right, that the chosen specification is not special, that dropouts do not matter, and that the mean summarizes the distribution well.

6.4.1 Redone random draws

The HC2 standard error relies on approximations that hold in large samples. **Randomization inference** (Section 3.7) does not. The idea is to ask: if the tools had no effect at all, on any student, how large would the difference between treated and control be just because of the random draw?

> **Analogy: the shuffled name tags**
>
> Imagine that each student wears a name tag with the name of their group, and that each one's October saving is written on a slip of paper they hold. If the tools made no difference at all, the slips would be the same whatever the tag. So I collect the tags, shuffle them within each grade and hand them out again, without touching the slips. For each shuffle, I compute the difference between those who got a tool tag and those who got a control tag. After many shuffles, I know what size of difference pure chance produces.

I did this 20,000 times. Figure 6.6 shows the result. The differences produced by chance cluster around zero, and the largest of them, across all redone draws, is

21.6 points. The observed difference, without controls, is 52.5 points, more than twice the largest simulated one. None of the 20,000 redone draws came close. The randomization *p*-value is therefore the smallest the test can give, one divided by the number of redone draws plus one: 0.00005. The same holds for each tool separately.

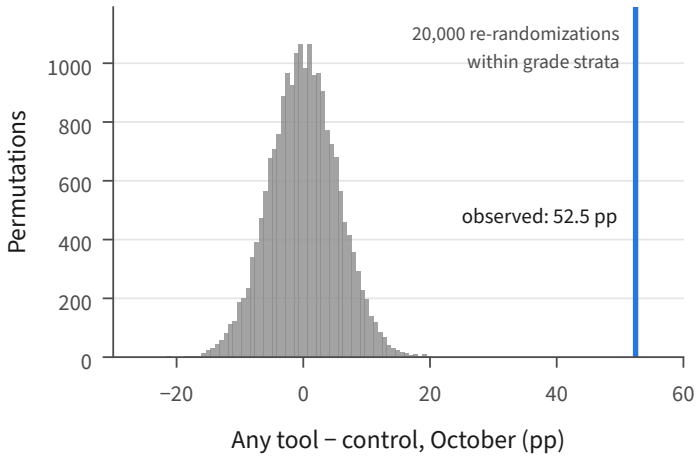


Figure 6.6. Randomization inference. Histogram of the difference between the October mean for any tool and the control mean in 20,000 redone draws of the groups, run on the computer within the grade strata, assuming the tools had no effect. The blue line is the observed difference (52.5 points, without controls).

Source: study data, estimated (`code/07_robustness.py` and `13_figures.py`).

6.4.2 Other ways of doing the calculation

Every analysis involves choices: how to measure the outcome, which controls to include, which estimator to use, which students to keep. A result that appears only with one particular combination of those choices is suspect. The **specification curve** (Simonsohn et al., 2020) estimates the effect under every reasonable combination and shows the whole set.

I used six versions of the outcome (share of residual income, capped at 100 %, winsorized at the 5th and 95th percentiles, in reais, as a percentile rank, and as a share of the stipend), three sets of controls (none, initial saving only, full), two estimators (OLS and quantile regression at the median) and three samples (completers, completers without extreme values, and everyone assigned with worst-case values for dropouts). That makes 108 specifications. To make them comparable, all are expressed in control standard deviations.

Figure 6.7 shows that the effect ranges from 1.2 to 3.8 standard deviations, with a median of 2.0. In 100 % of the specifications the effect is significant at 5 %, and the largest *p*-value among all of them is 8.6×10^{-15}. The smallest estimates combine the sample with worst-case values for dropouts with the outcome in reais or as a share of the stipend. These two versions of the outcome always give the same

estimate in standard deviations, because one is the other divided by three, and so these specifications appear in pairs in the figure. No combination of choices comes close to making the effect disappear.

6.4.3 Attrition

Five students dropped out, and I do not have their October saving. If those who left were different in a way that depended on group, the comparison among completers could be biased. Section 5.7 showed no sign of this, but it is better to know how much the estimate could change in the worst case.

Lee bounds (Lee, 2009) answer this with a weak assumption: that receiving the tool can change dropping out in one direction only. Either it makes some students stay who would have left, or it makes some leave who would have stayed, but not both at once. The group with fewer dropouts, here the treated, is trimmed until both have the same share of completers. Trimming the highest values gives the lower bound of the effect; trimming the lowest gives the upper bound. I had to discard 2.0 % of the treated group, and the bounds are [51.4, 53.8] points. With sampling uncertainty, computed by resampling, the interval runs from 42.5 to 62.0.

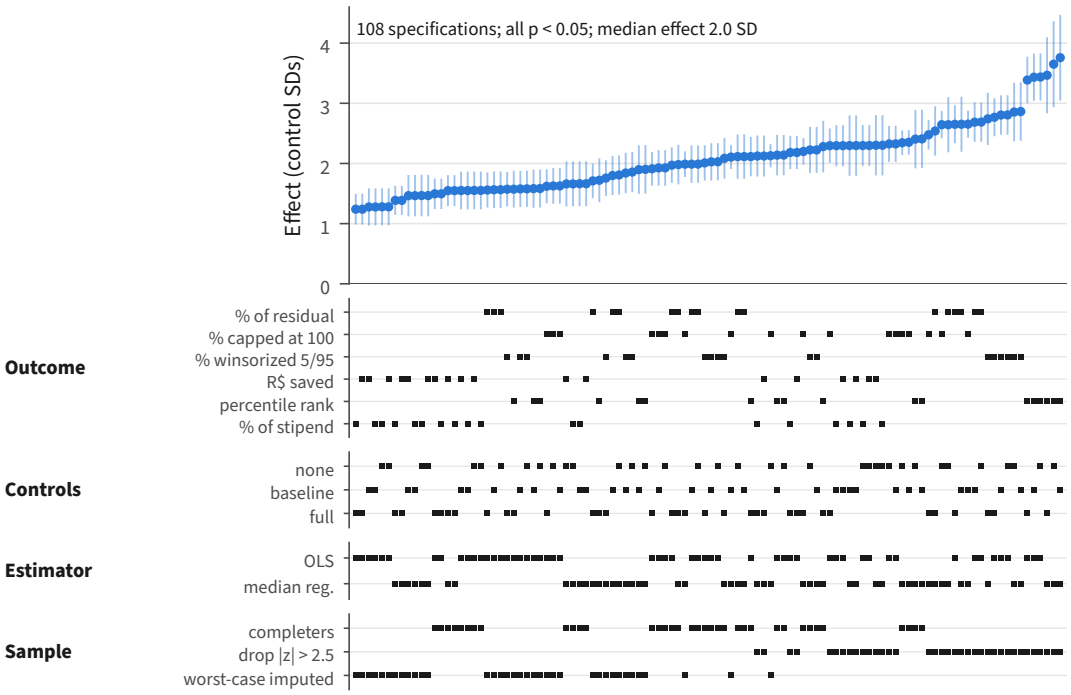


Figure 6.7. Specification curve for the effect of any tool on October saving. Top: estimate and 95 % confidence interval for each of the 108 specifications, in control standard deviations, in increasing order. Bottom: the choices behind each estimate (a dot marks the option used): outcome, controls, estimator and sample.

Source: study data, estimated (`code/07_robustness.py` and `13_figures.py`).

Lee bounds still assume some regularity. **Worst-case** imputation assumes nothing: I give zero saving to every treated student who dropped out, as if they had not put aside a single real, and the highest saving observed in the control group to every control dropout. This is the scenario least favorable to the effect that the data allow. Even so, the estimated effect is 47.5 points, with the lower limit of the 95 % interval at 38.5. With only five dropouts, there is no way for attrition to explain the result.

6.4.4 Across the whole distribution

Lastly, the mean can summarize a distribution badly. The average effect would be the same if the tools had lifted a few students a lot and left the rest where they were, or if they had lifted everyone a little. **Quantile effects** tell these cases apart: they compare the 10th percentile of the treated with the 10th percentile of the control group, then the 20th with the 20th, and so on.

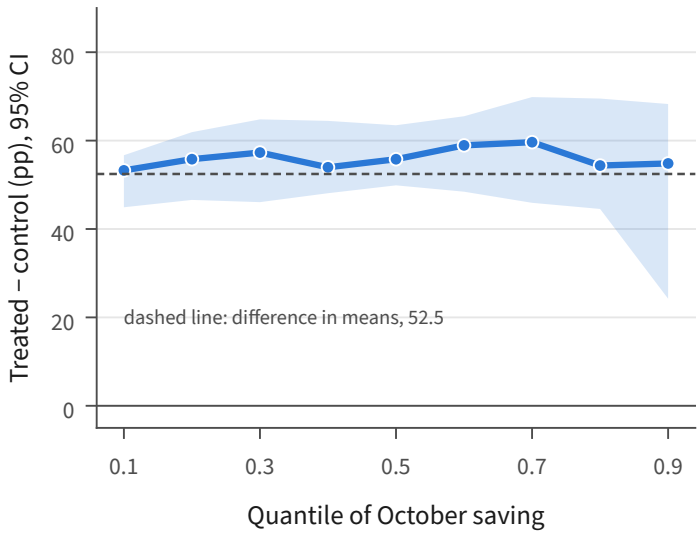


Figure 6.8. Quantile effects: difference between each quantile of October saving among the treated and the same quantile in the control group, from the 10th to the 90th percentile, in percentage points, with a 95 % resampling confidence interval (band). The dashed line is the difference in means.

Source: study data, estimated (`code/07_robustness.py` and `13_figures.py`).

Figure 6.8 shows nearly constant effects, between 53.3 and 59.6 points from the 10th to the 90th percentile: 53.3 at the 10th and 54.8 at the 90th. The whole distribution shifted up, by a similar distance at every point. The interval widens at the top, where there are few students, but even there the lower limit stays well above zero.

Two measures summarize the overlap between groups. The **probability of superiority** is the chance that a randomly picked treated student reported more saving than a randomly picked control student: 92 %. If the tools made no difference, it would be 50 %. And only 6 % of control students reported saving more than the treated median.

6.4.5 What the checks do not check

All the checks in this section have one thing in common: they use the same number reported by the student. They show that the difference between groups is not a fluke of the random draw, does not depend on a particular choice of specification, is not produced by dropouts, and does not come from a few extreme students. They do not show, and could not show, that this number measures saving well. If all the treated had reported, say, twenty points more than they actually saved, every check would come out the same, because a shift common to the whole treated group behaves, statistically, exactly like a true effect. Randomization inference, for example, tests the hypothesis that the assignment changed nothing; it cannot tell whether the assignment changed saving or changed the report. That is the reason Chapter 8 exists.

6.5 What these results establish

The results of this chapter establish one thing with considerable confidence. Being assigned to receive a budgeting tool, with the training that came with it and, in the groups with the app, the reminders, raised the saving the scholars *reported* by about 52.0 percentage points of residual income, about R$ 71 a month. This effect is causal in the sense of Chapter 3: random assignment guarantees the groups were comparable, and the result survives redone random draws, 108 specifications and worst-case dropouts, and it appears across the whole distribution. Internal validity, for what was measured, is good.

There are three things these results do not establish.

The first is that the students saved more *money*. The outcome is a number the student wrote on a form, unchecked. If the treated, trained to record expenses, with a goal on the screen and knowing the study was about saving, were more inclined to report a high number, the figures in this chapter would look exactly as they do. Chapter 8 examines this possibility with what the data themselves allow: the persistence of reports from one month to the next, values above 100 %, the rounding of amounts, and what a skeptical reader should conclude.

The second is *which* tool works best and *why*. The three have similar effects, and the differences among them are small relative to the uncertainty. Chapter 7 asks whether they can be ranked, whether they can be considered equivalent, and whether those who used the tool more saved more, which is the most direct test of the mechanism in Chapter 4.

The third is whether the effect lasts. Three months show an effect that grows and then slows, but they say nothing about the fourth month or the following year.

The result of this chapter shows, with little room for statistical doubt, that the treated students' reports changed. Chapters 7 and 8 deal with what changed behind those reports.

CHAPTER 7

Which tool, for whom and why

Chapter 6 showed that the students assigned to receive any of the three tools reported saving, in October, 52.0 pp more of their residual income than the control group. That number answers the study's first question, whether receiving a tool changes anything. It does not say which tool works best, nor why it works, nor for whom. This chapter takes up those three questions.

They are not idle questions. A program like the PIC or Pé-de-Meia that wanted to adopt the intervention would have to choose a tool, and the cheapest choice is not the same as the most expensive one. It would also need to know what to monitor: if the effect runs through use, the number of logs in the app would be a good gauge of the program; if it does not, it would be a misleading one. And the question of why is the one that links the data to the model of Chapter 4, which turned the intuitive explanation (the tool makes spending visible) into predictions the data can contradict.

The first version of the study (Ribeiro, 2025a) already answered all three questions, and with confidence: the hybrid was the best, with a "synergy" between the tools; each additional weekly log raised saving by about 12 points; girls, students aged 15 to 17, and rural students with the spreadsheet benefited most. In this chapter I redo each of these calculations with the microdata and the public code. I will give away the result now: none of these claims holds up with statistical support, and the most important of them, the one about use, is contradicted by the data.

One warning about method applies to the whole chapter. No analysis plan was published before data collection. What the first version describes in its methods section (the outcome, the regression on use, and the comparisons by gender, age and origin) is the closest thing to a plan, and I repeat all of those analyses. The rest (the equivalence tests, the model, the predictions about scarcity and room to improve) was added after seeing the data and should be read as exploratory (Section 5.6).

7.1 The ranking of the tools

The October averages put the tools in a clear order. The spreadsheet group reported saving 91.40 % of residual income, the app group 97.82 % and the hybrid group 101.07 %. This is the same order as in the first version, which is no surprise, since the means match to the second decimal place (Table C.1). The question is whether this order is a fact about the tools or a fluke of the random draw.

To answer it, I compare the four groups two at a time. With four groups there are six pairs: three of each tool against the control group and three of the tools against each other. For each pair, I use Welch's *t*-test, which compares two means without assuming the two groups have the same variance. That matters here, because the October standard deviations range from 20.3 points in the spreadsheet group to 26.4 in control.

Six tests bring a problem that one test does not. If each has a 5 % chance of raising a false alarm, the chance that at least one of them does is well above 5 %. It is like buying six raffle tickets instead of one. The correction of Holm (1979) (Section 3.6) handles this in a simple way: it sorts the *p*-values from smallest to largest and requires the first to pass a threshold six times stricter, the second five times, and so on, stopping at the first one that fails. The result is a p_{Holm} value for each pair, which can be read against the usual 5 %.

The last column of Table 7.1 gives the size of each difference in standard deviations, Hedges' *g*, with a confidence interval computed by resampling the students (the bootstrap). It is there for comparison with other studies, which measure saving on different scales.

Table 7.1. Pairwise comparisons of groups on October saving, in percentage points of initial residual income. Diff.: difference between the means of the first and the second group. Welch *t*-tests, which do not assume equal variances. p_{Holm}: *p*-value adjusted for the six comparisons by the method of Holm (1979). Hedges' *g*: the difference in pooled standard deviations, with a 95 % percentile bootstrap CI.

Contrast	Diff. (pp)	Welch *t*	*p*	p_{Holm}	Hedges' *g* [95% CI]
Spreadsheet vs Control	47.15	10.10	<0.001	<0.001	2.00 [1.45, 2.77]
App vs Control	53.57	10.38	<0.001	<0.001	2.05 [1.46, 2.89]
Hybrid vs Control	56.82	11.42	<0.001	<0.001	2.27 [1.71, 3.08]
App vs Spreadsheet	6.42	1.41	0.161	0.322	0.28 [−0.09, 0.72]
Hybrid vs Spreadsheet	9.67	2.23	0.028	0.083	0.44 [0.06, 0.84]
Hybrid vs App	3.25	0.67	0.505	0.505	0.13 [−0.26, 0.51]

Source: study data, estimated (`code/03_main_effects.py`).

The first three rows confirm the result of Chapter 6. Each tool beats the control group by about two standard deviations (2.00, 2.05 and 2.27), and no adjustment for multiple comparisons comes close to changing that: all three *p*-values are below one in a thousand, with or without Holm.

The last three rows tell another story. The app is 6.4 pp above the spreadsheet ($p = 0.16$; with Holm, 0.32). The hybrid is 3.2 pp above the app ($p = 0.50$). Only the largest difference, the hybrid over the spreadsheet, of 9.7 pp, clears the 5 % line when looked at on its own ($p = 0.03$). After Holm's correction it becomes $p = 0.08$ and no longer clears it.

One detail in the table can be confusing. The interval for Hedges' *g* for hybrid against spreadsheet runs from 0.06 to 0.84 and does not include zero, which seems to contradict a p_{Holm} above 5 %. It does not. The interval belongs to a single test, with no adjustment for the six pairs, and it agrees with the unadjusted *p*, which is below 5 %.

Someone who chooses to look at this comparison after seeing that it is the largest is doing exactly what Holm's correction exists to prevent.

The first version's "synergy"

The first version read the hybrid's first place as a sign of a "synergistic effect between the tools" (Ribeiro, 2025a). The word needs care, because it can mean two things.

In the strong sense, **synergy** means the whole is greater than the sum of the parts: the effect of receiving both tools together would be larger than the effect of the spreadsheet plus the effect of the app. The data are far from this. The hybrid's effect over the control group (56.8 pp) is much smaller than the sum of the spreadsheet's effect (47.1 pp) and the app's (53.6 pp). Nor should this result be read as proof that the tools get in each other's way. The outcome has a practical ceiling (saving more than 100 % of residual income requires cutting essential spending or having other income, Section 8.3), and two effects of this size could hardly add up on top of the control average of 44.25 %.

In the weak sense, synergy just means the combination is better than the best single tool. That is the test in the last row of Table 7.1: hybrid against app, 3.2 pp, with $p = 0.50$. A difference of that size easily shows up by pure chance between two groups of about 50 students. Synergy may exist, but this experiment provides no evidence of it.

7.2 Equal, or just indistinguishable?

The previous section concluded that the tools cannot be told apart. It is tempting to read this as "the tools are the same", and that reading is wrong. Not finding a difference is not the same as showing there is none. A metal detector that stays silent on the beach may mean there are no coins or that the detector is too weak to find them. To claim that the tools are similar, one needs a test that has similarity as a possible conclusion.

That test exists and is called TOST, short for "two one-sided tests" (Lakens, 2017; Schuirmann, 1987) (Section 3.8). The idea is to reverse the burden of proof. First, I choose an **equivalence margin** m: the largest difference I would consider negligible in practice. Then I run two tests. The first asks whether the true difference is above $-m$; the second, whether it is below $+m$. I declare the tools equivalent only if both tests reject, each at 5 %. In practice, this is the same as requiring the 90 % confidence interval of the difference to fall entirely within the band from $-m$ to $+m$.

There is no obvious margin. I use three: 5, 10 and 15 points. To get a sense of what they mean in money: 10 points is a tenth of residual income, and average residual income is R$ 143.68, so the 10-point margin is worth about two cafeteria snacks a

month. Someone who thinks two snacks a month make no difference would accept that margin; someone who thinks they do would stick with 5.

Table 7.2. Are the tools equivalent? Difference between the tools in October saving (percentage points of initial residual income), with Welch 90 % CI, and the result of the two one-sided tests (TOST) for margins of 5, 10 and 15 points. "Yes": both tests reject at 5 %, and the pair is equivalent within the margin. In parentheses, the TOST *p*-value.

Difference	Estimate [90% CI]	±5 pp	±10 pp	±15 pp
App – Spreadsheet	6.42 [–1.13, 13.97]	no (0.62)	no (0.22)	yes (0.03)
Hybrid – Spreadsheet	9.67 [2.48, 16.86]	no (0.86)	no (0.47)	no (0.11)
Hybrid – App	3.25 [–4.81, 11.31]	no (0.36)	no (0.08)	yes (0.01)

Source: study data, estimated (`code/12_equivalence_bayes_design.py`).

Table 7.2 and Figure 7.1 show the result. With the 15-point margin, two of the three pairs pass: app and spreadsheet ($p = 0.03$) and hybrid and app ($p = 0.01$). The third, hybrid and spreadsheet, does not pass even with this wide margin, because its 90 % interval reaches 16.9 points. With the 10-point margin, no pair passes; the closest is hybrid and app, with $p = 0.08$. With 5 points, none comes close. The smallest margin at which all three pairs would be declared equivalent at once is 16.9 points.

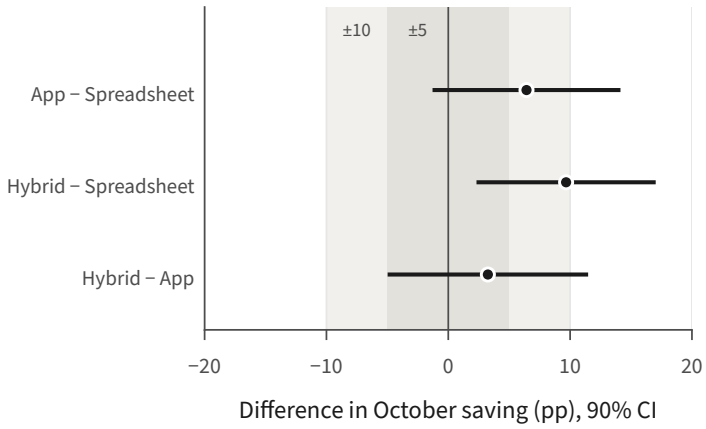


Figure 7.1. Differences between the tools in October saving, in percentage points of initial residual income. The dot is the estimate and the bar the Welch 90 % CI. The shaded bands mark margins of ±5 (darker) and ±10 points. A pair is equivalent within a margin only if its whole bar fits inside the band.

Source: study data, estimated (`code/13_figures.py`).

Putting the two sections together, the conclusion is uncomfortable. The tools cannot be told apart, and they are not demonstrably equivalent either. They might differ by a few points or by more than ten, and this experiment cannot say which is true.

Why the experiment could not rank the tools

This stalemate is not bad luck. It was written into the size of the experiment before any student was assigned. The **power** of a test is the probability that it detects a difference when one really exists (Section 3.5). It depends on three things: the size of the true difference, the number of students in each group, and how much students vary within each group.

For two means with n students per group and standard deviation σ within groups, the standard error of the difference is $\sigma\sqrt{2/n}$. The smallest difference that a two-sided test at 5 % detects with 80 % power is approximately

$$\text{MDE} \approx (z_{0.975} + z_{0.80})\,\sigma\sqrt{\frac{2}{n}} \approx 2.8\,\sigma\sqrt{\frac{2}{n}}, \tag{7.1}$$

where MDE is the minimum detectable effect and the two z values are the points of the normal distribution that correspond to the test level and the desired power (1.96 and 0.84). Here there are about 51 students per group, and the October standard deviation within groups is 24.0 points. The exact calculation, with the t distribution, gives a minimum detectable effect of 13.5 points between two tools.

Figure 7.2 shows the whole curve. The three observed differences between the tools are all below the minimum detectable effect. If the true difference between hybrid and app were exactly the observed one, 3.2 points, the experiment would detect it only 10 % of the time. For app against spreadsheet, power would be 27 %; for hybrid against spreadsheet, 52 %, practically a coin toss. Detecting a difference the size of the one observed between hybrid and app with 80 % power would require about 855 students per group, against the study's 52.

The figure also shows something that comes back in Chapter 8. The orange line marks the typical effect of financial education interventions on behavior, about 0.1 standard deviation (Kaiser et al., 2022), which here corresponds to 2.6 points. If the tools' true effect were that size, not even the comparison of all treated students against the control group would detect it: power would be 10 %. The first version reports no power calculation, and the experiment ended up sized to detect only effects much larger than those in the literature. It found one, and that is one of the reasons to look at the result with care.

Another way to read the same estimates is to ask how likely it is that one tool is better than another. With a flat prior (Section 3.9), that is, with no prior opinion about which is better, uncertainty about the true difference is described by a normal distribution centered on the estimate, with the standard error as its width. The probability that the hybrid is better than the app is then 75 %: closer to a coin toss than to certainty. The probability that the hybrid is better than the spreadsheet is 99 %, a number that looks strong. But it comes from the comparison I picked because it was the largest of three, it answers whether the difference is positive, and it says

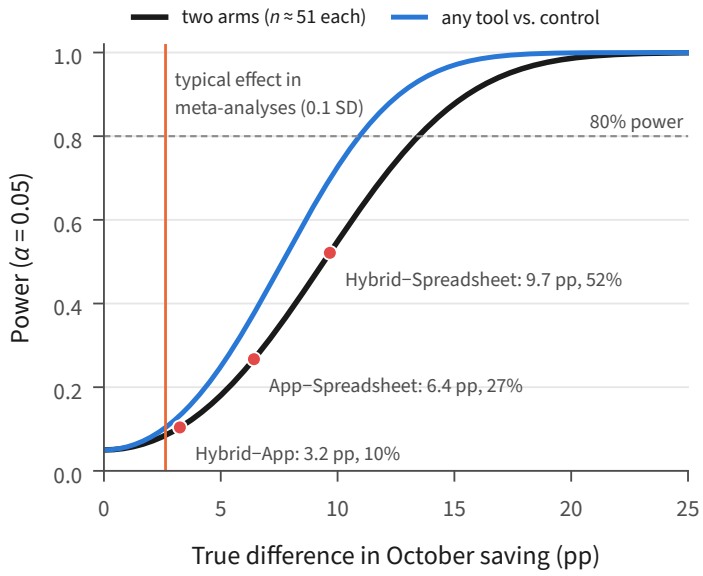


Figure 7.2. Power of a two-sided test at 5 % as a function of the true difference in October saving, with the observed within-group standard deviation. Black line: comparison between two groups of about 51 students (two tools). Blue line: any tool against the control group. Red dots: the three observed differences between the tools. Vertical orange line: the typical effect of financial education interventions in meta-analyses, 0.1 control standard deviation (2.6 points).

Source: study data, estimated (`code/09_power.py`; figure in `13_figures.py`).

nothing about how large it is. The 90 % interval for the same difference runs from 2.5 to 16.9 points, that is, from almost nothing to a large advantage.

The conclusion of this section is about the design, not about the tools. With four groups of 52, the experiment had power to spare for comparing any tool with the control group and no useful power for comparing the tools with each other. It could never have ranked them, unless they differed by more than 13.5 points. For a public program, the practical consequence is the one already stated in the revised paper: the data give no reason to prefer one tool over another, and a program or a replication can start with the simplest.

7.3 Does using more mean saving more?

This is the question that interested me most when I planned the study. If the tools work because they make spending visible, those who used them more should have saved more. In the model of Chapter 4, the student's attention to small expenses grows with intensity of use, u_{it}, and the tool's effect (Equation (4.8)) has a part proportional to it. That gives Prediction P2: among the treated, saving grows with use, and those who never use the tool gain only through the salience of the goal, so they should be closer to control than active users.

In statistics, this relation between "how much treatment" and "how much result" is called **dose-response**, by analogy with medicine: if one pill helps, two should help more, at least up to a point. Here the dose is the number of logs per week, counted from the screenshots or summaries the treated students sent with the monthly form (Section 5.4).

What the first version reported

The first version estimated a simple regression and reported the line

$$\begin{aligned} &\text{"\% change in saving"} \\ &\quad = 10.5 + 12.1 \times \text{records/week}, \end{aligned} \tag{7.2}$$

with a significant coefficient ($p < 0.01$; v1 values). It read this as "each additional weekly record is associated with an increase of roughly 12 percentage points in the saving rate", and concluded that the tracking routine was the central mechanism of the change (Ribeiro, 2025a). The v1 figure that went with this regression carried the label "illustrative data".

The problem is that the first version does not say what the "% change" is. It could be the change in percentage points of the saving rate, from baseline to October: a student who went from 40 % to 90 % had a change of 50. Or it could be the **relative change**, the change divided by the starting value: the same student had a relative change of 125 %, because 50 is 125 % of 40. The two readings give very different regressions, so I estimate both.

What the microdata show

Table 7.3 brings together five models. Model (A) is the reading in points, from baseline to October. Model (B) reproduces the first version's equation as it was specified, with the change from August to October and group indicators. Model (C) looks directly at October saving and adds initial saving, the covariates and grade fixed effects. Model (D) is the relative reading, from August to October, which the raw file also contains. Model (E) is the same as (D) without the top 5 % of values.

In the reading in points, the slope is –1.33 per weekly log, with a 95 % interval from –3.19 to 0.54. The sign is the opposite of what was expected, but the most important thing is the upper limit of the interval. The data rule out any slope above about 0.5 point per weekly log, a small fraction of the first version's 12.1. Measuring the change from August, with the group indicators (model B), or adding initial saving and the covariates (model C) does not change the picture: the slope stays negative, and the upper limit of the interval stays far below 12.1.

In the relative reading (model D), the slope is positive, 33.57, but with a huge interval, from –30.21 to 97.36, and $p = 0.30$. This model's intercept is 13.0, close to the

Table 7.3. Reported saving and frequency of tool use among the 153 treated students. $\hat{\beta}$: slope, in units of the outcome per weekly log. The first row is what v1 reported, with an outcome it does not define. (A) and (D) are the two possible readings of the v1 "% change": in percentage points and relative. (B) is the v1 specification: change from August to October on logs and group indicators. (C) adds initial saving, the covariates and grade fixed effects. (D) and (E) exclude students with zero saving in August, for whom the relative change does not exist; (E) also excludes the top 5 % of values. HC2 standard errors.

Model	Outcome	$\hat{\beta}$ (logs/week)	95% CI	p	n
v1 (reported)	"% variation" (undefined)	12.10	–	< 0.01	–
(A)	Change baseline→Oct (pp)	−1.33	[−3.19, 0.54]	0.16	153
(B)	Change Aug→Oct (pp)	−0.79	[−3.11, 1.54]	0.51	153
(C)	October saving (%)	−0.87	[−2.47, 0.73]	0.29	153
(D)	Relative change Aug→Oct (%)	33.57	[−30.21, 97.36]	0.30	152
(E)	Same, top 5% trimmed	−3.25	[−6.47, −0.03]	0.05	144

Source: study data, estimated; first row, v1 (`code/05_use_and_saving.py`).

first version's 10.5, which is consistent with v1 having used this measure, although the slope is far from 12.1. The relative change has a serious flaw: it divides by August saving, which for some students is close to zero, and a hypothetical student who went from R$ 1 to R$ 50 would show up with a change of 4,900 %. These few students dominate the regression. Without the top 5 % of values (model E), the slope turns negative, −3.25, with $p = 0.05$. In none of the readings is there a significant positive slope.

Figure 7.3 shows the data behind these regressions. In the left panel, each dot is a treated student; the black line is the fit to the data and the dashed red one the line reported in v1. The cloud of dots does not rise from left to right. The Spearman rank correlation between use and the change in saving, which does not depend on a straight line and is not pulled by extreme values, is −0.15 ($p = 0.06$).

The students who never logged

The right panel of Figure 7.3 holds the comparison that, to me, is the most revealing in the study. Of the 153 treated students, 25 (16 %) have no logs at all. They received the tool, were invited to the same training as the others and, as far as the file shows, never used it. In October they reported saving 96.6 % of residual income. The students who did log reported 96.7 %. The difference is practically nil ($p = 0.97$). Compared with the control group, those who never logged are 52.3 points above, the same size as the average effect for all treated students.

Among those who logged, I split the students into usage terciles. The least active tercile logged on average 0.7 times a week; the middle one, 1.8; the most active, 4.8. Their October saving was 101.9 %, 96.2 % and 92.1 %. The most active tercile reported *less* than the least active. By the first version's line, the most active tercile should have reported $12.1 \times (4.8 - 0.7)$ points more than the least active, several dozen points.

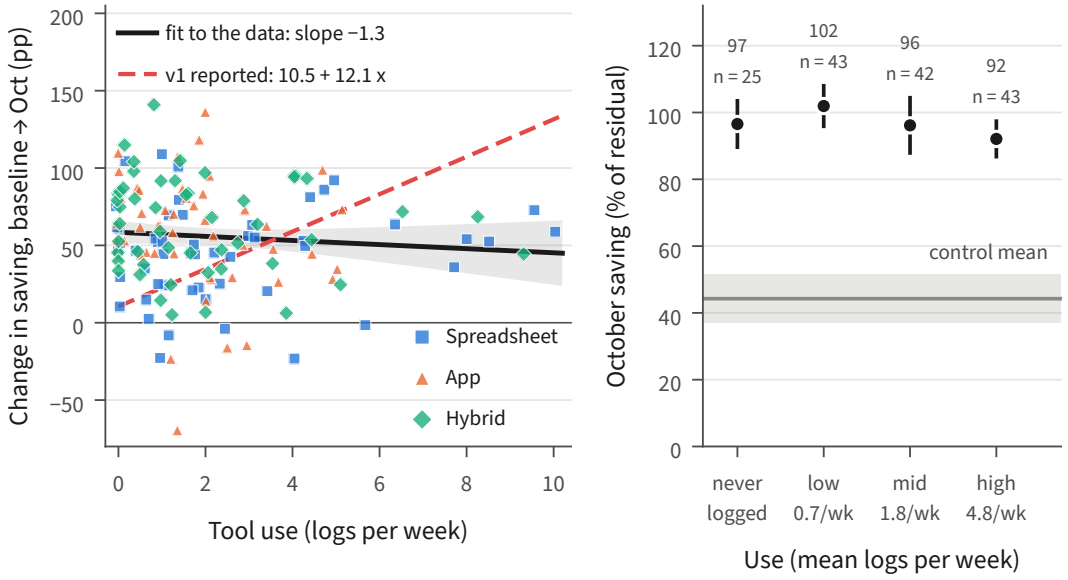


Figure 7.3. Tool use and reported saving among the 153 treated students. Left: change in saving from baseline to October (percentage points of residual income) against logs per week, by group (spreadsheet in blue, app in orange, hybrid in green), with the OLS fit to the data (black line, 95 % band) and the line reported in v1 (dashed red). Right: October saving of those who never logged and of the usage terciles among those who did, with 95 % CI; the gray band is the control mean with its 95 % CI.

Source: study data, estimated; v1 line (`code/05_use_and_saving.py`; figure in `13_figures.py`).

A description of use helps put these numbers in perspective. The average number of logs per week among the treated was 2.0, and the median 1.4: half the students logged fewer than two times a week, far from the "every expense" the training asked for. Use was similar in the three groups (2.5 logs per week for the spreadsheet, 1.7 for the app and 1.9 for the hybrid; $p = 0.33$ in a Kruskal and Wallis (1952) test). The app, with its weekly reminders, did not lead to more records than the spreadsheet.

Why this is not causal, and what it still says

None of these comparisons is an experiment. The random draw decided who got the tool; how much each student used it was their own decision. That opens the door to **selection**: students who used it more may differ from those who used it less in ways that also affect saving. If the students who found it hardest to save were precisely the ones who logged most, for example because they worried more about money, that would hide a real effect of use. There is also a measurement problem. Logs were counted from screenshots, and for the 25 students with zero the file does not distinguish a true zero from a missing screenshot. A measure of use with a lot of error flattens any slope.

These caveats are serious, but they have limits. To hide a slope the size the first version reported, selection would have to be enormous: the most active students would have to start with a propensity to save tens of points lower than the least active.

Model C controls for initial saving and baseline characteristics, and the slope stays negative; selection of that size would have to come from something the baseline does not capture.

And there is something these comparisons say even though they are not causal, when read within the model of Chapter 4. By Equation (4.8), the tool's effect has two parts: attention, $\kappa\, u_{it}\, m_i$, which is zero for those who do not use it ($u_{it} = 0$), and salience, which applies to all treated students. The students who never logged could gain only through salience. They gained as much as the rest. So either salience does all the work and attention none, or the effect runs through something all the treated students had in common that the model did not include: the training in the separate room, the feeling of being part of an intervention, or the expectation of reporting progress to the researcher who trained them.

The data in this chapter do not decide among these possibilities; Chapter 8 tries to go further. But they do not support the first version's claim that "engagement was the main driver of success" (Ribeiro, 2025b). And they have an immediate practical consequence: since logged use did not track reported saving, an app's usage statistics (how many students opened it, how many times they logged) are a poor way to judge whether a program of this kind works.

7.4 Who benefits?

The last question is whether the effect was larger for some students than for others. In economics this is called **heterogeneity** of the effect. It matters for two reasons. For public policy, an effect concentrated among certain students would allow the program to be targeted. For the model, two predictions of Chapter 4 are precisely about heterogeneity: the one on scarcity and the one on room to improve.

There is a well-known risk in this search. Someone who looks across many splits ends up finding some "significant" effect by pure chance, just as someone who tosses a coin twenty times has a good chance of seeing, at some point, four identical results in a row. So I fixed six splits, those of the first version plus three characteristics that were in the baseline: gender, origin (rural or urban), age (15 to 17 or not), school level (lower secondary or high school), laptop ownership, and essential spending (above or below the median). For each one, I estimate the effect of any tool within each subgroup and test whether the difference between the two subgroups, the **interaction**, is different from zero.

With six tests, I apply the false discovery rate control of Benjamini and Hochberg (1995). The idea is similar to Holm's but less strict: instead of guarding against any false alarm, it limits the share of false alarms among the discoveries declared. It is the usual adjustment when exploring splits.

Six splits, no interaction

Figure 7.4 shows the result. No interaction is significant. The smallest p-value is 0.07, for essential spending, and after the Benjamini and Hochberg adjustment all p-values are 0.37 or higher. The other five are further away: gender ($p = 0.84$), origin (0.62), age (0.70), school level (0.12) and laptop (0.69). In every subgroup the effect is close to the estimate for all students, the blue band in the figure.

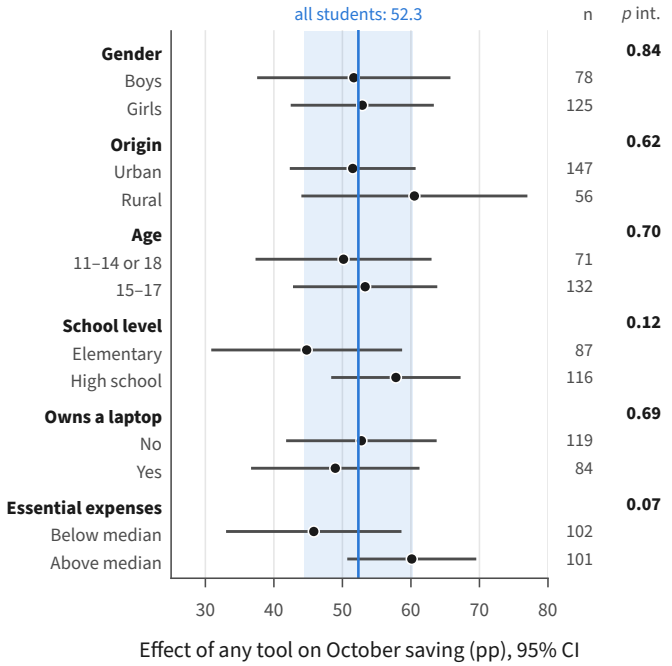


Figure 7.4. Effect of any tool on October saving within subgroups (percentage points of initial residual income), controlling for initial saving and grade fixed effects, 95 % CI. The blue line and band are the estimate for all students and its CI. In the right-hand columns, the p-value of the interaction (on each split's row, without adjustment for multiple tests) and the number of students in each subgroup.

Source: study data, estimated (`code/06_heterogeneity.py`; figure in `13_figures.py`).

The first version's three claims

The first version made three claims about subgroups. I check each one.

Gender. v1 reported that, in October, girls in the treated groups saved 4.1 points more than boys (v1). In the microdata, the raw difference between treated girls and boys is 3.8 points; with controls for group, initial saving and grade, it is 4.7 points, with a 95 % interval from –3.8 to 13.2 and $p = 0.28$. The direction holds, without statistical support. There are two further problems. First, this comparison is not about the tool's effect: girls might save more in the control group as well, and the interaction, which is the right test, has $p = 0.84$. Second, v1 spoke of a larger "increase" in girls' saving, but the gain from baseline to October for treated girls minus that for treated boys is –2.6 points, that is, a difference in the opposite direction.

Age. v1 reported the largest impact among students aged 15 to 17 and suggested this age range might be a "sweet spot" for financial education. The effect in this range is 53.3 points, against 50.2 at other ages ($p = 0.70$). The difference goes in the reported direction and is small next to the uncertainty.

Origin and tool. v1 reported that urban students did better with the app and rural students with the spreadsheet. The data point in the direction of half the story, without statistical support. With the spreadsheet, the effect is 56.9 points for rural students and 44.2 for urban ones ($p = 0.25$). With the app, the two are practically equal (54.7 and 54.6; $p = 0.99$). The exact v1 question is whether the app's advantage over the spreadsheet is larger among urban students. That difference in differences is 12.5 points, with $p = 0.26$.

The power calculation explains why none of this could have worked. There are only 11 rural students in the spreadsheet group. An interaction between origin and spreadsheet would have to exceed about 31 points to be detected with 80 % power, a number larger than half the main effect. Interaction tests need much larger samples than tests of average effects, because they compare differences between small subgroups.

Scarcity: the gradient the outcome manufactures

Prediction P3 is the most interesting from a theoretical point of view, because the model does not give the sign of the result. If scarcity narrows attention (Mani et al., 2013; Mullainathan and Shafir, 2013), students with tighter budgets pay less attention to small expenses at baseline and have more to gain from a tool. If, on the other hand, leaks m_i grow with the budget (those with more left over spend more on unplanned things), there is more to gain when residual income is large. The sign of the gradient would say which force dominates.

Measured as a share of residual income, which is the study's outcome, the effect falls as residual income rises: –7.3 points per standard deviation of residual income (R$ 29), with an interval from –14.2 to –0.3 and $p = 0.04$. Read carelessly, this seems to support the scarcity hypothesis: those with less gained more.

The problem is in the denominator. The outcome divides the reais saved by each student's residual income (Equation (5.1)). So the same gain in reais looks larger, as a percentage, for someone with a small budget.

Example 7.1 The same gain in two budgets

Think of two students who both start saving the same R$ 70 more a month. The first has residual income of R$ 114, the average of the bottom tercile; the second, R$ 174, the average of the top tercile. As a percentage, the first student's gain is 70/114 ≈ 61.4 points, and the second's 70/174 ≈ 40.2 points. The effect in reais is identical, and the effect on the study's outcome falls by 21 points from the first to the second. This is illustrative arithmetic: the R$ 70 is a round number chosen for the example, of the order of the effects in reais that appear just below, and not an estimate.

With the same yardstick, in reais, the gradient disappears. The effect rises by R$ 6.5 per standard deviation of residual income, with an interval from −5.0 to 18.0 and $p = 0.27$. Figure 7.5 shows the two panels side by side. As a percentage (panel a), the effect is 56.4, 61.1 and 39.8 points in the three terciles of residual income. In reais (panel b), it is about R$ 67, R$ 86 and R$ 70, with no trend at all. The values in Example 7.1 have the same shape as those in panel (a): the division alone produces a percentage gradient of a size similar to the one observed.

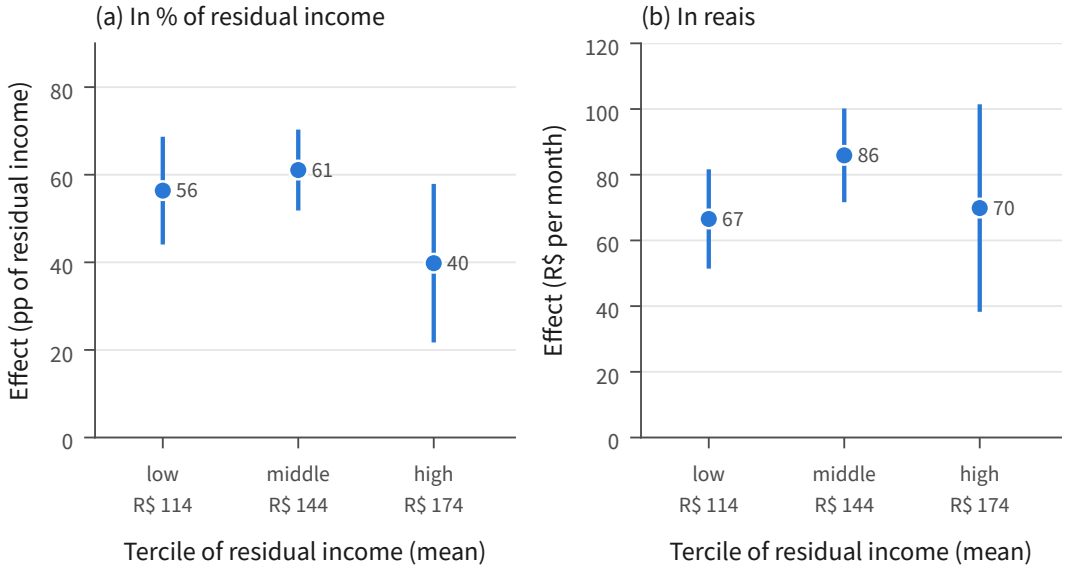


Figure 7.5. Effect of any tool on October saving by tercile of initial residual income; below each label, the tercile's average residual income. (a) In percentage points of residual income, the study's outcome. (b) In reais per month. 95 % CI. The same effect in reais looks smaller, as a percentage, for those with a larger budget.

Source: study data, estimated (`code/10_mechanisms.py`; figure in `13_figures.py`).

The lesson goes beyond this study. Whenever an outcome is a ratio, a constant effect in the numerator turns into a gradient in the denominator. To test Prediction P3, the right outcome is in reais, because the prediction is about the effect in reais. On that yardstick, the data show no gradient at all, and the interval is wide enough to contain a moderate gradient in either direction. They do not separate the scarcity story from the story about the size of the leaks.

Room to improve

Prediction P4 says that students who already saved a large share of residual income at baseline have less room to improve, so the effect should be smaller. The prediction is almost arithmetic: a student who saved 80 % can only rise to 100 % without cutting essential spending, while one who saved 20 % has 80 points of slack.

The data go in the predicted direction, without statistical support. Splitting the students into terciles of initial saving (averages of 21 %, 41 % and 62 %), the effect is

54.2, 57.7 and 43.0 points. In a regression with a continuous interaction, the effect falls by –2.3 points for every 10 extra points of initial saving, with $p = 0.34$. The top tercile's effect is the smallest, as predicted, but the difference is within what chance produces.

The five predictions against the data

Table 7.4 lists the five predictions of Chapter 4 with the outcome for each. The first, that receiving a tool increases saving, is confirmed as far as reported saving goes. The other four are not: the one on use is contradicted, the ones on scarcity and room to improve are not detected, and the one on persistence, which Chapter 8 examines in detail, fails in a way that says more about the reports than about saving.

Table 7.4. The predictions of Chapter 4 against the data. P1: Prediction P1; P2 and P2': Prediction P2; P3: Prediction P3, measured in reais; P4: Prediction P4; P5: Prediction P5. Effects in percentage points of initial residual income unless stated otherwise, with 95 % CI in brackets. ICC: intraclass correlation of each student's saving across the four months, net of group means (Section 8.2).

	Prediction	Result	Holds?
P1	Assignment raises saving	+52.0 pp [44.2, 59.8]	yes
P2	More use, more saving	slope –1.33 [–3.19, 0.54]	no
P2'	Never-loggers look like controls	97% vs. control 44%	no
P3	Effect (R$) larger for tighter budgets	6.5 R$ per SD of residual (p = 0.27)	not detected
P4	Smaller effect for high baseline savers	–2.3 pp per 10 pp of baseline (p = 0.34)	not detected
P5	Individual saving persists across months	ICC –0.05; adjacent-month r = 0.00	no

Source: study data, estimated (`code/10_mechanisms.py`).

Read all at once, the table shows a pattern. Everything that depends only on comparing averages between the randomly assigned groups works: the effect is large, shows up early, and survives every check. Everything that depends on looking inside the groups, student by student (who used more, who had less, who saved a lot, who saved one month and the next), does not show what the model predicted. This pattern can be read in two ways. In the first, the model is wrong, and the tools act through a route that runs neither through use nor through each student's budget. In the second, the individual reports are too noisy or too skewed to show any relation within the groups, and what shifts is the average of the treated. Chapter 8 examines this second reading.

CHAPTER 8

The reports under doubt

The two previous chapters arrived at a result that looks contradictory. Chapter 6 showed a large effect, one that shows up in the very first month and survives every statistical check I could think of: new random draws, 108 specifications, bounds for attrition. Chapter 7 showed that almost nothing the model predicted within the groups appears in the data. Using the tool more did not go together with saving more, the effect in reais did not vary detectably with each student's budget, and a student's saving in one month says nothing about the next month.

This chapter takes one explanation for that combination seriously. The experiment is internally valid for what it measured: the draw was random and balanced, almost nobody dropped out, and the difference between the groups is not chance. The doubt is about what it measured. The outcome is the saving the student *reported* on a form. If that number is a good measure of saving, the effect is real. If it is not, the effect may be, in part or entirely, a change in the way students report.

The model in Chapter 4 already had an equation for this (Equation (4.10)):

$$r_{it} = s_{it} + b\,T_i + \eta_{it}, \tag{8.1}$$

where r_{it} is what student i reported in month t, s_{it} is what they actually saved, T_i equals 1 for treated students and 0 for the control group, b is a shift in the reports of the treated, and η_{it} is a reporting error that can have either sign and changes from month to month. The two parts of the error do different things. The shift b moves the mean of the treated, and on its own it would produce an "effect" without anyone saving a single real more. The noise η moves no mean at all, but it scrambles the relations among the reports of the same student and between the reports and the use of the tool.

In the following sections I look at five signals about these two terms: who reported and to whom, the persistence of the reports, the reports above 100 %, what the values look like, and the size of the effect next to the literature. Then, in Section 8.6, I run the test I consider the most informative in the chapter: I manufacture data in six different worlds, where I know what is true, and see which of them looks like the real data.

8.1 Who reported, and to whom

At the end of August, September and October, each student filled in an online form with the amount saved that month (Section 5.4). The treated students also sent a screenshot or a summary from the tool, which is where the number of logs per week used in Chapter 7 comes from. There was no measure of saving that did not pass through the student. The screenshots were requested "for verification", but the first version reports no comparison between the reported saving and the amounts that appear in them (Ribeiro, 2025a).

The conditions under which students answered were very different across groups. The treated students had been trained, in a separate room, to record all their spending. The app and hybrid groups got weekly reminders to record. All the treated students knew the study was about saving, and they had at hand a tool that showed a goal and, in the spreadsheet's case, a "thermometer" of progress toward it (Section 5.3). The control group had only the common part of the workshop. And every student reported the amount directly to the researcher who had run the training, that is, to me.

Under these conditions there are two mechanisms that could produce a difference between the groups without the tool, as a tool, having done anything.

The first is the **experimenter demand effect**: participants pick up what the researcher expects and adjust their answers in that direction, out of politeness, out of a wish to show progress, or out of embarrassment at admitting they did not save. The first version listed one form of this, social desirability bias, among its limitations, but did not test it. The evidence on how large these effects are is mixed. de Quidt et al. (2018) deliberately induced demand effects in eleven classic experimental tasks, by telling participants what the researcher expected, and used the reaction to put bounds on the size of these effects; they concluded that typical demand effects are probably modest. But the participants in those experiments were adults with no personal relationship to the researcher. Here they were teenagers reporting their saving to the researcher who had trained them, on a topic where there was a clearly "good" answer.

The second mechanism is that being followed changes behavior for real. Zwane et al. (2011) randomized who would be surveyed in five field experiments and later measured behavior with data that did not depend on what participants reported. In the three on health, being surveyed changed later behavior; in the two on microcredit, it did not. The systematic review by McCambridge et al. (2014) on the so-called Hawthorne effect concluded that taking part in research can change the behavior being studied, but that little is known with confidence about when this happens, through which mechanisms and how large it is. And Levitt and List (2011), going back to the original data from the Hawthorne plant, showed that the famous patterns attributed to those data do not exist, although they did find subtler signs of a possible effect. Being observed does not change behavior automatically.

In this study both groups were followed equally: the control group also answered three monthly forms. What only the treated students got was the separate training, the tool with its goal and, for two groups, the reminders. If that extra attention made the treated students save more, the effect is real, but it does not come from the tool; it would be the same with any intervention that gave the same attention. If it made the treated students *report* more, the effect is the b in Equation (8.1). The design of this study cannot separate either of these possibilities from the story in which the tool works the way the model predicted. Chapter 9 proposes an attention placebo group for exactly this reason.

> **Analogy: the thermometer read by the patient**
>
> Imagine a trial of a fever drug in which the doctor does not take the temperature but hands the thermometer to the patient and asks what it says. The patients who took the drug know they are supposed to get better, they like the doctor and they do not want to let the doctor down. If they say 37 degrees and the others say 38, the drug may have worked. But it may also be that the first group rounded down, or read the thermometer before it finished rising. Looking only at what the patients said does not settle the question, however many patients there are and however well the random assignment was done. Someone has to read the thermometer without relying on the patient.

8.2 Saving does not persist

The first signal comes from Prediction P5. In the model, a student's saving depends on how much they value the present (β), on the size of their leaks (m_i) and on their initial attention (a_{0i}). These are features of the person, and they do not change from one month to the next. So someone who saved a lot in August should, on average, also save above the mean in September. You do not need a model to expect this: spending habits change slowly.

In these data there is no persistence at all. To measure it, I subtract from each report the mean of its group in that month, so that the treatment effect and each group's trend do not create an artificial correlation. What remains is each student's deviation from their group. Figure 8.1 shows these deviations in consecutive months, one dot per student. The clouds are round, with no slope at all.

The mean of the correlations between adjacent months is 0.00, with a 95 % interval from –0.08 to 0.08. The **intraclass correlation** (ICC) sums up the same thing across the four months at once (Shrout and Fleiss, 1979). It answers the question: what fraction of the total variation in the reports lies between students, rather than within each one? If each student had their own level of saving and only wobbled a little around it, the ICC would be high. If the four months of one student are no more alike

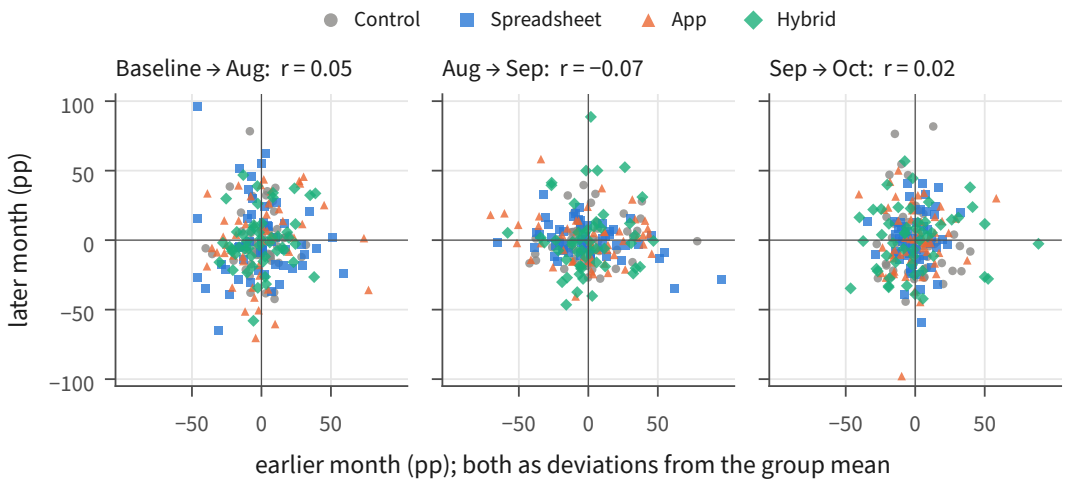


Figure 8.1. Persistence of reported saving from one month to the next. Each dot is a student, measured as the deviation from the mean of their group in the earlier month (horizontal axis) and in the later month (vertical axis), in percentage points of residual income. Control in gray, spreadsheet in blue, app in orange, hybrid in green. The title of each panel gives the correlation.

Source: study data, estimated (`code/11_measurement.py`; figure in `13_figures.py`).

than they are like the months of any other student, it is zero. Here it is –0.05, with an interval from –0.10 to 0.01. A student's reports do not resemble one another.

A property of all the reports

This result is not about the treated students. If I compute the correlation between adjacent months within each group, there are twelve correlations (four groups and three pairs of months). Their mean is –0.00, and they range from –0.41 to 0.18. The three in the control group are no farther from zero than those of the treated groups, and the ICC for the control group alone is –0.07. The correlation between the baseline and October, averaged over the four groups, is –0.04.

This matters for how the result is read. The control group got no tool, was not trained to record and had no goal on a screen. Even so, its reports do not persist. So the lack of persistence is not an effect of the tool, and it is not the shift b of the treated students either. It is a property of all the reports in this study, the noise η in Equation (8.1), or else a property of these students' true saving.

How much of the report can be signal

The lack of persistence has a precise consequence, and it is worth deriving. Suppose true saving has a correlation ρ from one month to the next and that the report adds independent noise, with $r_{it} = s_{it} + \eta_{it}$ within each group. The covariance between two consecutive reports is that of true saving, because one month's noise has no relation

to the next month's: $\mathrm{Cov}(r_t, r_{t+1}) = \rho\ \mathrm{Var}(s)$. The variance of each report is $\mathrm{Var}(s) + \sigma_\eta^2$. Therefore

$$\mathrm{corr}(r_t, r_{t+1}) = \rho\,\lambda, \qquad \lambda = \frac{\mathrm{Var}(s)}{\mathrm{Var}(s) + \sigma_\eta^2}, \tag{8.2}$$

where λ, the **reliability** of the report, is the fraction of the variance of the reports that is signal (full derivation in Appendix A.4). The observed correlation is the true persistence multiplied by the reliability.

Equation (8.2) gives a bound. Taking the top of its 95 % interval, 0.08, as a ceiling for the observed correlation, we get $\lambda \le 0.08/\rho$. I do not know the true persistence, but I can look at plausible values. If $\rho = 0.3$, the reliability is at most 0.27; if $\rho = 0.5$, at most 0.16; if $\rho = 0.7$, at most 0.12 (Figure 8.2). Put another way, the variance of the noise would be at least 2.7, 5.2 or 7.7 times the variance of the signal in the three cases.

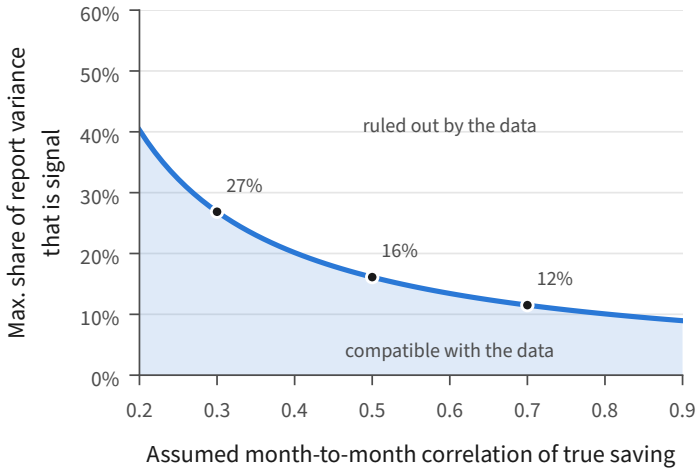


Figure 8.2. Largest fraction of the variance of the monthly reports that can be true signal, as a function of the assumed correlation of true saving across months. The curve is the upper limit of the 95 % CI of the observed correlation divided by the assumed correlation; combinations above it are ruled out by the data. The dots mark assumed correlations of 0.3, 0.5 and 0.7.

Source: study data, estimated (`code/11_measurement.py`; figure in `13_figures.py`).

There is a way out of this conclusion, and it is only honest to point it out: these students' true saving may be close to random from one month to the next. With a stipend of R$ 300 and expenses that come in lumps (a birthday one month, a phone charger the next, an extra bus ticket to get to an exam), a student can save a lot in August and almost nothing in September without anything changing in the way they handle money. The model treats these lumps as the shock ε_{it} in Equation (4.2). If they dominate, ρ is small and Equation (8.2) says nothing about the report. But for the reports to be almost all signal, the true ρ would have to be around 0.08 or lower, which means the student's own traits would carry almost no weight in their saving. I find that unlikely, but the data do not let me rule it out.

A side effect of the same fact appears in Chapter 6. Controlling for initial saving (the ANCOVA in Equation (5.2)) usually makes the estimate more precise, because

the baseline predicts part of the outcome. Here that does not happen, and now we can see why: the baseline predicts nothing.

> **Analogy: the scale that jumps by kilos**
>
> Think of a bathroom scale that, each time you step on it, shows your weight plus or minus three kilos at random. Stepping on twice in a row, you read 61 and then 66. If you weigh a hundred people once each, the average still comes out close to the true average, because the errors up and down cancel out. But one person's reading says almost nothing about that same person's next reading, and the scale cannot tell you who lost weight. A noisy scale does not mislead you about a group's average. The problem would be a different one: a scale that showed less only for the people on a diet. The lack of persistence shows that the scale in this study jumps by kilos. On its own, it does not show that the scale favors the treated.

8.3 Saving more than the whole budget

The second signal is in the upper tail. In October, 43 % of the treated students and 4 % of the control group, 68 students in all (66 treated and 2 in the control group), reported saving more than 100 % of their initial residual income. For these students, the amount reported as saved in the month was larger than everything left of the stipend after essential expenses.

This is not impossible. Residual income is computed with the essential expenses from the baseline, and the study did not measure them again. A student who started walking to school, or stopped helping out at home that month, could save more than their initial residual income. Another might have received money beyond the stipend. The first version read the hybrid group's mean above 100 % in exactly this way, as a sign that students "cut back on essential expenses or found other sources of income" (Ribeiro, 2025a). But neither thing was measured, and the fact that almost half of the treated students and almost nobody in the control group went above 100 % calls for an explanation tied to the treatment.

One concrete possibility is that some of the treated students reported the *accumulated balance* in the tool rather than the month's saving. According to the first version, the spreadsheet showed the month's balance and a thermometer of progress toward the month's goal, so the right number was at hand for anyone who used it. Still, a student filling in the form in a hurry could have reported the total put aside since August.

This hypothesis has a testable consequence. If a student reports the accumulated balance, their number goes up every month, because the balance only grows as long as they save something. Individual trajectories would be increasing: August lower than September, September lower than October. Among the treated, 43 % have this

trajectory (48 % in the spreadsheet group, 39 % in the app group and 42 % in the hybrid group), against 24 % in the control group.

This number needs a reference point. With the treated groups' means rising month by month, many increasing trajectories would show up even without any accumulated balance. To find out how many, I shuffled the months within each group: one student's August value is paired with another student's September value, drawn at random. This keeps the mean of each month and breaks any link between the months of the same student. In these shuffles, 42 % of the treated have an increasing trajectory, practically the same as in the real data. The trajectories do not rise more than chance would produce, and the accumulated balance hypothesis, as a general explanation, finds no support.

Nor is there any relation between going above 100 % and the size of the budget. In a regression among the treated, one more standard deviation of residual income changes the chance of going above 100 % by 1 point ($p = 0.77$); in the bottom third of residual income, 45 % go above 100 %, and in the top third, 49 %. If the reports above 100 % came from students with budgets so tight that any cut to an essential expense weighs heavily in the sum, they would be concentrated in the bottom third. They are not. Use of the tool does not predict going above 100 % either (–2.7 points per weekly log, $p = 0.16$).

The evidence in this section therefore comes out negative in both directions. The simplest explanation of the values above 100 % as a reporting error is not confirmed. The benign explanations (cuts to essential expenses, other income) were not measured and leave no trace in the data. What remains is an unexplained fact: almost half of the treated students reported having saved, in a single month, more than was left of their stipend.

8.4 What the values look like

The third signal is what the numbers look like. When people report from memory how much they spent or saved, they tend to give round values: R$ 150, R$ 160, R$ 200. This phenomenon, which statisticians call **heaping** (the values heap up on round numbers), is so common in survey answers that its absence stands out.

These data have almost no heaping. The essential expenses at baseline are recorded to the cent, and only 4 of the 208 are whole reais, or 1.9 %. If the cents were spread at random between .00 and .99, the expected fraction of whole reais would be 1 %, that is, one cent value in a hundred. The left panel of Figure 8.3 shows the cents of the 208 expenses: they spread over the whole range, with no spike at .00 or .50.

The monthly amounts saved need one more step. The data file stores each month's saving as a percentage of residual income, not in reais. To recover the reais, I multiply

the percentage by each student's residual income, which brings in a rounding error of about one cent. So I count as round any value within one cent of a whole real or of a multiple of R$ 5. With this tolerance, the chance that an arbitrary value is counted as a whole real is 3 in 100 (the cents .99, .00 and .01), or 3 %; the chance that it is counted as a multiple of R$ 5 is 3 in 500, or 0.6 %.

In the data, leaving out months with zero saving, 2.2 % of the monthly amounts saved are whole reais and 0.1 % are multiples of R$ 5 (right panel of the figure). Both fractions are at the level of chance or below it. In answers given from memory, one would expect the opposite: many more multiples of R$ 5 and R$ 10 than chance would give.

One explanation would be that the treated students copied the exact amount from the tool, which computes to the cent. But that does not hold for the control group, which had no tool to copy from, and whose values are whole reais in 3.5 % of cases, at the level of chance. Nor does it hold for the essential expenses, which were reported before any tool existed.

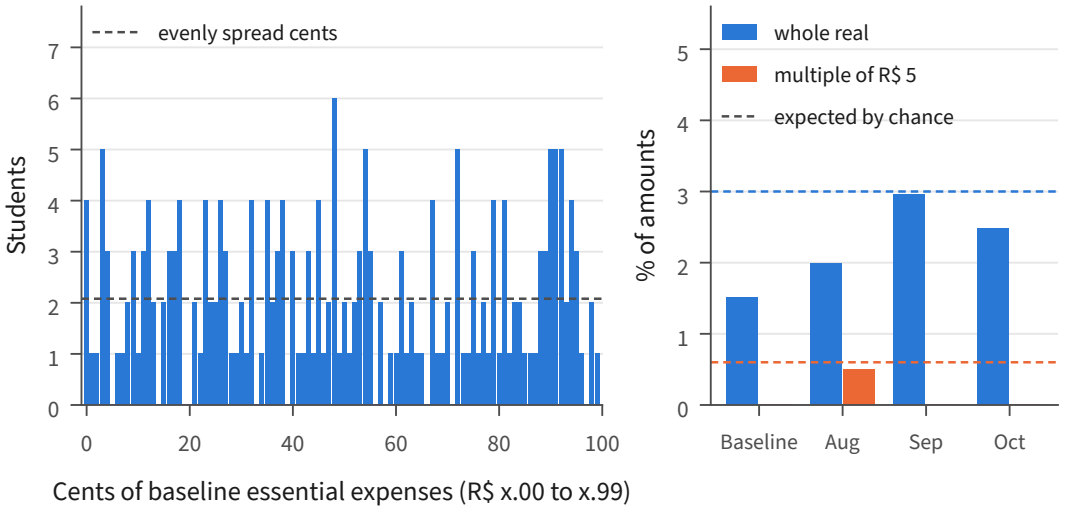


Figure 8.3. Left: number of students by the cents of their initial essential expenses (from R$ x.00 to x.99), for the 208 students; the dashed line is what evenly spread cents would give. Right: fraction of the monthly amounts saved that lie within one cent of a whole real (blue) or of a multiple of R$ 5 (orange), by month, with the fractions expected by chance under the same tolerance (dashed).

Source: study data, measured and estimated (`code/11_measurement.py`; figure in `13_figures.py`).

Precise values are not impossible. A student who checked a bank statement, or who writes everything down in a notebook, can give an amount with cents. But it is rare for almost every student in a study to do that, the control group included, in a question about a whole month. This signal does not say whether the reports are too high or too low. It says they do not look like what one usually gets when asking people how much they put aside, and it is one more reason to want to see the original forms, and not just the final spreadsheet.

8.5 What effect size should we believe?

The fourth signal is the size of the effect. In control-group standard deviations, receiving a tool raised reported saving by 2.0. Meta-analyses of financial education find effects on financial behavior of about a tenth of a standard deviation, and sometimes smaller (Fernandes et al., 2014; Kaiser et al., 2022). The effect in this study is about twenty times that. It is also larger than the effects of commitment savings products and of reminders in field experiments (Ashraf et al., 2006; Karlan et al., 2016), which measured saving through account balances and not through what participants reported.

An effect twenty times larger than the typical one is not impossible. The population is special (olympiad medalists with a small, regular income), the intervention was tailored to them, and the outcome is a fraction of a small budget, which moves easily. But a careful reader should be suspicious of a result so far off the curve, and Bayesian statistics (Section 3.9) offers a way to turn that suspicion into a calculation.

The idea is a skeptical reader who, before seeing the data, believes the effect is probably similar to the one in the literature. I write this belief as a normal distribution, the **prior**, centered at 0.1 control-group standard deviations, which here is 2.6 points, with a width τ that says how sure the reader is of it. The data bring the estimate $\hat{b} = 52.0$ points, with a standard error of 4.0; in the formula below I write EP, the standard error. When both are normal, the belief after the data, the **posterior**, is also normal, and its mean is a weighted average of the two:

$$\bar{b} = \frac{\hat{b}/\mathrm{EP}^2 + \mu_0/\tau^2}{1/\mathrm{EP}^2 + 1/\tau^2}, \tag{8.3}$$

where μ_0 is the center of the prior. Each source is weighted by the inverse of its variance: the surer it is, the more weight it gets.

With a narrow prior, 0.1 standard deviation wide, the reader ends up with a posterior mean of 17.7 points (interval from 13.4 to 22.0), about R$ 25 a month. With a width of 0.25 standard deviation, the mean is 38.8 points (R$ 56); with 0.5, it is 47.9 points (R$ 69). Figure 8.4 shows the whole curve. A prior 0.15 standard deviation wide is enough for the reader to end up believing at least half of the estimate.

Even a committed skeptic ends up believing in an effect several times larger than the one in the literature. That result seems to close the discussion, but it does not, and the reason is the central point of this chapter. The calculation in Equation (8.3) takes the reports as they are. It protects against luck, against a draw that happened to put the biggest savers in the treated groups. It does not protect against the shift b in Equation (8.1), which moves all the treated students in the same direction and does not shrink with the sample size. With more students, the standard error would shrink and the posterior would move even closer to the estimate, biased or not.

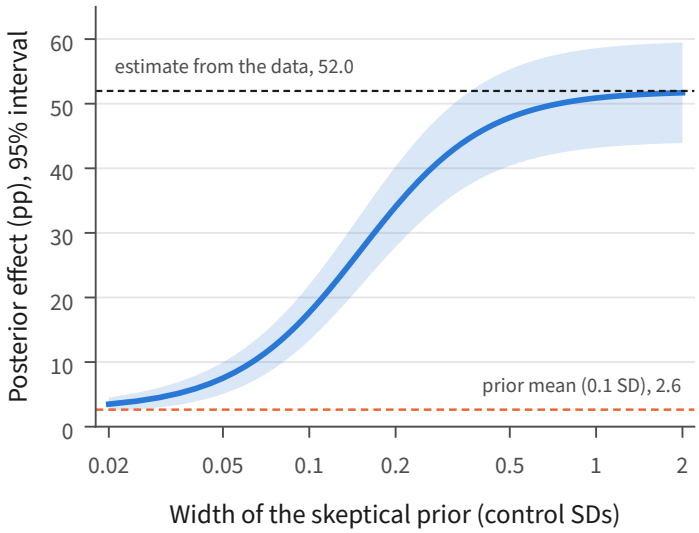


Figure 8.4. Posterior mean and 95 % interval of the pooled effect of the tools on reported saving in October (percentage points of residual income), for a normal prior centered on the typical effect in the literature (0.1 control-group standard deviation, dashed orange line) and different widths (horizontal axis, log scale). The dashed black line is the estimate from the data.

Source: study data, estimated; illustrative prior (`code/12_equivalence_bayes_design.py`; figure in `13_figures.py`).

8.6 Six synthetic worlds

The four previous signals share a limitation: each was examined on its own, and for each there is a possible benign explanation. True saving may be close to random across months; the values above 100 % may come from cuts in spending; the cents may come from bank statements; the effect may simply be large. What is missing is to look at all the patterns at once and ask: what kind of world would produce *these* data?

To answer, I manufactured data. The idea is simple. I write, in code, several possible worlds, each with a different answer to the question of what the tool did. In each world I generate, many times over, the reports of 203 students, as if the experiment had been run there. Then I run on these synthetic reports the same calculations I ran on the real data. If the real data look like what a world produces, that world is compatible with them. If they do not, that world, with the assumptions I wrote for it, becomes hard to defend. Everything in this section is *simulated*: these are synthetic data, generated by the program `code/15_synthetic_worlds.py`, not measurements.

How the worlds were built

The worlds combine two choices. The first is the *channel* through which the tool acts, with three options.

- In the *attention* world, each treated student's gain is proportional to their use of the tool, as in the attention part of Equation (4.8). Someone who logs nothing gains nothing.
- In the *equal effect* world, every treated student gains the same amount, whether or not they use the tool. This is the salience part of the model, or any other channel that all the treated students had in common.
- In the *reporting only* world, nobody saves a single real more. The treated students just report more; this is the shift b in Equation (8.1), with no true effect at all.

The second choice is the *reliability* of the reports, with two options. With *faithful* reports, the report is the true saving. With *noisy* reports, only a fraction $\lambda = 0.10$ of the variance of the report is signal, and the rest is noise η that changes every month. Three channels times two reliabilities give six worlds.

To make the comparison fair, all the worlds were calibrated on the real data. They have the same 203 students in the same four groups. True saving each month starts from the real means of the control group. The mean effect for the treated, month by month, is the observed difference between treated and control, net of the baseline difference (zero at baseline, growing until October), so that all six worlds reproduce, on average, the main result of Chapter 6. The standard deviation of the reports is the observed one, about 22 points. And each synthetic treated student's use of the tool is drawn from the real distribution of logs per week, including those who never logged anything.

In every world, true saving persists: each student has a trait of their own, and the correlation of true saving from one month to the next is $\rho = 0.5$. This value and the λ of the noisy world are illustrative, chosen to explain, not estimated. For each of the six worlds I generated 1,000 synthetic samples.

The four diagnostics

In each synthetic sample, and also in the real data, I compute four numbers.

1. The *slope* of the change in saving, from baseline to October, on logs per week, among the treated. This is model (A) in Table 7.3. I also note whether the 95 % interval of the slope lies entirely above zero, that is, whether the sample "sees" a positive slope.
2. The difference, in October, between the treated students who never logged anything and those who did.
3. The *intraclass correlation* of the reports across the four months, net of the group means, as in Section 8.2.
4. The *quantile spread*: the effect at the 90th percentile minus the effect at the 10th. If the effect is the same for everyone, the distribution of the treated is the control

distribution shifted over, and the spread is close to zero. If some gain a lot and others nothing, the distribution of the treated fans out, and the spread is large.

Figure 8.5 shows the result. Each row is a world; each panel, a diagnostic. The dot is the mean of the 1,000 samples and the bar is the range where 95 % of them fall. The dashed black line is the value in the real data. A world is compatible with the data on a diagnostic when the dashed line passes through its bar.

What the worlds show

Let me start with the two attention worlds, the first two rows of the figure. In them, the mean slope is 25.8 points per weekly log. The number is large for a simple reason: the mean effect has to match the observed one, and mean use is only 2.0 logs per week, so each log has to be worth a lot. With this slope, the synthetic samples see a positive slope 100 % of the time with faithful reports and 100 % of the time with noisy reports. In the real data, the slope is –1.3. The difference between those who never logged and those who did is, on average, –63.0 points, because the first group gained nothing; in the real data, the two groups reported the same (96.6 % and 96.7 %). And the quantile spread is 81.2 points, because some treated students gain a lot and others nothing; in the real data, it is 1.6.

The ICC of the attention worlds is also high, 0.58 with faithful reports and 0.45 with noisy ones. The second number is the more interesting one. Even with 90 % noise, the attention channel creates persistence, because each student's gain depends on their use, which is the same every month: a heavy user sits above their group in every month after the baseline. The noise cannot hide this.

The real data fall outside the range of the attention worlds on all four diagnostics, and by a wide margin: the dashed line passes far from the orange bars in every panel. This is the firmest result of the simulation, with the caveat that it holds for the attention world as I wrote it: a gain proportional to logs per week, counted without error, and use drawn with no relation to the saving the student would have had anyway. The channel that motivated the study, in which the tool works because the student records and starts to see their spending, produces nothing like these data.

The other four worlds split by reliability. In the equal effect and reporting only worlds, the slope is zero on average, and the samples see a positive slope only 3 % of the time, close to the 2.5 % of false alarms expected when a 95 % interval is required to lie entirely above zero. The difference between those who never logged and those who did stays close to zero, and the quantile spread stays close to the observed 1.6 (1.6 in the equal effect world with faithful reports, 1.4 in the reporting only world with noisy reports). On these three diagnostics, all four worlds are compatible with the data.

The ICC splits them into two pairs. With faithful reports, it is 0.50 in the equal effect world and 0.50 in the reporting only world. This is the true persistence I put

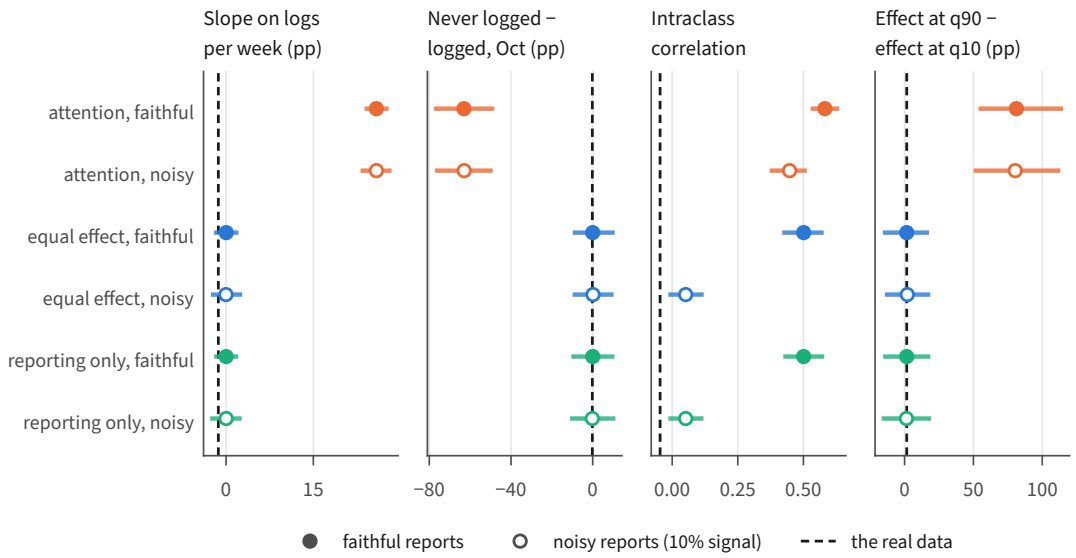


Figure 8.5. The four diagnostics in six synthetic worlds (synthetic data, simulated). Rows: channel of the effect (attention through use in orange, equal effect in blue, reporting only in green) and reliability of the reports (filled dot: faithful reports; hollow dot: noisy reports, with 0.10 of the variance as signal). Dot: mean of 1,000 synthetic samples; bar: range of 95 % of the samples. Dashed: the value in the real data. Panels, from left to right: slope of the change in saving on logs per week; October difference between those who never logged and those who did; intraclass correlation; effect at the 90th percentile minus effect at the 10th. All worlds have the observed mean effect, control means and standard deviation, and true persistence $\rho = 0.5$ (illustrative).

Source: simulated, calibrated on the study data (`code/15_synthetic_worlds.py`).

into the worlds, passing whole into the reports. The real data, with an ICC of –0.05, sit far below the range of these two worlds. With noisy reports, the ICC drops to 0.05 and 0.05, as Equation (8.2) predicts: $0.5 \times 0.10 = 0.05$. These values are much closer to the observed one, but the real ICC is still below the 95 % range of these two worlds, which runs from –0.01 to 0.11: in their synthetic samples, the fraction with an ICC equal to or below the real one was 0.0 %. By the criterion of the figure, none of the six worlds is compatible with the data on the ICC; the two noisy ones come closest. The remaining gap suggests that, with $\rho = 0.5$, the fraction of signal in the real reports would be even smaller than the illustrative 0.10, or that true saving persists less than I assumed.

What the simulation does and does not allow us to conclude

The simulation leads to three conclusions.

The first is that, under the assumptions of the simulation, the data are not compatible with the world in which the effect works through use of the tool. This holds with faithful or noisy reports, and it backs up what Chapter 7 found with the regressions. The worlds also let me put a number on this. Suppose the real effect is a mixture: a fraction α of it comes from attention, proportional to use as in the attention world,

and the rest, 1 – α, from a channel that is the same for everyone. Only the first part varies with use, so the expected slope is α times the 25.8 points of the pure attention world. The 95 % interval of the real slope goes up to about 0.5 point (Section 7.3). For the expected slope to fit below that limit, α would have to be at most the ratio of 0.5 to 25.8, about 2 %. (The script uses the exact upper limit of the interval, 0.54, not the rounded value.) This calculation rests on three assumptions, the same ones as the attention world. The first is that the gain from attention grows in a straight line with logs per week and is zero for someone who does not log. The second is that the logs measure use without error; if the count from the screenshots is very noisy, the real slope would be flattened, and the fraction could be larger. The third is that use has no relation to the saving the student would have had without the tool, as in the worlds, where it is drawn at random; if the students who would have saved least were the ones who logged most, the observed slope underestimates that of the attention channel (Section 7.3). And the limit comes from a 95 % interval, not from certainty.

The second is that, to get close to the observed ICC, a world with true persistence of 0.5 needs noisy reports. No world with faithful reports comes close to the data, and even the noisy ones, with λ = 0.10, still sit a little above them. Either these students' true saving barely persists, or most of the variation in individual reports is noise. This holds for all groups, the control group included (Section 8.2).

The third, and the most important, is what the simulation *cannot* do. The rows "equal effect, noisy" and "reporting only, noisy" in Figure 8.5 are practically identical in all four panels. In one of these worlds, every treated student really saves a few dozen reais more each month; in the other, nobody saves a cent more. The two are equally close to the real data (and equally a little above them on the ICC), and no calculation done with the reports alone can tell them apart.

The reason is structural, not a shortage of students. The four diagnostics look only at the reports. In both worlds, the reports of the treated shift by the same amount and carry the same noise. The only thing that distinguishes them is true saving, which never appears in the data. Doubling the sample, or making it ten times larger, would narrow the bars in the figure, but the two rows would stay in the same place. This is why the next study needs a measure that does not pass through the student's report, such as the balance of a bank account, and an attention placebo group that gets the same attention without the tool (Chapter 9).

The simulation has its own limits too, and they should be stated. It examines only six worlds, and the real world may be a mixture of them: part of the effect true and equal for everyone, part reporting. The values of ρ and λ are illustrative. The attention world was written in its most extreme form, in which someone who does not log gains nothing. And the diagnostics are the ones I chose; others might separate worlds that these do not. But the main conclusion does not depend on these details. Two worlds in which the reports of the treated shift in the same way and carry the same noise produce reports with the same distribution, whether the shift comes from saving or from reporting.

8.7 What follows from this

None of the signals in this chapter shows that the effect is false. The lack of persistence holds for the control group too and, on its own, creates no difference between the groups. The values above 100 % and the cents have possible benign explanations, though none was measured. The effect is much larger than those in the literature, but a skeptical reader would still end up believing in a large effect, if they took the reports as they are. And in the simulation, a true effect equal for all treated students, with noisy reports, is as close to the data as any other world.

What the signals show, taken together, is that the reports do not behave like careful records of real behavior. They do not persist, they have almost no round numbers and, among the treated, they often exceed the whole budget. And what the data can say with confidence stays tied to these reports. The simulation makes this clear: the world in which every treated student saved more and the world in which the treated students only reported more produce reports with the same distribution.

So the claim the evidence supports is narrower than the one in the first version. It is not "the tools increased the students' saving". It is: *receiving a budgeting tool greatly increased the saving that students reported.* That claim is solid. It survives chance, the specifications and attrition, and any explanation for it has to involve something all the treated students had in common, because those who never used the tool reported the same as those who did.

Knowing whether the tools put more money in the students' pockets requires an outcome that does not depend on the student's own report. This is not a technical detail. It is the difference between a study showing that an intervention changes what people say and one showing that it changes what people do. Chapter 9 describes a design that can answer the second question.

CHAPTER 9

Proposal for a new experiment

The experiment in the previous chapters answers one question well and another one badly. It shows, by a wide margin, that the students drawn to receive a budgeting tool *reported* saving much more than the control group (Chapter 6). It cannot say how much of that difference is money actually put aside and how much is a change in the way students answered the form (Chapter 8). Nor can it separate the tool from everything that came with it: the separate room at the workshop, the weekly reminders in the groups with the app, the feeling of being part of a special group, and the wish to show progress to the person who ran the training.

This chapter describes the experiment I would run to answer those two questions. It changes three things relative to the first one. The main outcome becomes the balance of a bank account, a number that does not depend on what the student says. An **attention placebo** group is added, which gets the same kind of contact as the tool group, but on a subject unrelated to money. And the sample size is calculated in advance, for the effects the literature makes plausible, and not for the huge effect the reports showed. On top of that, the whole analysis plan is written down and registered before there are any data, and the program that will do the calculations already exists and has already been tested on synthetic data.

None of what follows has been done. It is a proposal, and the numbers in this chapter come from formulas and from simulations with parameters I chose for illustration; each one carries its label.

9.1 The design

9.1.1 Who takes part and how the draw works

The population stays the same: PIC scholars, who receive the same stipend of R$ 300 a month. Keeping the population has a clear advantage, because the new study tests the same intervention on the same kind of student, and any difference in results cannot be blamed on a change of audience. The entry criteria would also be those of the first study (Section 5.2): an active scholarship, internet access and a cell phone, no prior systematic use of budgeting tools and no prior formal financial education.

The draw would again be done within the 7 grade strata, as in the first study (Section 3.2), now for three groups instead of four. Drawing within strata makes sure

that each group has the same mix of 6th graders and high school students, and this cuts the noise of the comparisons a little (Bruhn and McKenzie, 2009).

9.1.2 The three groups

The **control group** gets what everyone gets: the study account, an explanation of how it works, and the monthly forms. It gets no tool, no budgeting workshop and no weekly messages.

The **tool group** gets a single budgeting tool, the simplest one possible. The data from the first study give no reason to prefer the combination of spreadsheet and app (Section 7.1), and a single tool is cheaper to offer and easier to explain. I would choose the app, for a measurement reason: it stores each entry with a date and time, which gives a much better measure of use than the screenshots in the first study (Section 7.3). If those records cannot be obtained with the students' consent, the spreadsheet would be the alternative, because the version history in Google Sheets also keeps the edits. The group has an online workshop in the same format as the first one, with a part in a separate room to learn the tool, and gets weekly reminders to *record* spending.

The **attention placebo** group gets the same kind of contact as the tool group, without the tool. It has a workshop of the same length, a part in a separate room and weekly messages at the same frequency, but the subject is a different one, unrelated to money, for example study habits for the olympiad. The same person should run both workshops, and the messages should be of similar length and tone.

> **Analogy: the sugar pill**
>
> In a drug trial, the group that gets the sugar pill is not there to show that sugar does not cure anything. It is there so that the drug is compared against everything that comes with it: the appointment, the doctor's attention, the expectation of getting better and the ritual of taking something every day. Only the difference between the drug and the sugar pill is the effect of the substance. In the proposed study, the attention placebo does this job: the workshop and the messages about study habits are the appointment and the ritual, and the budgeting tool is the substance.

Why three groups and not two? The comparison that matters most is tool against placebo, because it isolates the tool from the attention that comes with it. But the comparison between placebo and control also teaches something. If the placebo group saves more than the control group, part of the effect in the first study may have come from the contact, the messages and the feeling of being followed, and not from the tool. This kind of effect has an old name, the Hawthorne effect, and how strong it really is has been argued about for decades (Levitt and List, 2011; McCambridge et al., 2014). Being asked about a behavior again and again can also change it (Zwane et al., 2011). With three groups, the study measures these things instead of assuming they do not exist.

The design leaves out the spreadsheet and hybrid groups of the first study. This has a cost: the new study does not say which tool is better. But Chapter 7 showed that ranking the tools would take hundreds of students per group, and it makes no sense to spend the sample on that before knowing whether any of them increases measured saving.

9.1.3 The outcome: the balance in the bank

The most important change is the outcome. In the first study, the student said how much they had saved. In the new one, the main outcome is the balance of a savings account opened for the study or, if the student already has an account, the balance reported by the bank itself with the consent of the student and their parent or guardian. The balance would be measured at the start, at 3 months and at 12 months.

The balance has three advantages over the report. Nobody has to remember anything, so there is no rounding and no confusion between the month's saving and the accumulated balance (Sections 8.3 and 8.4). The student does not choose the number, so there is no way to report a nicer amount to please the researcher. And the same instrument measures all groups the same way, whereas in the first study the treated students had a tool in front of them with a balance panel and the control group did not.

The balance also has limits, and it is better to state them up front. A student can keep money at home, in a family account or with a friend, and that money does not show up in the study balance. If this happens equally in the three groups, the random draw takes care of the problem: the difference between groups in the balance is still a causal effect on the money kept in the account, even if not on all saving. The risk is that the tool changes *where* the student keeps money without changing *how much* they keep. For example, an app that shows the account might lead the student to move into it money that used to sit in a drawer. To reduce this risk, the account would be presented the same way to all three groups, with the same explanation and the same suggestion for its use, and the final form would ask everyone whether they kept money anywhere else.

9.1.4 The report stays, now as an object of study

The monthly form from the first study stays, with the same questions, in all three groups. It stops being the outcome and is now measured against the bank. For each student, the difference between what they say they saved and how much their balance changed is the **reporting gap**.

Example 9.1 a reporting gap

Suppose a student says on the forms that they saved R$ 120 over the three months, and that their account balance went from R$ 40 to R$ 120 over the same period. The balance rose by R$ 80. This student's reporting gap is 120 – 80 = 40 reais. One student's gap does not say much: they may have kept the other R$ 40 at home. But if the mean gap in the tool group is larger than in the control group, the difference measures how much the tool changed the report beyond the money. The values in this example are illustrative.

In the model of Chapter 4, this gap is the $b\,T_i + \eta_{it}$ part of Equation (4.10): the shift of the treated and the reporting error. The first study had no way to estimate b, because it only had the report. The new study estimates b directly, and that is what led me to keep the form.

9.1.5 Use, from the app's records

In the first study, use of the tool was counted from screenshots the students sent, and the file does not distinguish a student who never logged anything from one who did not send the screenshot (Section 7.3). In the new one, use would come from the app's own records: how many expenses the student entered, on which days, and when they last opened the app. With the student's consent, these records can be exported at the end of each month. This measure still does not say whether use *causes* saving, because each student decides how much to use the tool; but it stops being a crude measure.

9.2 How many students are needed

9.2.1 The calculation by formula

Section 3.5 explained what statistical power is: the probability that the test detects an effect when there is one. For the proposed study, the question is how many students each group needs so that the main test, tool against placebo, detects an effect of a given size with 80 % probability.

The answer depends on three things. The first is the size of the effect, which I measure in standard deviations (SD) of the balance, so as not to depend on how much the students save in reais. The second is the level of the test, two-sided at 5 %, as everywhere in this work. The third is the correlation ρ between the same student's initial and final balance. The larger ρ is, the more the initial balance helps predict the final one, and the more the ANCOVA, which controls for the initial balance, cuts the noise (McKenzie, 2012). The variance of the estimator falls to a fraction $1 - \rho^2$ of the variance of the simple difference in means (Section 3.3; the derivation is in Appendix A.6).

Table 9.1. Students per group needed for the main test (tool against placebo) to detect the effect shown in the first column, in standard deviations of the balance, with 80 % power. Each column assumes a different correlation ρ between the initial and final balance.

Effect (SD)	$\rho = 0$	$\rho = 0.5$	$\rho = 0.7$
0.1	1,745	1,309	890
0.2	437	328	223
0.3	194	146	99
0.5	70	53	36

Notes: two groups, two-sided test at 5 %, ANCOVA with the initial balance; the calculated size was divided by 0.9, a margin for 10 % attrition. The proposed study has three groups, so the total sample is three times each number.

Source: estimated with the ANCOVA power formula, with illustrative effects and correlations (`code/12_equivalence_bayes_design.py`).

Table 9.1 gives the result of the formula with a margin for attrition: the calculated size is divided by 0.9, so that enough students remain if 10 % of them leave the study. With $\rho = 0.5$, detecting an effect of 0.3 SD takes 146 students per group, and an effect of 0.2 SD takes 328. If the correlation is higher, $\rho = 0.7$, the numbers fall to 99 and 223. If there is no correlation at all, as in the reports of the first study, where the correlation between months was 0.00 (Section 8.2), they rise to 194 and 437.

Why aim at 0.2 or 0.3 SD, and not at the 2.0 SD effect the reports showed? Because I do not believe the true effect on money is that large, and Chapter 8 explains why. Evaluations of financial education find effects on behavior of about a tenth of a standard deviation (Kaiser et al., 2022). A study sized to detect 2.0 SD would need few students, but if the real effect is ten times smaller, it would end up with a wide confidence interval and no conclusion. Aiming at 0.2 or 0.3 SD is a bet that the tool has a real effect, larger than the typical one in financial education and much smaller than the reported one. Effects of 0.1 SD would require a study on another scale (first row of Table 9.1).

9.2.2 The same calculation by simulation

The formula behind Table 9.1 is an approximation. It assumes that balances follow a normal distribution, that the variance is the same in all groups and that the sample is large enough. To check it, I redid the calculation another way: I generated many synthetic studies and counted in how many the main test rejected the hypothesis of no effect.

Each synthetic study has three groups, with the draw done within grade strata that mimic the composition of the first study. The initial and final balances have a standard deviation of R$ 55 and a correlation of 0.5, both illustrative; the standard deviation was chosen close to that of the monthly saving in reais observed in the first study, only so that the figures are in reais. Since the effect is given in standard deviations, this value in reais does not change the power. The correlation is the one in the middle column of Table 9.1, and no synthetic student drops out. All three groups

are generated, but the main test uses only the students in the tool and placebo groups. For each group size and each effect, I generated 1,000 samples and analyzed each one with the same program that will be used on the real data, `code/analyze_followup.py` (Section 9.4). The simulated power is the fraction of samples in which the test of tool against placebo gave $p < 0.05$.

Figure 9.1 shows the result. With an effect of 0.3 SD, power goes from 41 % with 50 students per group to 66 % with 100, 86 % with 150 and 99 % with 300. With 0.2 SD, power is much lower: 22 % with 50, 53 % with 150, and 81 % only with 300 students per group.

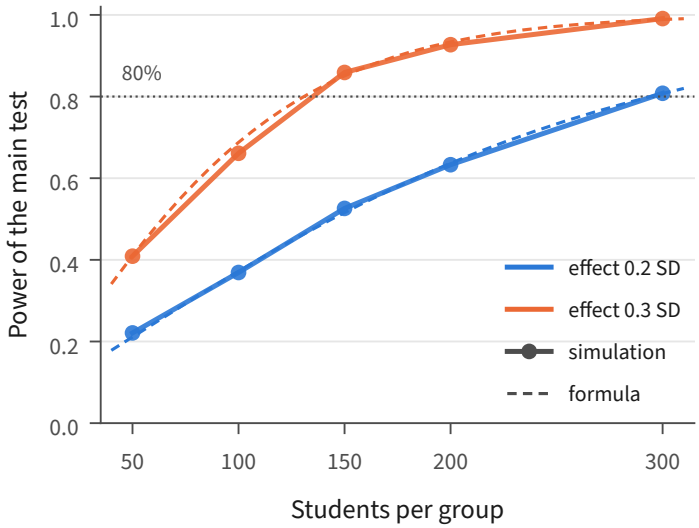


Figure 9.1. Power of the main test in the proposed study (tool against placebo, on the balance at 3 months) against the number of students per group. Dots and solid lines: fraction of 1,000 synthetic samples in which the test rejected at 5 %. Dashed lines: the ANCOVA formula, the same as in Table 9.1, with $\rho = 0.5$ and without the attrition margin. Blue: effect of 0.2 SD; orange: 0.3 SD. The dotted line marks 80 %. None of the curves includes attrition.

Source: simulated, with an illustrative standard deviation of R$ 55 and $\rho = 0.5$ (`code/16_followup_proposal.py`).

9.2.3 Why the simulation and the formula agree

The simulation's dots fall almost on top of the formula's dashed lines. This was expected, and it is good to understand why. The synthetic data were generated exactly the way the formula assumes: normal balances, the same variance in all groups, the same correlation ρ for all students, and no attrition. Under these conditions, the ANCOVA estimates the effect with a standard error that is, on average, the formula's, and the test statistic has a distribution very close to the normal, which is what the formula uses. The program's stratum fixed effects and HC2 standard errors cost very little with 50 or more students per group. The small differences that remain come

from the random draw of the samples themselves: with 1,000 samples, the simulated fraction has an error of a few percentage points.

The numbers in Figure 9.1 and in Table 9.1 should not be read as two estimates of the same thing. The table gives a sample size already divided by 0.9 to cover attrition, and for three values of ρ; the simulation gives the power for fixed sizes, with $\rho = 0.5$ and no attrition. For 0.3 SD and $\rho = 0.5$, the table calls for 146 students per group. Without the margin, the same formula would call for 0.9 times that number, and it is in that range, between 100 and 150 students per group, that the simulated 0.3 SD curve crosses 80 %. To plan the study I use the numbers in the table, which already allow for attrition; the simulation is there to check the formula and the program.

If the simulation only repeats the formula, what is it for? It does three things the formula does not. First, it runs the actual analysis program, with the draw within strata, the fixed effects and HC2; if the program had a bug, the simulated power would come out different from the formula, and I would see it. Second, it can be redone with data more like real balances, which probably have many zeros and a long tail of a few students with high balances; in that case the formula no longer holds and only the simulation gives the answer. Third, once the baseline balances are available, before the draw, the simulation can be rerun with the observed standard deviation and correlation instead of the illustrative values used now.

9.2.4 What this means for the size of the study

The most realistic number in Table 9.1, for an effect of 0.3 SD and $\rho = 0.5$, is 146 students per group. Three groups of that size add up to more students than the 389 scholars contacted for the first study, of whom 208 took part. A single recruitment round the size of the first one is therefore not enough, even if everyone contacted agreed to take part. The ways out are to recruit across more than one edition of the program, to open the study to other scholarship programs with similar income, or to settle for detecting only larger effects. I prefer the first: it keeps the population and only asks for more time.

9.3 The analysis plan

A **preregistered analysis plan** is a document, published before the data exist, that says which hypotheses will be tested, with which specification, on which sample and with which correction for multiple testing. The idea is simple: if the choices are made before seeing the results, they cannot have been made to produce a result.

The first study had no plan. The methods section of the first version describes the outcome, the regression on use and the comparisons by gender, age and origin, the closest thing there is to a plan, but none of it was published before data collection;

everything else, including the model in Chapter 4, was added after the data were collected (Chapter 7). I showed that the main effect survives 108 different specifications (Section 6.4), but that does not replace a plan: the other conclusions, about use, heterogeneity and mechanisms, should be read as exploratory. In the proposed study, the following analyses are fixed before data collection, in the order in which they would be reported.

1. Sample and estimand. All analyses are intention to treat: each student stays in the group to which they were assigned, whether or not they use the tool and whether or not they attend the workshop. Every randomized student whose balance has been measured at the time of the analysis is included.
2. H1, the main hypothesis. Among the students in the tool and placebo groups, the balance at 3 months is regressed on an indicator for the tool group, the initial balance and stratum fixed effects:

$$B_{i3} = \alpha + \tau_1 F_i + \gamma B_{i0} + \mu_{s(i)} + \varepsilon_i, \tag{9.1}$$

   where B_{i0} and B_{i3} are the initial balance and the balance at 3 months, in reais, F_i equals 1 in the tool group, and $\mu_{s(i)}$ is the fixed effect for the student's grade stratum. This is the same ANCOVA as in the first study (Equation (5.2)), without the other covariates. The standard errors are HC2, and the test of $\tau_1 = 0$ is two-sided at 5 %. H1 is the only main hypothesis; its result goes through no correction.
3. H2, the effect of attention. The same regression as in Equation (9.1), among the students in the placebo and control groups, with the placebo indicator in place of F_i. A positive τ_2 indicates that the contact, the workshop and the messages, without the tool, already raise the balance.
4. H3, reporting bias. For each student, the gap is $L_i = r_i - (B_{i3} - B_{i0})$, where r_i is the sum of what they reported having saved over the three months. Among the students in the tool and control groups,

$$L_i = \alpha + \tau_3 F_i + \mu_{s(i)} + u_i, \tag{9.2}$$

   with HC2 errors. The coefficient τ_3 estimates the shift b in Equation (4.10): how much more the tool group reports than its balance justifies, compared with the control group. The comparison is against the control group, and not against the placebo, because it is the same comparison as in the first study, whose result we want to explain.
5. Twelve months. H1, H2 and H3 are repeated with the balance at 12 months. These analyses say whether the effect lasts after the novelty wears off, but since the sample and the power were calculated for 3 months, they are secondary.

6. Use, from the app's records. Within the tool group, the change in balance is regressed on the mean number of logs per week, taken from the app's own records, with the initial balance and stratum fixed effects. This is the test of Prediction P2. Since each student chooses how much to use the tool, the slope is descriptive and not causal, and the plan says so in advance.
7. Heterogeneity by residual income, in reais. Equation (9.1) gains baseline residual income and its interaction with F_i. The outcome stays in reais, not as a percentage of residual income, because Section 7.4 showed that the percentage creates a gradient through the arithmetic alone. This is the test of Prediction P3.
8. Attrition. The plan reports the fraction of students without a measured balance in each group and tests whether it differs across groups. If it does, H1 to H3 come with the bounds of Lee (2009), which trim from the group with less attrition the fraction of students that balances the two and give the interval in which the effect can lie (Section 6.4).
9. Multiple testing. The tests are divided into families, and within each family the p-values are corrected by the method of Holm (1979) (Section 3.6). H1 forms a family on its own. H2 and H3 form the mechanisms family. The three hypotheses at 12 months form another, and items 6 and 7 form the family of secondary analyses.
10. Changes to the plan. Any analysis not on this list will be reported as exploratory, in a separate section. If a choice in the plan turns out to be impossible, for example because the bank did not report the balances of a group of students, the change and the reason will be reported.

This plan is shorter than the list of analyses in Chapters 6 to 8, and that is on purpose. A plan with many main hypotheses dilutes power and invites picking, afterwards, the one that worked.

9.4 A dry run with synthetic data

A written plan can still hide mistakes: a variable with the wrong name, a comparison between the wrong groups, a formula that does not do what the text says. To catch these mistakes before data collection, I wrote the analysis program for hypotheses H1 to H3 and ran it on synthetic data. The program is `code/analyze_followup.py`, reproduced in Appendix B. It takes a file with one row per student (group, stratum, initial balance, final balance and report) and returns, for each hypothesis, the estimate, the 95 % confidence interval and the p-value. The program does only H1, H2 and H3 with the balance at 3 months, items 2 to 4 of the plan. For item 5, it would be enough to run it on a file in which the final balance and the report were those at 12 months, but this has not been tested yet. Items 6 to 9 (use, heterogeneity, attrition

and the Holm correction) are not in the program; they would be added and tested the same way before registration.

The synthetic data have 150 students per group. In them, I chose the world: the tool raises the balance by 0.25 SD, or R$ 14; the placebo changes nothing; and the students in the tool group report 0.3 SD, or R$ 16.5, more than they saved, while the other two groups report without any shift. Everyone's reports also carry a random error. The true values are illustrative, chosen so that the effect and the bias have similar sizes and the test is demanding: if the design did not separate the two, this is the case in which the confusion would show up.

The result is in Figure 9.2. For H1, the program estimated an effect of R$ 20.3 on the balance, with an interval from 9.2 to 31.4 reais and $p = 3.5 \times 10^{-4}$; the interval contains the true value, R$ 14. For H3, the estimate of the reporting bias was R$ 16.0, with an interval from 8.7 to 23.4 and $p = 1.9 \times 10^{-5}$, close to the true value of R$ 16.5. For H2, the estimate was –9.4 reais, with an interval from –20.4 to 1.6 and $p = 0.093$.

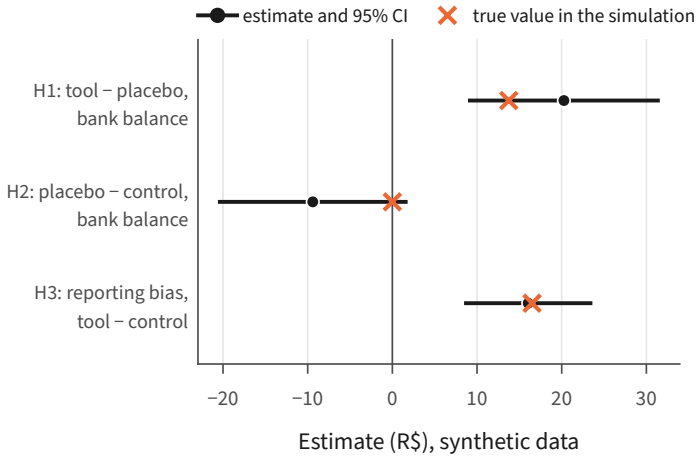


Figure 9.2. Dry run of the analysis program on synthetic data, 150 students per group. For each hypothesis in the plan, the black dot is the estimate and the line is the 95 % confidence interval, in reais; the orange x is the true value chosen in the simulation. The vertical line marks zero.

Source: simulated (synthetic data), with an illustrative effect of 0.25 SD and bias of 0.3 SD (`code/16_followup_proposal.py` and `analyze_followup.py`).

Three things come out of this dry run. The first is that the design separates true saving from reporting bias. In the synthetic world, the students in the tool group save more *and* report more than they saved. A study with only the reports would see the sum of the two and have no way to split it, which is exactly the situation in Chapter 8. With the bank balance, H1 recovers only the part that is money, and H3 recovers only the part that is reporting.

The second is about H2. The true value is zero, and the confidence interval contains zero. That is what should happen: a well-calibrated test, when there is no effect, does not reject the null hypothesis in 95 % of samples. But the estimate is not exactly zero, and the interval lies almost entirely on the negative side. This is not a sign that the placebo lowers the balance, because in the synthetic world it does

not. It is sampling noise: with 150 students per group, the balances of two randomly drawn groups differ a little by chance, and measuring the size of that chance is what the interval is for. Seeing a negative estimate in a world where the truth is zero helps to read the real result calmly, when it comes.

The third is a warning about the dry run itself. It is a single synthetic sample, and in it the H1 estimate came out above the true value. In another sample, it would come out below. The dry run does not prove that the program is always right; it proves that it runs from start to finish, compares the right groups and returns numbers close to the values I put in. The power simulation in Section 9.2, with 1,000 samples per point, is the test that it is right on average. And the real world may differ from the synthetic one in ways neither test covers: reporting bias may show up in the placebo group too, the balances may have a distribution very different from the normal, and money may move between the account and the drawer.

The synthetic data are in `data/simulation/followup_synthetic.csv`, whose first line says they are not measurements, and the result is in `followup_synthetic.json` in the same folder.

9.5 Ethics, registration, costs and timeline

9.5.1 Ethics committee

The first study does not report approval by a research ethics committee (Section 5.5). The new study would not start without it. In Brazil, research with human participants follows Resolution 466/2012 of the National Health Council (Resolução CNS 466/2012) (Brasil. Conselho Nacional de Saúde, 2012), and the project is submitted, through Plataforma Brasil, to a Research Ethics Committee (Comitê de Ética em Pesquisa, CEP), which can forward it to the National Research Ethics Commission (Comissão Nacional de Ética em Pesquisa, CONEP) when the rules require it. The submitted project would include the design, the consent forms, the analysis plan and the way the data would be stored, and recruitment would only start after approval.

This study has two points that call for special care. Most of the participants are minors, and the main outcome is banking data.

9.5.2 Consent and assent

For each student who is a minor, there would be two documents. The parent or guardian signs the free and informed consent form, and the student signs an **assent form**, written in language an 11- or 12-year-old understands, saying what will happen, what will be asked, who will see the data, and that they can leave at any time without losing anything in the PIC. Students who are of age sign their own consent form.

The forms need to state, separately and clearly, three things the first study did not have. That the study will see the account balance, on which dates, and nothing else; individual transactions and purchases are of no interest to the study and would not be requested. That the app's records will be exported, in the tool group. And that the data, without names or any information that identifies the student, will be published together with the code, like the data in this work. Consent to access to the balance must be revocable at any time, and the handling of the data must follow Brazil's data protection law, the Lei Geral de Proteção de Dados Pessoais.

There is also a precaution that comes from the results of the first study. If part of the reported effect was a wish to please the person who ran the training, the design should keep the reports away from that person. The forms would be collected through a form that does not carry the name of the researcher who ran the workshops, and the consent form would say that no answer has any consequence for the scholarship.

9.5.3 Trial registration

Before the draw, the study would be registered in a public trial registry, with the design, the hypotheses, the plan in Section 9.3 and the sample size. The analysis program would also be published, together with its SHA-256, a 64-character code computed from the contents of the file, which changes if a single letter of the program changes (Appendix B). That way, any reader can check that the program that analyzed the data is the same one that was registered before the data existed.

9.5.4 Costs and timeline

I do not have a detailed budget, and I am not going to make up prices. But I can say where the money would go. The tools are free, as in the first study. The largest costs would be the time of whoever runs the workshops, the messages and the data collection, for one year; the partnership with a bank to open the accounts or to report the balances; and, if the committee considers it appropriate, a small reimbursement for the students' time spent on the forms, the same in all three groups.

The timeline would be as follows, in order:

1. write the project, the consent forms and the plan, and submit them to the ethics committee;
2. once approved, agree with the bank and with the program on how to open the accounts and obtain the balances;
3. recruit, collect the consent forms and measure the baseline (initial balance, essential expenses and the characteristics from the first study);
4. rerun the power simulation with the baseline data, register the trial and the program, and only then do the draw;
5. run the workshops and follow the students for 3 months, with monthly forms;

6. measure the balances at 3 months and run the registered program;
7. measure the balances at 12 months and repeat the analysis;
8. publish the de-identified data, the code and the result, whatever it is.

The intervention and the data collection take one year; how long the earlier steps take depends on the committee and the bank, and I could not say before starting.

The last item on the list is the easiest to forget. A registered study that finds no effect has to be published too, or the registration loses its point. If the bank balance shows that the tool changes nothing, that answers the question the first study left open just as much as a positive effect would.

CHAPTER 10

Conclusion

10.1 What I set out to do

I started this work with a simple question: does a free tool for keeping track of money help a low-income student save? The PIC scholars seemed the right place to ask it. They all receive the same R$ 300 a month, and for some of them this may be the first regular income they manage on their own. There were good theoretical reasons to expect a yes. Scarcity narrows attention, present bias pushes saving off to later, and a spreadsheet or an app makes visible the small expenses that drain away from the budget without anyone deciding to spend them (Chapters 2 and 4).

To test it, I randomly assigned 208 scholars to a control group and three tools, followed them for three months, and asked them to report, every month, how much they had saved. The first version of this work, from October 2025, told that story with an ANOVA, a chart of means and a regression on use (Ribeiro, 2025a). This monograph went back to each student's data and redid everything with public code. It also wrote down the model that was implicit in the original explanation, to see whether the data agreed with it, and asked how far we can trust the numbers the students reported themselves.

10.2 What I found

The main result is large and survives every test I ran. In October, the control group reported saving 44.25 % of residual income, and the three tool groups reported between 91.40 % and 101.07 %. Receiving any tool raised reported saving by 52.0 pp (95 % CI from 44.2 to 59.8), about R$ 71 a month. None of the 20,000 redone random draws produced a difference that large, and the estimate holds up across the 108 specifications of the curve in Chapter 6, with Lee bounds, and under the worst assumption for the dropouts. As a statistical result, it is solid.

The rest of the evidence changed how I read that result. The data do not rank the tools: the hybrid had the highest mean, but its lead over the app, 3.2 pp, is far below the 13.5 pp this experiment could detect, and an equivalence test does not show they are the same either, except with a wide margin of 15 points, and only for two of the three pairs (Chapter 7). Use did not go together with saving. The 25 treated students who never logged a single expense reported 96.6 % in October, the same as those who

did log (96.7 %). In my model, this was the most direct test that the tool works by making spending visible, and it failed.

The effect also showed up quickly. Already in August, the first month, 55 % of the October effect was there (Section 6.3). This fits a tool that changes behavior right at the start, but it fits just as well a change in the way students answered the form, which takes no time at all to happen.

The reports themselves do not behave like measures of a habit (Chapter 8). Net of the group means, a student's saving in one month has no correlation with the next month's (0.00). The amounts of money are almost never round numbers, which is rare when people recall sums. And an effect of 2.0 standard deviations is many times larger than the typical effect of financial education on behavior (Fernandes et al., 2014; Kaiser et al., 2022).

The synthetic worlds in Section 8.6 helped me put these pieces together. I generated data in worlds where I knew the truth: in one, the tool works through attention, and those who use it more gain more; in another, every treated student gains the same, whether or not they use it; in yet another, nobody saves more and the treated students only report more. The real data do not look like the attention world, which would produce a steep slope of the gain on use (25.8 points per weekly log, against –1.3 in the data). But they look equally like the other two worlds, as long as the reports are very noisy. In other words, the diagnostics I had do not separate a real effect that does not run through use from an effect that exists only in the reports.

The conclusion, then, is narrower than the one in the first version. Receiving a budgeting tool greatly increased the saving that students *reported*. This experiment cannot say how much of that is money put aside.

10.3 What this means for public policy

Even with that caveat, there are three practical lessons. The first is about cost. The intervention costs almost nothing: free tools and a one-hour online workshop. If at least part of the reported effect is real saving, offering a tool and a short training in programs like the PIC, or in larger programs that support low-income students, such as Pé-de-Meia (Brasil, 2024), may be worth it. But that "if" is exactly what this study does not settle, and I would not recommend a large-scale policy before settling it.

The second is to start with the simplest tool. The data give no reason to prefer the combination of spreadsheet and app to a single tool. A single tool is cheaper to offer and easier to evaluate. Anyone who repeats the idea should start there and add complexity only when there is evidence that it helps.

The third is not to judge the program by engagement with the app. It is tempting to measure the success of a digital tool by the number of logins, entries or days of use, because those numbers show up for free on a dashboard. In this study they

did not go together with reported saving: those who never logged anything reported the same as those who did. A manager who looked only at engagement would reach conclusions that the saving data do not support. What matters is the money, and the money has to be measured directly.

10.4 What I learned about doing research

The part of this work that taught me the most was not in the plan. Redoing the first version from the microdata meant checking, one by one, the 24 numerical claims it made (Appendix C.1). Of the 24, 13 check out, 5 change value without changing the conclusion, 4 point in the right direction without statistical support, and 2 do not reproduce. The descriptive statistics were all right. The problems were in the inferences: a sum of squares that did not match its own table of means and, above all, a slope of saving on use that the data do not show under any of the possible readings.

Those mistakes were mine. The first version had no public code that redid each calculation from the spreadsheet, and so nobody, not even I, found them earlier. The lesson I took from this is about method. In this version, no result is typed by hand: each number in the text comes out of a script, is saved to a file and enters the text through a macro (Appendix B). A single command redoes everything. That does not stop me from making mistakes, but it means any mistake can be found by anyone, including me, a year from now.

I also learned to be suspicious of a good result. When I first saw an effect of 2.0 standard deviations, the natural reaction was to think I had discovered something big. The right reaction would have been to ask why it was so much larger than everything the literature finds. The questions in Chapter 8 about persistence, round values and students who reported saving more than they had left of their stipend should have been asked before any conclusion, not after.

I also learned the value of writing the model down. While the explanation was just a sentence, “the tool makes spending visible”, it seemed confirmed by the main result. Once it became an equation, it started to predict things that could go wrong, such as the relation between use and saving, and one of them did go wrong.

One last thing I learned was to use synthetic data to think. Before this work, I saw simulation as something you do when data are missing. The synthetic worlds in Chapter 8 showed me the opposite: only by generating data in worlds where I knew the answer could I see that my diagnostics did not separate the explanations. The dry run in Chapter 9 uses the same idea before data collection, to test whether the new design separates what the old one did not. It would have been better to do this before the first study, and not after.

10.5 Limitations

Besides the measurement problem, which runs through the whole work, there are five limitations the reader needs to keep in mind. The sample is made up of OBMEP medalists. They are public school students, but they are not typical students: they were selected for their performance in mathematics, and it is reasonable to think they handle numbers and spreadsheets more easily than average. An effect found in them may not repeat in other students.

The follow-up lasted three months. The reported effect was still growing in October, though more slowly, and nothing in the study says whether it would have stopped, continued or vanished once the novelty wore off. Financial habits matter for how long they last, and three months is too little to speak of a habit.

The model and most of the analyses were added after data collection. The first version defined the outcome and some analyses, and all of them were repeated here, but the predictions in Chapter 4, the reliability tests and the synthetic worlds were done already knowing the data. They should be read as exploratory, and I wrote them that way.

There is also a limitation of the design itself, which only became clear to me when I wrote the model. The control group attended the common part of the workshop, but the treated students had a separate room, a new tool and, in the app and hybrid groups, weekly reminders to record spending. The comparison between treated and control measures all of this together. With these data there is no way to know how much of the reported effect would come just from the extra contact with the researcher.

The usage logs are a crude measure. They were counted from screenshots sent by the students, and the file does not distinguish a student who never logged anything from one who did not send the screenshot. Part of the lack of relation between use and saving may come from this noise, although it is hard to explain in this way why those who logged expenses and those who logged nothing reported the same.

10.6 The next step

Chapter 9 describes the experiment that would answer the question this one left open. It has three groups, control, attention placebo and a single tool, and it measures saving through the balance of a bank account at 3 and at 12 months, without asking the student. It still collects the report, now to measure the distance between what the student says and what the bank shows. The analysis plan and the program that will do the calculations are ready before data collection, and the program has already been tested on synthetic data in which I knew the answer.

If that study shows that the balance of the students with a tool grows more than that of the students in the placebo group, the idea that motivated this work gains the

evidence it lacks today. If it shows that the balance does not change and only the report changes, that will be an answer too, and perhaps the most useful one for people who design financial education programs, because it would say that measuring the success of these programs by what participants tell us can lead to wrong conclusions.

I started out wanting to know whether a simple tool helps a student save. This work's answer is partial: the tool changed what the students reported, and the proposed study is the way to find out whether it also changes what they put aside.

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Glossary

The terms below are in alphabetical order. Each one points to the place in the main text where it is defined.

ANCOVA (analysis of covariance)
A regression of the outcome on the group indicators that includes the baseline, Y_{i0}, as a control. It is not needed for the estimate to be unbiased, but it takes the old variation out of the calculation and makes the estimate more precise: the variance falls to about $(1-\rho^2)$ times that of the difference in means (see Sections 3.3 and 5.6).

ANOVA, one-way (F)
A test that asks whether the means of several groups are equal, by comparing the variation between the groups with the variation within them. The result is the F statistic, which stays close to 1 if the means are equal (see Sections 3.4 and 6.2).

Assent form
A document signed by a student who is a minor, written in language they understand, saying what will happen, who will see the data and that they can leave at any time. The parent or guardian signs the consent form (see Section 9.5).

Attention (a_{it})
The fraction, between 0 and 1, of the unplanned purchases m_i that the student notices. The noticed part, $a_{it} m_i$, enters the plan as one more chosen expense. In the model, $a_{it} = a_{0i} + \kappa\, u_{it}$ (see Sections 4.2 and 4.4).

Attention placebo
A group in the proposed experiment that gets the same kind of contact as the tool group (workshop, weekly messages), without the tool and on a subject unrelated to money (see Section 9.1).

Attenuation
Reporting noise always pulls the observed correlation between two measures toward zero. With reports of reliability λ, the correlation between the reports of two months is $\rho\lambda$ (see Section 3.10).

Attrition
The loss of students during the study, which can spoil a well-randomized experiment. If, for example, the control students who saved little dropped out more often, the remaining control group would seem to save more than it really does (see Section 5.7).

Average treatment effect (ATE, τ)

The mean of the individual effects in a group of N students, $\tau = \frac{1}{N}\sum_i [Y_i(1) - Y_i(0)]$. Unlike the individual effect, it can be estimated (see Section 3.1).

Balance

Groups that look alike at baseline. Checking balance means looking at the differences the draw left and asking whether they are the size chance produces or larger (see Sections 3.2 and 5.7).

Baseline

The measures taken before the intervention: the student's characteristics, essential expenses and initial saving. It is used to check balance and to make the estimates more precise (see Section 5.4).

Benjamini and Hochberg (correction of)

A correction for many comparisons that controls the expected *fraction* of false discoveries among the results declared significant. It is less demanding than Holm's and is used in the subgroup breakdowns (see Sections 3.6 and 7.4).

***Bootstrap* (resampling)**

A way to compute a confidence interval by drawing the students of the sample again, with replacement, and redoing the calculation in each draw (see Sections 6.4 and 7.1).

Bounded rationality

People cannot consider every option or process all the available information, so they use simple rules and settle for solutions that are good enough (see Section 2.5).

Causal effect (τ_i)

The effect of the tool on student i, the difference between their two potential outcomes, $\tau_i = Y_i(1) - Y_i(0)$ (see Section 3.1).

Cohen's d

The effect divided by the standard deviation of the outcome, $d = (\bar{Y}_1 - \bar{Y}_0)/\sigma$. It allows comparing studies with different outcomes (see Section 3.5).

Commitment device

A rule or a product that a person chooses today to restrict their own future choices, such as an account from which money can only be taken out after a certain date (see Section 2.3.3).

Completers

The students with all three monthly reports. The estimates are computed on them (see Section 5.4).

Confidence interval (CI)

Combines the estimate and the standard error: the 95 % interval is $\hat{\tau} \pm 1.96 \times \mathrm{SE}(\hat{\tau})$. If the experiment were repeated many times, 95 % of these intervals would contain the true effect (see Section 3.4).

Consumption (c_t)

What a person spends in a period, on anything: a bus fare, a snack, phone credit, a book (see Section 2.1).

Control group

The group that gets no tool. In the proposed experiment, it gets only what everyone gets: the study account, an explanation of how it works and the monthly forms (see Sections 3.1 and 9.1).

Corner solution

The best choice lies at the edge of what is possible, not at the point where the two satisfactions are equal. In the model, the student spends everything they have and plans to save nothing (see Section 4.3).

Difference in differences

The difference between a group and the control group in a month, net of the difference that already existed at baseline (see Section 5.6).

Difference in means ($\hat{\tau}$)

The simplest estimator of the average effect: the mean of the treated group minus that of the control group, $\hat{\tau} = \bar{Y}_1 - \bar{Y}_0$ (see Section 3.3).

Discount factor (δ)

A number between 0 and 1 that gives each future date a smaller weight than the present. If δ is close to 1, the person is patient; if it is small, they are impatient (see Section 2.2).

Discretionary spending (c)

Purchases the student thinks about before making them: the Friday snack, the movie ticket, a game, a present (see Section 4.2).

Dose-response

The relation between "how much treatment" and "how much result", by analogy with medicines. Here the dose is the number of expense logs per week (see Section 7.3).

Effect heterogeneity

The effect being larger for some students than for others (see Section 7.4).

Equivalence margin (m)

The largest difference between two tools that would be considered of no practical importance, for example 10 percentage points of residual income (see Sections 3.8 and 7.2).

Essential expenses (e_i)

Expenses the student does not choose month by month and considers obligatory, such as the bus fare to school or helping out at home (see Sections 2.6 and 4.2).

Eta squared (η^2)
The fraction of the total variation that lies between the groups in an ANOVA (see Section 6.2).

Event study
A regression on the student-month panel that estimates the difference between each group and the control group in each month, to see how the effect built up over time (see Sections 5.6 and 6.3).

Experimenter demand effect
The participant tries to guess what the researcher expects and acts, or answers, accordingly (see Sections 3.10 and 8.1).

Exponential discounting
A model in which the weight of one period ahead is δ, that of two periods is δ^2, and so on: the weight always falls in the same proportion each period (see Section 2.2).

Financial literacy
Knowledge of basic concepts such as interest, inflation and risk diversification (see Section 2.7).

First-order condition
The equation obtained by differentiating satisfaction with respect to c and setting the result to zero. In the model it is $1/c = \beta v$: the student spends up to the point where the last real spent gives them as much as the last real saved (see Section 4.3).

Fixed effect
An intercept of its own for each stratum, student or month, which absorbs any constant mean difference between them (see Section 5.6).

Flow and stock
The month's saving is a flow: how much went into the reserve that month. The total put aside to date, adding up all the months, is a stock, the balance (see Section 2.1).

Fundamental problem of causal inference
For each student, one of the two potential outcomes is observed and the other never is. That is why the effect of a tool on a specific person cannot be observed (see Section 3.1).

Heaping
The tendency of people who report from memory how much they spent or saved to give round values, such as R$ 150, R$ 160 or R$ 200: the values heap up on the round numbers (see Section 8.4).

Heteroskedasticity
A situation in which the spread of the data, or of the error, is not the same for all groups or students. In that case the classical standard error can come out too small or too large (see Sections 3.3 and 5.6).

Holm (method of)

A correction for many comparisons that controls the chance of making any false positive in the set. It orders the m p-values from smallest to largest, compares the smallest with α/m, the next with $\alpha/(m-1)$, and so on, stopping at the first one that does not pass (see Sections 3.6 and 7.1).

Inattention

A situation in which a person takes into account only part of the variables relevant to a decision and gives more weight to the most visible ones. Small, frequent expenses are the typical case of something not very visible (see Section 2.5).

Income (y_t)

The money a person receives in a period, for example a month. For a PIC scholar, the most regular income is the R$ 300 stipend (see Section 2.1).

Intention to treat

An estimate that keeps each student in the group to which they were assigned, whether or not they use the tool. It measures the effect of *offering* the tool and the training, not of using them (see Section 3.3).

Interaction

The difference between the effect in two subgroups, for example girls and boys. Testing the interaction means asking whether this difference is different from zero (see Section 7.4).

Intertemporal choice

Any decision that compares something gained now with something gained later, such as the decision to save (see Section 2.2).

Intraclass correlation (ICC)

The fraction of the total variance that is due to stable differences between people. It answers the question: what fraction of the variation in the reports lies between students, rather than within each one? (see Sections 3.10 and 8.2).

Leak

The unnoticed part of unplanned purchases, $(1 - a_{it})m_i$, which leaves the budget without going through the plan (see Section 4.2).

Lee bounds

An interval that contains the true effect when there is attrition, under the weak assumption that the tool changes dropout in only one direction. The group with fewer dropouts is trimmed until it has the same fraction of completers, first from the top and then from the bottom (see Sections 3.3 and 6.4).

Mental accounting

The set of operations people use to organize, evaluate and keep track of their own money. They split it into accounts by source and by use, and they dislike taking from one account to cover another (see Section 2.4).

Minimum detectable effect (MDE)

The smallest effect the study can detect with 80 % power. With α = 5 %, it is about 2.8 standard errors (see Sections 3.5 and 9.2).

Naive and sophisticated

A naive person has present bias but does not know it and believes they will follow the plan. A sophisticated person knows that their tomorrow self will also prefer the present, and takes that into account when deciding (see Section 2.3.2).

Null hypothesis

The assumption that the true effect is zero, under which the *p*-value is computed (see Section 3.4).

Outcome (S_{it})

The variable the experiment tries to change. Here, the fraction of initial residual income that the student reported having saved in the month, S_{it} = 100 × (saved in month t)/(R$ 300 − e_i). It can exceed 100 (see Section 2.6.1 and Section 5.4).

***p*-value**

The probability of getting, by chance alone, an estimate at least as far from zero as the observed one, assuming the true effect is zero. It is not the probability that the null hypothesis is true, and it does not measure the size of the effect (see Section 3.4).

Posterior

In Bayesian statistics, the final belief, which combines the prior with the data. With a normal prior and a normal estimate, its mean is a weighted average of the two, with more weight on the more precise one (see Sections 3.9 and 8.5).

Potential outcomes ($Y_i(1)$, $Y_i(0)$)

$Y_i(1)$ is the outcome student i would have if they received the tool, and $Y_i(0)$ is the outcome they would have without it. Both exist before the draw; the draw only decides which of them will be seen (see Section 3.1).

Power

The probability that a test rejects the null hypothesis when the true effect has a specified size. It depends on the size of the effect, the spread of the outcome, the number of students and α (see Sections 3.5 and 7.2).

Preference reversal

Changing one's mind between two options only because the earlier one has reached the present, without the amounts or the gap between them having changed. Exponential discounting rules it out; quasi-hyperbolic discounting produces it (see Section 2.3.1 and Appendix A.3).

Preregistered analysis plan

A document, published before the data exist, that says which hypotheses will be

tested, with which specification, on which sample and with which correction for multiple testing (see Sections 3.6 and 9.3).

Present bias (β)

The tendency to give the current moment a disproportionate weight relative to all future moments. The difference between today and tomorrow weighs much more than the difference between a year from now and a year and a day from now. The smaller β, the greater the bias (see Sections 2.3 and 4.3).

Prior

In Bayesian statistics, the initial belief about the size of the effect, before seeing the data. Here, a normal distribution centered on the typical effect in the literature (see Sections 3.9 and 8.5).

Probability of superiority

The chance that a treated student drawn at random reported more saving than a control student drawn at random. It would be 50 % if the tools made no difference (see Section 6.4).

Quantile effects

They compare the 10th percentile of the treated with the 10th percentile of the control group, then the 20th with the 20th, and so on. They show whether the effect lifted a few students a lot or all of them a little (see Section 6.4).

Quasi-hyperbolic discounting (beta-delta model)

A model of present bias that adds to exponential discounting a parameter β, between 0 and 1, that multiplies the whole future. Between the present and the next period the weight falls from 1 to $\beta\delta$; between two future periods, only by δ (see Section 2.3).

Randomization inference

Using the random draw itself as the source of uncertainty: redoing the draw many times on the computer, with each student's saving held fixed, and seeing what fraction of the differences is at least as large as the observed one, $p = (1 + K)/(N + 1)$ (see Sections 3.7 and 6.4).

Regression to the mean

The tendency of people who reported extreme values in one month to report values closer to the mean in the next. It affects treated and control students alike (see Section 4.5).

Relative change

The change divided by the starting value. A student who went from 40 % to 90 % had a change of 50 points and a relative change of 125 % (see Section 7.3).

Reliability (λ)

The fraction of the variance of the reports that comes from the true signal, λ =

$\mathrm{Var}(s)/[\mathrm{Var}(s) + \mathrm{Var}(\eta)]$. It equals 1 if there is no reporting error and approaches 0 when the error dominates (see Sections 3.10 and 4.6).

Reported saving (r_{it})
What the student wrote on the monthly form as the amount saved. The study observes this report, not true saving: $r_{it} = s_{it} + b\,T_i + \eta_{it}$ (see Sections 3.10 and 4.6).

Reporting bias (b)
A shift, in the same direction for all treated students, between what they saved and what they reported. It enters the difference in means in full, but it does not change the slope on use (see Section 4.6).

Reporting error (η_{it})
The random difference between the true value and the reported one, coming from poor memory, a guess or a wrong calculation. It has mean zero and does not move the means, but it reduces the persistence of the reports (see Section 4.6).

Reporting gap
In the proposed experiment, the difference between what the student says they saved and how much their bank balance changed (see Section 9.1).

Residual income (R_i)
What is left of the stipend after the essential expenses at baseline, $R_i = R\$\,300 - e_i$. It is the money the student actually makes decisions about (see Sections 2.6 and 4.2).

Salience (Δv)
The tool's second path in the model: seeing the goal every week makes the future feel closer and raises the value of each real put aside, from v to $v + \Delta v$. It does not depend on the student recording anything (see Section 4.4).

Sampling variability
The variation of the estimate from one draw to another. If the draw were redone, the groups would have other students and the difference in means would come out a little different (see Section 3.4).

Saving (s_t)
What is left of income after consumption in the period, $s_t = y_t - c_t$. The equation is an accounting identity and says nothing about behavior (see Section 2.1).

Scarcity
The lack of a resource, whether money, time or food, captures attention. People with little money focus their minds on the most urgent problem, and less mental capacity is left for everything else, including planning (see Section 2.5.1).

Selection bias
How much more those who choose a treatment would have than the others *even without it*, $\mathbb{E}[Y_i(0) \mid \text{uses}] - \mathbb{E}[Y_i(0) \mid \text{does not use}]$. The random draw brings it to zero; the choice of how much to use the tool brings it back (see Sections 3.1 and 7.3).

Shock (ε_{it})

A variation in the month, positive or negative, that the student does not control: a birthday with a present to buy, a phone that breaks, money a grandmother gave. In the model it has mean zero and is independent from one month to the next (see Section 4.2).

Social desirability bias

The tendency of people answering questions about their own behavior to give the answer that seems most acceptable. Since saving is seen as a virtue, a student may round their saving up (see Section 3.10).

Specification curve

Estimates the effect under every defensible combination of analysis choices and shows the distribution of the estimates. If the effect shows up in almost all of them, the result does not depend on a hand-picked path (see Sections 3.6 and 6.4).

Standard deviation (SD)

Measures how much the values of a variable spread around the mean. Measuring an effect in standard deviations allows comparing studies with different outcomes (see Section 2.7).

Standard error (SE)

The standard deviation of sampling variability: it says how far, typically, an estimate lies from the true value. For the difference between two means, $\mathrm{SE}(\hat{\tau}) = \sqrt{\sigma_1^2/n_1 + \sigma_0^2/n_0}$. Some formulas in the text write EP, the standard error, in place of SE (see Section 3.4).

Standard error clustered by student

A standard error that treats the several observations of the same student as one block, rather than as independent pieces of information (see Section 5.6).

Standard error, robust (HC2)

A standard error that estimates the spread from each student's own residuals, without assuming it is the same for everyone. The HC2 version is a correction for small samples (see Sections 3.3 and 5.6).

Standardized difference

The difference between the mean of a treated group and that of the control group, divided by the standard deviation of the characteristic. It does not depend on sample size; values below 0.1 in absolute value are usually treated as negligible (see Section 5.7).

Statistically significant

By convention, a result with $p < 0.05$ (see Section 3.4).

Stratification

Splitting the sample into blocks of similar students before the draw and drawing

within each block. In the study, the blocks are the school grades (see Sections 3.2 and 5.3).

Synergy
In the strong sense, the whole being larger than the sum of the parts: the effect of receiving both tools together would be larger than that of the spreadsheet plus that of the app (see Section 7.1).

Synthetic data
Data manufactured by a program, in a world where the researcher chooses the answer. They are used to see what each explanation would produce and to test an analysis program before data collection. They are simulated, not measurements (see Sections 8.6 and 9.4).

***t* test**
Compares the means of two groups: it divides the difference by its standard error and asks whether the ratio is too large to be chance. Welch's version does not assume equal variances (see Sections 3.4 and 7.1).

Test-retest correlation
The correlation between two measures of the same thing. It is used to measure reliability without knowing the true value (see Section 3.10).

Time consistency
A property of exponential discounting: all that matters is the distance between two dates, not how far they are from today. That is why the plan a person makes today is the same one they will want to follow when the day comes (see Section 2.2).

TOST (two one-sided tests)
An equivalence test: one test asks whether the true difference is greater than $-m$, the other whether it is smaller than $+m$, and equivalence is declared only if both reject. It is the same as requiring the 90 % interval to fit entirely between $-m$ and $+m$ (see Sections 3.8 and 7.2).

Type I and type II errors
A type I error is declaring an effect where there is none; its probability is the level of the test, α. A type II error is failing to find an effect that exists (see Section 3.5).

Unbiased
An estimator whose mean, if the experiment were repeated many times with new draws, would equal the true value (see Section 3.3).

Use intensity (u_{it})
How much the student uses the tool. In the study, expense logs per week (see Sections 4.4 and 7.3).

Worst case (imputation)
Giving zero saving to every treated student who dropped out and the highest saving observed in the control group to every control dropout. It is the scenario least favorable to the effect that the data allow (see Section 6.4).

APPENDIX A

Derivations

This appendix shows, step by step, the calculations that the main text only cites or summarizes: those of the model in Chapter 4, the preference reversal in Chapter 2, the persistence and reliability of the reports in Chapter 8, and the statistics in Chapters 3 and 9. Every step was checked with symbolic algebra in *sympy*, Python's symbolic mathematics library, by the script `code/17_check_derivations.py`; the output, with the expression *sympy* obtained at each step, is in `data/checks/appendix_a_derivations.txt`. The script's blocks run from A.1 to A.10 and do not match the sections of this appendix one to one, because some sections combine two blocks; so I give the block in parentheses, as in "(block A.4)", at the point where each one starts. The numbers that appear here follow from the formulas and from parameters chosen for illustration; they are not data from the study.

A.1 Planned spending

(Block A.1.) The student chooses discretionary spending $c > 0$ to maximize the month's satisfaction, Equation (4.3), with saving given by Equation (4.2). Substituting one into the other,

$$U(c) = \log c + \beta v \left[R - c - (1 - a)\, m + \varepsilon \right], \tag{A.1}$$

where I drop the indices i and t and assume $\beta > 0$ and $v > 0$. The terms R, $(1 - a)m$ and ε do not depend on c. Differentiating with respect to c, the derivative of $\log c$ is $1/c$ and that of $-\beta v\, c$ is $-\beta v$:

$$\frac{\mathrm{d}U}{\mathrm{d}c} = \frac{1}{c} - \beta v. \tag{A.2}$$

This is the expression the script obtains, `-beta*v + 1/c`. Setting it to zero gives $1/c = \beta v$, and inverting both sides,

$$c^{*} = \frac{1}{\beta v}. \tag{A.3}$$

The solution is unique, because $1/c$ is strictly decreasing for $c > 0$ and crosses the constant βv only once.

To know whether c^{*} is a maximum, I differentiate again. The derivative of $1/c$ is $-1/c^2$, and that of $-\beta v$ is zero:

$$\frac{\mathrm{d}^2 U}{\mathrm{d}c^2} = -\frac{1}{c^2}. \tag{A.4}$$

It is negative for every $c > 0$, so U is concave and the point where the first derivative vanishes is a maximum. At c^*, $1/c^2 = (\beta v)^2$, and

$$\left.\frac{\mathrm{d}^2 U}{\mathrm{d}c^2}\right|_{c=c^*} = -(\beta v)^2 < 0, \tag{A.5}$$

as in the script's output (`-beta**2*v**2`).

Two remarks. First, R does not appear in Equation (A.3): since utility is linear in saving, each extra real of income goes entirely into s. Second, the calculation assumes the student can spend c^*, that is, that $R-(1-a)m \geq c^*$. If $R-(1-a)m < c^*$, the derivative in Equation (A.2) is positive over the whole feasible range, $0 < c \leq R-(1-a)m$, because there $1/c > 1/c^* = \beta v$. Satisfaction keeps rising up to the edge, and the best choice is to spend everything available, $c = R - (1 - a)m$, with zero planned saving. This is the corner solution of Section 4.3.

A.2 The effect of the tool and the role of present bias

Saving at the optimum (block A.2). Plugging Equation (A.3) into Equation (4.2), with attention a_0 and value of saving v, saving without the tool is

$$s(a_0, v) = R - \frac{1}{\beta v} - (1 - a_0)\, m + \varepsilon. \tag{A.6}$$

With the tool, attention becomes $a_0 + \kappa u$ (Equation (4.7)) and the value of saving becomes $v + \Delta v$. With the same shock ε, and assuming the student stays away from the corner,

$$s(a_0 + \kappa u,\ v + \Delta v) = R - \frac{1}{\beta(v + \Delta v)} - (1 - a_0 - \kappa u)\, m + \varepsilon. \tag{A.7}$$

Subtracting. In the difference between Equation (A.7) and Equation (A.6), R and ε cancel. The planned spending terms give

$$-\frac{1}{\beta(v + \Delta v)} + \frac{1}{\beta v}, \tag{A.8}$$

and the leak terms give

$$-(1 - a_0 - \kappa u)\, m + (1 - a_0)\, m = \kappa u\, m. \tag{A.9}$$

Adding the two parts,

$$\Delta s = \kappa\, u\, m + \left[\frac{1}{\beta v} - \frac{1}{\beta(v + \Delta v)}\right], \tag{A.10}$$

which is Equation (4.8).

Closed form of the salience term. Putting the two fractions in the brackets over the common denominator $\beta v(v + \Delta v)$,

$$\frac{1}{\beta v} - \frac{1}{\beta(v + \Delta v)} = \frac{(v + \Delta v) - v}{\beta v(v + \Delta v)} = \frac{\Delta v}{\beta\, v\,(v + \Delta v)}. \tag{A.11}$$

With β, v and Δv positive, the term is positive. Combining the two parts of Equation (A.10) over the same denominator gives the form the script prints for Δs:

$$\Delta s = \frac{\Delta v + \beta\kappa\, m\, u\, v\,(v + \Delta v)}{\beta\, v\,(v + \Delta v)}, \tag{A.12}$$

since the first term of Equation (A.10), multiplied and divided by $\beta v(v + \Delta v)$, gives $\beta\kappa muv(v + \Delta v)/[\beta v(v + \Delta v)]$.

Slope on use. Only the first part of Equation (A.10) depends on u, and it is linear in u:

$$\frac{\partial \Delta s}{\partial u} = \kappa\, m, \tag{A.13}$$

which is positive if $\kappa > 0$ and $m > 0$. This is Prediction P2. At $u = 0$, the first part vanishes and Δs reduces to the salience term in Equation (A.11).

Salience and present bias (block A.3). I write the salience term as β^{-1} times a quantity that does not depend on β, $K = \Delta v/[v(v + \Delta v)]$. The derivative of β^{-1} is $-\beta^{-2}$, so

$$\frac{\partial}{\partial \beta}\left[\frac{\Delta v}{\beta\, v\,(v + \Delta v)}\right] = -\frac{\Delta v}{\beta^2\, v\,(v + \Delta v)} < 0. \tag{A.14}$$

The salience term falls as β rises; that is, it is larger for people with more present bias. Since it is proportional to $1/\beta$, the dependence is easy to read: going from $\beta = 0.8$ to $\beta = 0.6$ multiplies the term by $0.8/0.6 = 4/3$, which is the ratio between the R$ 20 and the R$ 15 in Example 4.1. The attention term, κum, does not depend on β.

A.3 Preference reversal in the β-δ model

(Block A.4.) In the discounting model with present bias (Laibson, 1997) (Section 2.3), an amount z received k periods from now is worth today

$$V(z, k) = \begin{cases} z, & k = 0, \\ \beta\, \delta^k\, z, & k \geq 1, \end{cases} \tag{A.15}$$

where $\delta < 1$ is the per-period discount and $\beta < 1$ is the extra discount applied to everything that is not immediate. Consider the choice between x reais in one period and $y > x$ reais one period later. Here the period is a week.

Both dates near. If the first option is today, it is worth x, and the second, received a week from now, is worth

$$V(y, 1) = \beta\,\delta\,y. \tag{A.16}$$

Both dates far. If the first option is $s \geq 1$ weeks from now and the second is $s + 1$ weeks from now, both get the factor β, and the ratio of the values is

$$\frac{V(y, s+1)}{V(x, s)} = \frac{\beta\,\delta^{s+1}\,y}{\beta\,\delta^{s}\,x} = \frac{\delta\,y}{x}. \tag{A.17}$$

The β cancels, and $\delta^{s+1}/\delta^{s} = \delta$, so the ratio depends neither on β nor on how far off the dates are.

With numbers. Take $x = R\$\,100$, $y = R\$\,110$, $\beta = 0.7$ and $\delta = 0.99$ per week (illustrative). Today, Equation (A.16) gives

$$\beta\,\delta\,y = 0.7 \times 0.99 \times 110 = 0.693 \times 110 = 76.23 < 100, \tag{A.18}$$

and the student keeps the R$ 100 today. With both dates far off, Equation (A.17) gives

$$\frac{\delta\,y}{x} = \frac{0.99 \times 110}{100} = \frac{108.9}{100} = 1.089 > 1, \tag{A.19}$$

and the same student, looking from a distance, prefers to wait for the R$ 110. The choice flips when the nearer date arrives at today, without any new information having appeared. This reversal is what makes a student with present bias plan to save "starting next month" and then, when the month comes, put it off again. The script checks both numbers as exact fractions, 7623/100 and 1089/1000.

A.4 Persistence of the reports and the bound on reliability

The model (block A.5). I measure each student relative to the mean of their group in the month, which removes the differences between groups and between months, including the reporting bias b in Equation (4.10), which is constant within the treated group. The deviation of true saving is

$$s_{it} = \theta_i + \varepsilon_{it}, \tag{A.20}$$

where θ_i sums up in one number the effect of the student's fixed characteristics (residual income, present bias, leaks and initial attention, which in Equation (4.6) enter as $R_i - 1/(\beta_i v) - (1 - a_{0i})m_i$), with variance σ_θ^2, and ε_{it} is the month's shock, with variance σ_ε^2. The deviation of the report is

$$r_{it} = s_{it} + \eta_{it} = \theta_i + \varepsilon_{it} + \eta_{it}, \tag{A.21}$$

with a reporting error of variance σ_η^2. I assume that θ_i, ε_{it} and η_{it} are independent of one another, and that shocks and errors in different months are too.

Variances. Variances of independent terms add up:

$$\mathrm{Var}(s_{it}) = \sigma_\theta^2 + \sigma_\varepsilon^2, \qquad \mathrm{Var}(r_{it}) = \sigma_\theta^2 + \sigma_\varepsilon^2 + \sigma_\eta^2. \tag{A.22}$$

Covariance between two months. For $t \neq t'$, I expand $\mathrm{Cov}(r_{it}, r_{it'})$ into nine terms. Only the term $\mathrm{Cov}(\theta_i, \theta_i) = \sigma_\theta^2$ survives, because all the others involve at least one shock or error that is independent of the rest:

$$\mathrm{Cov}(r_{it}, r_{it'}) = \mathrm{Cov}(s_{it}, s_{it'}) = \sigma_\theta^2. \tag{A.23}$$

Correlations. Dividing the covariance by the variance (which is the same in both months), the persistence of true saving and that of the reports are

$$\rho \equiv \mathrm{corr}(s_{it}, s_{it'}) = \frac{\sigma_\theta^2}{\sigma_\theta^2 + \sigma_\varepsilon^2}, \qquad \mathrm{corr}(r_{it}, r_{it'}) = \frac{\sigma_\theta^2}{\sigma_\theta^2 + \sigma_\varepsilon^2 + \sigma_\eta^2}. \tag{A.24}$$

Reliability is the fraction of the variance of the reports that comes from true saving:

$$\lambda \equiv \frac{\mathrm{Var}(s_{it})}{\mathrm{Var}(r_{it})} = \frac{\sigma_\theta^2 + \sigma_\varepsilon^2}{\sigma_\theta^2 + \sigma_\varepsilon^2 + \sigma_\eta^2}. \tag{A.25}$$

The product. Multiplying the two, the factor $\sigma_\theta^2 + \sigma_\varepsilon^2$ cancels:

$$\rho\,\lambda = \frac{\sigma_\theta^2}{\sigma_\theta^2 + \sigma_\varepsilon^2} \cdot \frac{\sigma_\theta^2 + \sigma_\varepsilon^2}{\sigma_\theta^2 + \sigma_\varepsilon^2 + \sigma_\eta^2} = \frac{\sigma_\theta^2}{\sigma_\theta^2 + \sigma_\varepsilon^2 + \sigma_\eta^2} = \mathrm{corr}(r_{it}, r_{it'}). \tag{A.26}$$

This is Equation (4.13); the script checks that $\mathrm{corr}(r) - \rho\lambda$ simplifies to zero. An example with round numbers helps to see the size of the effect. If $\sigma_\theta^2 = \sigma_\varepsilon^2 = 1$ and $\sigma_\eta^2 = 8$ (illustrative), true saving has $\rho = 1/2$, reliability is $\lambda = 2/10$ and the correlation of the reports is 1/10: an individual trait that explains half of the variation in saving shows up in the reports as an almost invisible correlation. Note that the derivation holds for any θ_i: Prediction P5 only needs students to differ, $\sigma_\theta^2 > 0$, for $\rho > 0$.

The bound (block A.6). If I observe a correlation r between the reports and assume a value for ρ, Equation (A.26) gives $\rho\lambda = r$, and solving for λ,

$$\lambda = \frac{r}{\rho}. \tag{A.27}$$

If the true correlation of the reports does not exceed the upper limit $\bar{r}$ of its 95 % confidence interval, which is the case in 97.5 % of samples if the interval is well calibrated, reliability satisfies

$$\lambda \leq \frac{\bar{r}}{\rho}. \tag{A.28}$$

For example, if $\bar{r}$ were 0.1 and $\rho = 0.5$, at most 20 % of the variance of the reports would be signal. With the observed limit, 0.08, the bounds for $\rho = 0.3$, 0.5 and 0.7 are 0.27, 0.16 and 0.12 (estimated; Section 8.2). The more persistent true saving is assumed to be, the lower the reliability compatible with reports that do not persist.

A.5 What reporting bias does to the effect and to the slope

(Block A.7.) I use Equation (4.10), $r = s + b\,T + \eta$, with $\mathbb{E}[\eta \mid T, u] = 0$. For a treated student with use u, true saving is that of the control group plus the effect of the tool. I write the effect as $\tau + \kappa_m u$, where τ is the part that does not depend on use (salience) and $\kappa_m = \kappa m$ is the attention slope (Equation (A.10)):

$$s = s_0 + (\tau + \kappa_m u)\, T, \tag{A.29}$$

where s_0 is the saving the student would have without the tool. I assume that the mean of s_0 is the same among treated students with use u and in the control group, and I normalize that mean to zero. Then

$$\mathbb{E}[r \mid T = 1, u] = (\tau + \kappa_m u) + b, \tag{A.30}$$

$$\mathbb{E}[r \mid T = 0] = 0, \tag{A.31}$$

and the difference is

$$\mathbb{E}[r \mid T = 1, u] - \mathbb{E}[r \mid T = 0] = b + \kappa_m u + \tau. \tag{A.32}$$

Differentiating with respect to each parameter,

$$\frac{\partial}{\partial u}\left(b + \kappa_m u + \tau\right) = \kappa_m, \qquad \frac{\partial}{\partial b}\left(b + \kappa_m u + \tau\right) = 1. \tag{A.33}$$

The first says that b does not enter the slope on use; the second, that each real of bias enters the difference between treated and control in full. With $\tau = \kappa_m = 0$, no extra saving at all, Equation (A.32) still shows an effect of b.

Two assumptions hold the calculation up. The first is that b does not depend on u; if it did, say $b = b_0 + b_1 u$, the slope would become $\kappa_m + b_1$, and a reporting bias could create a relation with use without any saving at all. The second is that the mean of s_0 is the same at every level of use. The random draw guarantees this for T, but not for u, which each student chose; that is why the slope on use is not a causal estimate (Section 7.3).

A.6 How much the baseline reduces the variance

(Block A.8.) Let Y be the outcome (October saving) and X the same measure at baseline, with the same variance σ^2 in both and correlation ρ between them. Here ρ is the correlation between the baseline and the outcome, not the persistence of Appendix A.4, although the two are close relatives. With n students in each of two groups, the simple difference in means has variance

$$\operatorname{Var}\left(\bar{Y}_1 - \bar{Y}_0\right) = \frac{\sigma^2}{n} + \frac{\sigma^2}{n} = \frac{2\sigma^2}{n}. \tag{A.34}$$

The ANCOVA, a regression of the outcome on the group and the baseline (Equation (5.2)), compares the groups using the part of Y that X does not explain. The coefficient of the regression of Y on X is $\operatorname{Cov}(X, Y)/\operatorname{Var}(X) = \rho\sigma^2/\sigma^2 = \rho$, and the residual $Y - \rho X$ has variance

$$\operatorname{Var}(Y - \rho X) = \sigma^2 - \frac{\operatorname{Cov}(X, Y)^2}{\operatorname{Var}(X)} = \sigma^2 - \frac{(\rho\sigma^2)^2}{\sigma^2} = \sigma^2(1 - \rho^2). \tag{A.35}$$

The second equality uses $\operatorname{Cov}(X, Y) = \rho\sigma^2$. The script prints the same result in the factored form $-\sigma^2(\rho - 1)(\rho + 1)$, which equals $\sigma^2(1 - \rho^2)$. Replacing σ^2 by this variance in Equation (A.34), in large samples,

$$\operatorname{Var}_{\text{ANCOVA}} = \frac{2\sigma^2(1 - \rho^2)}{n}, \qquad \frac{\operatorname{Var}_{\text{ANCOVA}}}{\operatorname{Var}\left(\bar{Y}_1 - \bar{Y}_0\right)} = 1 - \rho^2. \tag{A.36}$$

With $\rho = 0.5$, the variance falls to 0.75 of the original, and the standard error to $\sqrt{0.75} \approx 0.87$. With ρ near zero, the ratio is near 1 and the baseline does not help. That is what happens when the reports do not persist: controlling for initial saving barely changes the precision (Section 8.2).

A.7 Minimum detectable effect

(Block A.9.) Two groups of sizes n_1 and n_2 have an outcome with standard deviation σ. The difference in means has standard error

$$\text{EP} = \sqrt{\frac{\sigma^2}{n_1} + \frac{\sigma^2}{n_2}} = \sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}. \tag{A.37}$$

In this section I write EP, the standard error, as in the formula above. The test rejects the hypothesis of zero effect when the estimate divided by the standard error exceeds a critical value z_α (1.96 for a two-sided test at 5 %; I ignore the tail on the opposite side, which weighs almost nothing). I want the smallest true effect, the minimum detectable effect (MDE), that the test rejects with probability equal to the desired

power. I call z_β the quantile of the standard normal associated with that power (0.84 for 80 %); the β here is the traditional name for the probability of a type II error, not present bias. If the true effect is D, the estimate divided by the standard error is normal with mean D/EP and variance 1, and the power is

$$P\left(Z > z_\alpha - \frac{D}{\mathrm{EP}}\right), \tag{A.38}$$

with Z standard normal. For this probability to equal the desired power, $P(Z > -z_\beta)$, the two thresholds must coincide: $z_\alpha - D/\mathrm{EP} = -z_\beta$. Solving for D,

$$\mathrm{MDE} = (z_\alpha + z_\beta)\,\sigma\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}. \tag{A.39}$$

The script checks that, with this value, $z_\alpha - \mathrm{MDE}/\mathrm{EP} + z_\beta$ is zero. With $n_1 = n_2 = 52$, 5 % and 80 %, $z_\alpha + z_\beta = 2.80$ and $\sqrt{2/52} \approx 0.196$, which gives an MDE of about $0.55\,\sigma$. The values computed with the standard deviation of the data are in Sections 3.5 and 9.2. Since the MDE falls with the square root of the group size, halving it takes four times as many students.

A.8 Posterior mean with a skeptical prior

(Block A.10.) A reader starts out believing that the effect θ is normal with mean μ_0 and standard deviation s_0, the prior (Section 3.9). The study gives an estimate $\hat{b}$ with standard error se, and I treat $\hat{b}$ as normal around θ. The notation θ here is that of the effect, not of the individual trait in Appendix A.4. By Bayes' rule, the posterior density is proportional to the product of the prior and the likelihood, and its logarithm is, up to a constant,

$$\ell(\theta) = -\frac{(\theta - \mu_0)^2}{2s_0^2} - \frac{(\hat{b} - \theta)^2}{2\,\mathrm{se}^2}. \tag{A.40}$$

Since ℓ is a parabola in θ, the posterior is normal, and its mean coincides with the maximum point. Differentiating,

$$\ell'(\theta) = \frac{\mu_0 - \theta}{s_0^2} + \frac{\hat{b} - \theta}{\mathrm{se}^2}. \tag{A.41}$$

Setting this to zero and collecting the terms in θ,

$$\theta\left(\frac{1}{s_0^2} + \frac{1}{\mathrm{se}^2}\right) = \frac{\mu_0}{s_0^2} + \frac{\hat{b}}{\mathrm{se}^2}, \tag{A.42}$$

and dividing,

$$\mu_1 = \frac{\mu_0/s_0^2 + \hat{b}/\mathrm{se}^2}{1/s_0^2 + 1/\mathrm{se}^2} = \frac{\hat{b}\, s_0^2 + \mu_0\, \mathrm{se}^2}{s_0^2 + \mathrm{se}^2}. \tag{A.43}$$

The second form, which is the script's, comes from multiplying numerator and denominator by $s_0^2\,\mathrm{se}^2$. It shows that μ_1 is a weighted average: the estimate gets weight $s_0^2/(s_0^2 + \mathrm{se}^2)$ and the prior gets the rest. The narrower the prior (smaller s_0) or the less precise the estimate (larger se), the closer the posterior stays to μ_0.

The second derivative of ℓ gives the precision, the inverse of the posterior variance:

$$-\ell''(\theta) = \frac{1}{s_0^2} + \frac{1}{\mathrm{se}^2} = \frac{1}{s_1^2}. \tag{A.44}$$

Precisions add up: the posterior is always narrower than either the prior or the estimate alone. The 95 % posterior interval is $\mu_1 \pm 1.96\, s_1$. The calculations with the skeptical prior centered at 2.6 points and the estimate of 52.0 points (standard error 4.0) are in Section 8.5. As there, this calculation takes the estimate as it is: a reporting bias b (Appendix A.5) shifts $\hat{b}$ and, with it, the whole posterior, and no prior on θ corrects that.

APPENDIX B

Code

Every number, figure and table in this work that comes from the data, from an estimate or from a simulation is produced by a Python script in the folder `code/`. This appendix is the manual for that code. It explains how to redo everything from the original spreadsheet (Appendix B.1), where each file lives (Appendix B.2), how a computed number reaches the text without being typed (Appendix B.3), what each script reads and writes (Appendix B.4) and, at the end, shows two pieces line by line: the analysis program for the study proposed in Chapter 9 and the core of the regressions in Chapter 6 (Appendix B.5). The code, its file names, variable names and comments are in English, and the listings below are reproduced as they are. The commands below are run from the root of the repository, the folder that contains `code/`, `data/` and `monograph/`.

B.1 How to reproduce everything

B.1.1 The environment

I ran the code with Python 3.13 (3.13.17). The libraries and their versions are pinned in the file `requirements.txt`, at the root: numpy 2.5.3, pandas 3.0.5, scipy 1.18.1, statsmodels 0.15.0, matplotlib 3.11.2, openpyxl 3.1.5 (which reads the Excel spreadsheet) and sympy 1.14.0 (which checks the derivations in Appendix A). An identical environment can be set up with

```
python3 -m pip install -r requirements.txt
```

No script uses the internet. The figures use the Source Sans Pro font, which comes with TeX Live, the same typeface as the headings and tables; if it is not installed, matplotlib uses DejaVu Sans and only the look of the figures changes. I compiled the monograph with TeX Live 2023 and `latexmk` 4.83, with the KOMA-Script class `scrbook`.

B.1.2 One command

The script `code/reproduce_all.py` redoes, in order, everything the other scripts generate:

```
python3 code/reproduce_all.py
```

It runs each script as a new process, in the order of Table B.2, and stops at the first one that fails, showing the end of the error message. The order matters in three places. The analysis scripts read the cleaned data that `01` writes, so it comes first. Script `13` draws the figures from the files that scripts `02` to `12` leave in `results/csv/`, so it comes after them. And `11` runs before `10`, because `10` builds the table of predictions with numbers that `11` computes. The last step is `build_numbers.py`, which gathers all the numbers into a single file for LaTeX.

The whole run takes a few minutes. The slow part is the random draws: randomization inference, the *bootstraps* and the simulations in Chapters 8 and 9. To test the code quickly, for example after changing a figure, there is a fast mode:

```
QUICK=1 python3 code/reproduce_all.py
```

With `QUICK=1`, each count of draws falls to a twentieth of its normal value, with a minimum of 200. The figures and tables come out with the same shape, but the numbers that depend on random draws change a little, so this run is not fit for the final text. Each count can also be changed on its own through an environment variable; for example, `NSIMFU` sets the number of synthetic samples per point in the power simulation of Chapter 9.

B.1.3 Seeds

Every random draw in the code uses a random number generator with a fixed seed. The base seed, defined in `code/common.py`, is `20250801`, the date of the first day of the intervention; each script adds its own offset to it (for example, the function `rng(60)` in script `16`), so that no two scripts use the same sequence. This is why running `reproduce_all.py` twice on the same machine gives the same files, byte for byte. On another machine, with other library versions, the last digits of some results may change; the numbers in the text, which have at most two decimal places, should not.

B.1.4 The manifest

At the end of the run, `reproduce_all.py` writes the file `data/sha256_outputs.txt`. Each line has the SHA-256 of a file and its path. The SHA-256 is a fingerprint of the contents: a code of 64 hexadecimal characters, computed from every byte of the file, that changes completely if a single byte changes. The manifest covers the original spreadsheets, the CSV, JSON and TXT files in `data/processed/`, `data/simulation/` and `data/checks/` (the Markdown reports are left out), the CSV, JSON, `.tex` and `.npz` files in `results/` (including the numbers and the tables) and the figures in PNG. The PDF versions of the figures are left out; the figures are checked through the PNG files.

To check a reproduction, it is enough to run, from the root,

```
sha256sum -c data/sha256_outputs.txt
```

The program `sha256sum` recomputes the fingerprint of each file and says which ones changed. On the same machine, none should. The first line of the manifest is that of the original spreadsheet, `data/raw/dados_PIC_208.xlsx`; if it does not check out, the problem is in the input data, not in the code.

B.1.5 The monograph

With the numbers, tables and figures ready, the Portuguese monograph compiles from the folder `monograph_pt/`, and this English version from `monograph/`, both with `latexmk -pdf main.tex`. It reads `../figures/`, `../results/numbers.tex`, `../results/tables/` and, in this appendix, `../code/`. A chapter can be compiled on its own with `python3 code/compile_chapter.py` followed by the file name, such as `ch09_proposal`.

B.2 Map of the folders

Table B.1 shows where everything lives. The rule is simple: `data/` holds data, input or generated; `results/` holds what comes out of the analyses; `figures/` holds the figures. Nothing in `data/processed/`, `results/` or `figures/` should be edited by hand, because the next run wipes out the edit.

B.3 How the numbers reach the text

No result in this work was typed into the text by hand. A number's path has four steps, which I follow here with an example: the pooled effect of the tools on reported saving in October, which appears in the text as 52.0 pp.

The first step happens inside the script that does the calculation. In `03_main_effects.py`, right after the regression, the line

```
record(S, "bPooled", pooled.params["treated"], ".1f")
```

stores the coefficient under the name `bPooled`, with one decimal place. The function `record`, defined in `common.py`, writes the value and the formatted text to the file `results/numbers/03_main_effects.json`, one per script. At the start of each run, the script deletes its own numbers file (`reset_numbers`), so that a number that is no longer computed is not left behind.

The second step is `build_numbers.py`. It reads all the files in `results/numbers/`, checks that no key was written by two different scripts (if one was, it stops with

Table B.1. Map of the repository folders and what each one holds.

Folder	What it holds
`code/`	the numbered scripts `01` to `18`, the shared module `common.py`, the program `analyze_followup.py`, `build_numbers.py`, `reproduce_all.py` and the tools for the monograph
`data/raw/`	the study spreadsheet (`dados_PIC_208.xlsx`, one student per row, identified only by a code) and the tables of the first version transcribed into a spreadsheet, kept for reference; the analysis does not read the second one
`data/processed/`	`students.csv` (one student per row) and `students_panel.csv` (student by month, completers only), written by script `01`
`data/simulation/`	synthetic data and simulation summaries from scripts `14`, `15` and `16`; the CSV files say on their first line that they are not measurements
`data/checks/`	the report checking the derivations in Appendix A (script `17`), the report of the audit of the first version and the notes from checking the bibliography
`data/sha256_outputs.txt`	the manifest described in Appendix B.1
`results/numbers/`	one JSON file per script, with the numbers it computed
`results/numbers.tex` and `.json`	all the numbers together, as LaTeX macros and as JSON
`results/csv/`	intermediate results (coefficients, curves, draws) that the figures read, and `master_table.csv` with every number and its origin
`results/tables/`	the tables `tab_*.tex` in English; in `pt/`, the same tables in Portuguese
`figures/`	the figures in English, in PDF (for LaTeX) and PNG, for this monograph, the article and the slides
`figures/pt/`	the same figures, in Portuguese
`monograph/`	this monograph (`monograph_pt/`, in Portuguese); `paper/`, the short article
`build_test/`	the test PDFs of a single chapter, outside version control

Source: prepared by the author.

an error that says which) and writes `results/numbers.tex`, with two definitions per number: one with a decimal point, for English, and another with a decimal comma and a period as thousands separator, for Portuguese. Very small values, such as some p-values, come out in scientific notation, $a \times 10^{b}$.

The third step is in the monograph's preamble, which reads `numbers.tex` and makes the macro `\R` use the version with the decimal comma; in this English version, it uses the one with the decimal point. The fourth is the text itself: where a result appears, the chapter file has `\R{bPooled}`, not the number. If a key does not exist, the PDF shows its name in bold, between question marks, and LaTeX warns about it in the compilation log; a typo in a key name does not go unnoticed.

Tables follow a similar path. Each script that produces a table calls `write_table`, which writes the English version to `results/tables/` and the Portuguese version to `results/tables/pt/`, with the numbers converted to the Brazilian format. The text includes the table with `\tab{name}`, which reads `tab_name.tex`. The figures are drawn twice, once in each language, by the function `in_both_languages`; the

function `save_fig` writes the PDF and the PNG to `figures/` or to `figures/pt/` and, in Portuguese, swaps the decimal point on the axes for a comma.

The script `18_master_data_table.py` closes the loop: it reads every recorded number and gives each one an origin label (measured, estimated, simulated, illustrative or v1), the same labels used in the sources of the figures and tables. The result is in Appendix C.

B.4 What each script reads and writes

Table B.2 lists the scripts in the order in which `reproduce_all.py` runs them, followed by the program for the proposed study and the three tools I use to build and check the monograph. To avoid repetition, the table leaves out two things that hold for all the numbered scripts from `01` to `16`: each one writes its numbers to `results/numbers/` under the same name as the script, and scripts `02` to `15` read `data/processed/students.csv`. Paths in `results/csv/` appear with the file name only. The module `common.py` is not in the table, because it does not run on its own: it holds the folder paths, the seed, the group colors, the figure style, the functions `record`, `write_table` and `save_fig`, and the statistical power formulas.

Table B.2. Scripts in `code/`, in the order in which they run, with the figures and tables of the monograph they produce, what they read and what they write.

Script	Figures and tables	Inputs	Outputs
`01_prepare_data.py`	none; checks the columns the spreadsheet computes (residual income, amount saved) and stops if any does not match	`data/raw/dados_PIC_208.xlsx`	`data/processed/students.csv`, `students_panel.csv`
`02_balance_attrition.py`	`tab_balance` (Table 5.2); data for the figure `res_balance`	`students.csv`	`balance_std_diff.csv`
`03_main_effects.py`	`tab_main` (Table 6.2), `tab_pairwise` (Table 7.1); ANOVA, regressions, pairs and randomization inference	`students.csv`	`means_by_arm_month.csv`, `main_coefficients.csv`, `pairwise_comparisons.csv`, `ri_permutations.npz`

continued

Table B.2. Scripts in code/ (continued).

Script	Figures and tables	Inputs	Outputs
04_panel.py	data for the figure res_event_study; event study and persistence numbers (the figure meas_persistence is computed again by script 13)	students.csv, students_panel.csv	event_study.csv, persistence.csv
05_use_and_saving.py	tab_use (Table 7.3); saving and logs per week	students.csv	use_saving.csv, use_terciles.csv
06_heterogeneity.py	data for the figure mech_heterogeneity; subgroups and the claims of the first version	students.csv	heterogeneity.csv
07_robustness.py	data for the figures res_specifications and res_quantiles; Lee bounds and worst case	students.csv	specification_curve.csv, quantile_effects.csv
08_audit_v1.py	tab_audit (Table C.1)	students.csv; numbers from scripts 02 to 07	v1_audit.csv
09_power.py	none; minimum detectable effect and power of the comparisons between tools	students.csv	numbers only
11_measurement.py	data for the figures meas_reliability and meas_round_values; intraclass correlation and bound on reliability	students.csv	reliability_bound.csv, round_values.csv, expense_cents.csv
10_mechanisms.py	tab_predictions (Table 7.4); data for the figure mech_scarcity	students.csv, event_study.csv; all numbers already written	scarcity_terciles.csv
12_equivalence_bayes_design.py	tab_equivalence (Table 7.2), tab_design (Table 9.1); data for the figures mech_equivalence and meas_skeptic	students.csv; numbers from script 03	equivalence.csv, skeptic_posterior.csv, followup_design.csv

continued

Table B.2. Scripts in code/ (continued).

Script	Figures and tables	Inputs	Outputs
13_figures.py	the 20 data figures: exp_flow, exp_sample, nine res_*, five mech_* and four meas_*	students.csv and the files in results/csv/ written by scripts 02 to 12	figures/*.pdf | png and figures/pt/*.pdf | png
14_foundations.py	the eight basics_* figures in Chapters 2 and 3	students.csv (only in basics_budget and basics_randomization); the rest is illustrative or formula	figures; data/simulation/basics_draws.json
15_synthetic_worlds.py	worlds_diagnostics; six synthetic worlds (Section 8.6)	students.csv; numbers from script 03	figure; data/simulation/synthetic_worlds.csv, synthetic_worlds_summary.json
16_followup_proposal.py	followup_power (Figure 9.1), followup_rehearsal (Figure 9.2)	no file; imports analyze_followup.py; illustrative parameters in the script itself	figures; data/simulation/followup_power.csv, followup_synthetic.csv, followup_synthetic.json
17_check_derivations.py	none; checks the derivations in Appendix A with sympy and stops if any fails	no file	data/checks/appendix_a_derivations.txt
18_master_data_table.py	tab_master (Table C.2)	results/numbers/*.json	master_table.csv
build_numbers.py	none; the \R macros throughout the text	results/numbers/*.json	results/numbers.tex, numbers.json
analyze_followup.py	none; the analysis program for the proposed study (H1 to H3, Section 9.3)	a CSV with one row per student (synthetic, for now)	the estimates on screen and, if requested, in a JSON file
compile_chapter.py	compiles a single chapter, with the monograph's preamble	monograph/<name>.tex (or monograph_pt/, with --folder); main.aux, if it exists	build_test/ (PDF and log); summary of errors and warnings

continued

Table B.2. Scripts in `code/` (continued).

Script	Figures and tables	Inputs	Outputs
`audit_layout.py`	looks for gaps at the end of columns, and for figures and tables placed far from their first citation	a PDF of the monograph or of a test, and the `.aux` file	report on screen
`check_revision.py`	checks whether a text revision changed any fact	two versions of a `.tex` file	report on screen; exit code 1 if numbers, keys, citations, labels, cross-references, equations, sources or figures changed

Source: prepared by the author, from the scripts themselves.

Two remarks about the table. Script 13 also does some calculations for the figures, such as the fitted lines in the figure on use, and writes a single kind of number of its own: each group's mean saving in reais at baseline and in October, which the figure `res_reais` shows. And script 18 writes no numbers of its own: it only reorganizes those that already exist, which is why it runs at the end.

B.5 Key excerpts

B.5.1 The analysis program for the proposed study

The program below is the one Chapter 9 proposes to register before data collection. It is complete. The function `analyze` takes a table with one row per student, computes the change in balance and the reporting gap (lines 33 and 34) and runs the three regressions in the plan (lines 36 to 47). The first two, H1 and H2, control for the initial balance; the third, H3, has the gap as its outcome and does not need that control. All of them have stratum fixed effects, written as `C(stratum)`, and HC2 errors. The final block makes it possible to run the program straight from the command line on a CSV file; script 16 imports it and calls `analyze` on each synthetic sample.

```
"""Analysis program for the follow-up study proposed in chapter 9.

It is ready before data collection, as part of the pre-registered plan. It takes a
CSV with one row per student and these columns:

    id student code
    group control, placebo or tool
    stratum school grade (randomization is done within each grade)
    initial_balance savings account balance at the start (R$)
```

```
    final_balance savings account balance at 3 months (R$), reported by the bank
    final_report how much the student says they saved over the 3 months (R$)

Usage:
    python3 code/analyze_followup.py file.csv [output.json]

Tests, in the order of the plan:
    H1 (primary) tool against placebo, on the bank balance: ANCOVA on the initial
                 balance, stratum fixed effects, HC2 errors, two-sided at 5%.
    H2 placebo against control, on the bank balance (attention effect).
    H3 reporting bias: (report - change in balance) in the tool arm minus
                 in the control arm.
So far, this program has only run on synthetic data (16_followup_proposal.py).
"""
import json
import sys

import pandas as pd
import statsmodels.formula.api as smf


def analyze(d: pd.DataFrame) -> dict:
    d = d.copy()
    d["change"] = d.final_balance - d.initial_balance
    d["gap"] = d.final_report - d.change
    out = {"n": {g: int((d.group == g).sum()) for g in ("control", "placebo", "
        tool")}}
    tests = {
        "H1_tool_minus_placebo": ("final_balance", ("tool", "placebo")),
        "H2_placebo_minus_control": ("final_balance", ("placebo", "control")),
        "H3_reporting_bias": ("gap", ("tool", "control")),
    }
    for name, (y, (a, b)) in tests.items():
        s = d[d.group.isin([a, b])].assign(treat=lambda x: (x.group == a).astype(
            int))
        controls = " + initial_balance" if y == "final_balance" else ""
        f = smf.ols(f"{y} ~ treat{controls} + C(stratum)", s).fit(cov_type="HC2")
        lo, hi = f.conf_int().loc["treat"]
        out[name] = {"estimate": float(f.params["treat"]), "ci95": [float(lo),
            float(hi)],
                     "p": float(f.pvalues["treat"]), "n": int(f.nobs)}
    return out


if __name__ == "__main__":
    if len(sys.argv) < 2:
        print(__doc__)
        sys.exit(1)
    res = analyze(pd.read_csv(sys.argv[1], comment="#"))
    text = json.dumps(res, indent=1, ensure_ascii=False)
    print(text)
    if len(sys.argv) > 2:
```

```
        open(sys.argv[2], "w").write(text + "\n")
```

B.5.2 The core of the main regressions

The excerpt below, from `03_main_effects.py`, estimates the four columns of Table 6.2 and the pooled effect cited throughout the text. The dictionary `specs` holds the four formulas: the group indicators only; with initial saving (the ANCOVA); with the covariates and grade fixed effects; and the same thing in reais. All are estimated by least squares with HC2 errors (`cov_type="HC2"`). Then come the model with a single treatment indicator, which gives the pooled effect `bPooled`, the same calculation in reais, and the effect in control-group standard deviations. Each result goes through `record`, as described in Appendix B.3.

```
specs = {
    "(1)": ("m3 ~ T_G2 + T_G3 + T_G4", "m3"),
    "(2)": ("m3 ~ T_G2 + T_G3 + T_G4 + base", "m3"),
    "(3)": ("m3 ~ T_G2 + T_G3 + T_G4 + base + age + female + rural + notebook"
            " + essential + C(grade)", "m3"),
    "(4)": ("brl_m3 ~ T_G2 + T_G3 + T_G4 + brl_base + age + female + rural"
            " + notebook + essential + C(grade)", "brl_m3"),
}
fits = {k: smf.ols(f, d).fit(cov_type="HC2") for k, (f, _) in specs.items()}
for k, ft in fits.items():
    tag = "abcd"[int(k[1]) - 1] # specification (1)..(4)
    for a in ("G2", "G3", "G4"):
        b, se = ft.params[f"T_{a}"], ft.bse[f"T_{a}"]
        lo, hi = ft.conf_int().loc[f"T_{a}"]
        record(S, f"b{L[a]}{tag}", b, ".1f")
        record(S, f"se{L[a]}{tag}", se, ".1f")
        record(S, f"lo{L[a]}{tag}", lo, ".1f")
        record(S, f"hi{L[a]}{tag}", hi, ".1f")
    record(S, f"rsq{tag}", ft.rsquared, ".2f")
    record(S, f"pEq{tag}", float(ft.f_test("T_G2 = T_G3, T_G3 = T_G4").pvalue), "
        .2f")
    record(S, f"pHybApp{tag}", float(ft.t_test("T_G4 - T_G3 = 0").pvalue), ".2f")
    record(S, f"pHybSheet{tag}", float(ft.t_test("T_G4 - T_G2 = 0").pvalue), ".2f"
        )

pooled = smf.ols("m3 ~ treated + base + age + female + rural + notebook +
    essential"
                 " + C(grade)", d).fit(cov_type="HC2")
record(S, "bPooled", pooled.params["treated"], ".1f")
record(S, "loPooled", pooled.conf_int().loc["treated", 0], ".1f")
record(S, "hiPooled", pooled.conf_int().loc["treated", 1], ".1f")
pooled_brl = smf.ols("brl_m3 ~ treated + brl_base + age + female + rural +
    notebook"
                     " + essential + C(grade)", d).fit(cov_type="HC2")
record(S, "bPooledBRL", pooled_brl.params["treated"], ".2f")
record(S, "loPooledBRL", pooled_brl.conf_int().loc["treated", 0], ".2f")
```

```
record(S, "hiPooledBRL", pooled_brl.conf_int().loc["treated", 1], ".2f")
record(S, "bPooledBRLyear", 12 * pooled_brl.params["treated"], ".0f")
record(S, "bPooledBRLr", pooled_brl.params["treated"], ".0f")
record(S, "loPooledBRLr", pooled_brl.conf_int().loc["treated", 0], ".0f")
record(S, "hiPooledBRLr", pooled_brl.conf_int().loc["treated", 1], ".0f")
# effect relative to the control mean
ctrl3 = d.loc[d.arm == "G1", "m3"].mean()
record(S, "relPooled", 100 * pooled.params["treated"] / ctrl3, ".0f")
record(S, "ctrlMeanThree", ctrl3, ".1f")
record(S, "ctrlMeanBRL", d.loc[d.arm == "G1", "brl_m3"].mean(), ".2f")
# effect in control-group SDs
record(S, "effSD", pooled.params["treated"] / d.loc[d.arm == "G1", "m3"].std(), "
    .1f")
```

APPENDIX C

Tables

This appendix brings together two tables that are too long for the chapters. The first checks, one by one, the numerical claims of the first version of this work against the microdata (Appendix C.1). The second gathers the numbers the text uses most, each with its origin and the script that computed it (Appendix C.2). Both are generated by code and enter the text without hand editing, as Appendix B explains.

C.1 The audit of the first version

The first version of this work consists of two texts: the October 2025 paper (Ribeiro, 2025a) and the December 2025 extended abstract (Ribeiro, 2025b). The script `08_audit_v1.py` takes each numerical claim in these two texts, redoes the calculation with the microdata and sorts the result into one of four categories. "Match" means the data give the same value, to the precision at which the first version reported it. "Differs" means the value changes but the conclusion does not. "Not significant" means the direction holds, without statistical support. "Not reproduced" means the data do not show what the claim says.

Of the 24 claims checked, 13 match, 5 differ in value without changing the conclusion, 4 point in the right direction but are not statistically significant, and 2 do not reproduce. Table C.1 shows all of them, with the section of the first version in which each one appears.

The pattern across the rows is clear. All the descriptive statistics match: the sample size, the dropouts, the breakdown by gender, origin, age and grade, the essential expenses and the sixteen monthly means by group, down to the second decimal place. This confirms that the first version and this one use exactly the same data. The differences lie mostly in the inferential results. The between-group sum of squares of the ANOVA reported in the first version, 88,541.3, cannot be right, because the first version's own table of means and group sizes implies 106,151, the same value the data give (Section 6.2). The slope of saving on use of the tool does not reproduce under either of the two possible readings of the outcome, and the students who never logged an expense reported the same as those who did (Section 7.3). The three claims about subgroups and the one about the hybrid's "synergy" have no statistical support; the direction holds, except for the part of the claim about rural and urban students that concerns the app, where the two groups are tied (Sections 7.1 and 7.4).

Two typos in the first version are not in the table, because they are not claims about the data, and they were corrected in this version without further comment: the year of the work by Lusardi and Mitchell (2014), written as 2012 in the text, and the sentence that announced “six” sections when the paper had seven. The audit report, with the calculations for each row, is in `data/checks/v1_audit.md`, and the table in an open format is in `results/csv/v1_audit.csv`.

Table C.1. Claims of the first version checked against the microdata.

v1 §	Claim	v1 says	Microdata give	Status
3.2	Students randomized / analyzed	208 / 203	208 / 203	match
3.4.1	Dropouts by arm (G1/G2/G3/G4)	2 / 0 / 1 / 2	2 / 0 / 1 / 2	match
3.2.1	Girls / boys	129 / 79	129 / 79	match
3.2.1	Urban / rural	151 / 57	151 / 57	match
3.2.1	Notebook / tablet / all three	86 / 23 / 6	86 / 23 / 6	match
Tab. 1	Age distribution, 11 to 18	4, 14, 14, 26, 50, 46, 38, 16	4, 14, 14, 26, 50, 46, 38, 16	match
Tab. 2	Grade distribution, 6 to 12	8, 16, 24, 40, 52, 44, 24	8, 16, 24, 40, 52, 44, 24	match
3.2	Same grade mix in every arm	yes	yes	match
3.4.2	Essential expenses by arm (R$)	157.86 / 163.74 / 149.21 / 154.47	157.86 / 163.74 / 149.21 / 154.47	match
3.4.2	Mean residual income (R$)	143.68	143.68	match
3.4.2	Essential expenses balanced across arms	yes	F test: $p = 0.04$ *chance imbalance; controlled for in Table 3, col. 3*	differs
Tab. 3	Monthly means by arm (16 cells)	Table 3	16 of 16 equal	match
5.1	Spreadsheet arm “more than doubled”	> 2×	1.99× *essentially double*	match
Abstr.	Hybrid: “>130% increase”	>130%	+169% *baseline to October*	match
Tab. 4	Between-group sum of squares	88,541.3	106,150.8 *the v1 means imply 106,151*	differs
Tab. 4	Within-group sum of squares	77,864.5	114,029.4	differs
Tab. 4	$F(3, 199)$	75.43	61.75 *both* $p < 10^{-25}$	differs
5.3	Slope of saving “variation” on logs per week	+12.1 ($p < 0.01$)	points: –1.33 ($p = 0.16$); relative: +33.6 ($p = 0.30$) *not significant under either reading*	not reproduced
6.1/7.1	Use of the tool goes with better results	yes	never logged 97%, logged 97%	not reproduced

Table C.1. Claims of the first version checked against the microdata (continued)

v1 §	Claim	v1 says	Microdata give	Status
5.4	Girls save more than boys (treated, October)	+4.1 pp	+4.7 with controls (p = 0.28) *direction holds*	not significant
5.4	Largest effect at ages 15 to 17	yes	53.3 vs. 50.2 (p = 0.70) *direction holds*	not significant
5.4	Rural students better with spreadsheet, urban with app	yes	sheet: rural 57, urban 44; app: 55, 55 (p = 0.26) *holds for the spreadsheet only*	not significant
6.1	Hybrid best (“synergy”)	yes	hybrid – app = 3.2 pp (p = 0.50) *highest mean, arms not distinguishable*	not significant
Fig. 3, 4	Distribution and usage plots	“illustrative values”	plots of the 203 records	differs

Notes: “v1 §” is the section, table or figure of the first version in which the claim appears (Abstr.: the paper’s abstract). In each row, the italic note below the value explains the classification.

Source: v1 (columns “Claim” and “v1 says”) and estimated with the study data (column “Microdata give”), `code/08_audit_v1.py`.

C.2 General table of data

Every number in this work that comes from the data, from an estimate or from a simulation is recorded by the script that computed it (Appendix B.3). The script `18_master_data_table.py` reads all these numbers and gives each one an origin label, the same one used in the sources of the figures and tables in the monograph:

- measured: taken straight from the spreadsheet, as the students reported it, or a simple mean of those values (for example, the mean saving a group reported in a month);
- estimated: computed from the data by a model or a test (a regression, a confidence interval, a p-value);
- simulated: comes from synthetic data generated by scripts `14` to `16`, and says nothing about the real students;
- illustrative: a parameter I chose to explain an idea, such as the standard deviation of the balance in the simulation of Chapter 9;
- v1: a value published in the first version of the work, kept for comparison.

The label is given by the step that produced the number, with exceptions listed one by one in the script itself. The numbers from script `01` are measured; those from scripts `02` to `12`, estimated; those from script `13`, which are each group’s means in reais, measured; those from scripts `14` to `16`, simulated. The main exceptions are

the monthly means by group (measured), the chosen parameters of the simulations (illustrative), the ANOVA values of the first version (v1) and some numbers from scripts `14` and `15` that come from the real data rather than from the simulation (measured or estimated).

Table C.2 shows only the numbers the text uses most, grouped by subject. The column "Step" is the number of the script that computed each one, and the column "Unit" uses pp for percentage points. The full list, with every recorded number, its key in the text, the script and the origin, is in the file `results/csv/master_table.csv`, which can be opened in any spreadsheet program.

Table C.2. Every number in the text that matters most, with its origin.

Quantity	Value	Unit	Origin	Step
Sample				
PIC scholars contacted	389		measured	01
students randomized	208		measured	01
students who finished	203		measured	01
share female	62.0	%	measured	01
share rural	27.4	%	measured	01
mean age	15.3	years	measured	01
essential spending at baseline, mean	156.32	R$	measured	01
residual income at baseline, mean	143.68	R$	measured	01
Reported saving, % of residual income				
control, baseline	41.27	%	measured	03
control, October	44.25	%	measured	03
spreadsheet, October	91.40	%	measured	03
app, October	97.82	%	measured	03
hybrid, October	101.07	%	measured	03
Main effects				
any tool vs control, ANCOVA	52.0	pp	estimated	03
the same effect in reais per month	71.29	R$	estimated	03
spreadsheet vs control	47.1	pp	estimated	03
app vs control	53.6	pp	estimated	03
hybrid vs control	56.8	pp	estimated	03
hybrid minus spreadsheet	9.7	pp	estimated	03
Holm-adjusted p, hybrid vs spreadsheet	0.08		estimated	03
smallest possible randomization p	0.00005		estimated	03
specifications with p < 0.05	100	%	estimated	07
Lee bound, lower	51.4	pp	estimated	07
worst-case imputation	47.5	pp	estimated	07
Use and mechanisms				
logs per week, treated, mean	2.0	per week	estimated	05
treated who never logged	16	%	estimated	05
slope of saving on logs per week	−1.33	pp	estimated	05
p, never-loggers vs loggers	0.97		estimated	05
effect gradient per SD of residual income	−7.3	pp	estimated	10
baseline × treatment interaction	−2.3	pp	estimated	10

Table C.2. Every number in the text that matters most, with its origin (continued).

Quantity	Value	Unit	Origin	Step
Measurement				
intraclass correlation of reports	−0.05		estimated	11
reliability bound if $\rho = 0.5$	0.16		estimated	11
monthly reports in whole reais	2.2	%	estimated	11
Power and design				
minimum detectable difference between two tools	13.5	pp	estimated	09
power for an effect the size of the literature	10	%	estimated	09
students per arm for 0.2 SD, $\rho = 0.5$	328		estimated	12
Synthetic worlds and the proposed study				
power to see the use slope, attention world	100	%	simulated	15
ICC, equal effect with noisy reports	0.05		simulated	15
power, 0.2 SD, 300 per arm	81	%	simulated	16
H1 estimate on synthetic data	20.3	R$	simulated	16
true H1 effect in the synthetic data	14	R$	illustrative	16
Illustrative parameters				
present-bias factor β in chapter 2	0.7		illustrative	14
weekly discount factor δ	0.99	per week	illustrative	14
persistence of true saving in the worlds	0.5		illustrative	15

Source: `code/18_master_data_table.py`, from the numbers recorded by scripts 01 to 16; the origin of each row is in the column “Origin”.